

The middle and the ends III.

Face selection for slowly switching Markov chains and the planar persistent walk

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Abstract

On Kagey's persistent Galton board each bounce repeats the previous one with probability p , and Kagey asked when the middle bin becomes as likely as an endpoint. We ask the same question for a Markov chain on d states run for N steps, which jumps from state i to state j with probability $q_{ij}T/N$. The bin is now the vector of visit counts. Write $T = \tau \log N$ and let λ_F be the rate at which the chain leaves a set of states F . To first order the most likely count vector is supported by a set F that minimizes $\tau\lambda_F + |F| - 1$: staying on F costs $N^{-\tau\lambda_F}$, and spreading over F costs a factor N per extra state. At second order a local limit theorem gives a crossing equation in which $\log \log N$ has the coefficient $\frac{1}{2}$ for strongly connected sets with start mass, and a directed 3-cycle shows that this can fail otherwise. Our main application is the persistent walk on the square lattice that turns with rate r to each side and reverses with rate s . The corner and the origin are equally likely exactly once, when $T/\log N$ is close to $\min(2/(2r+s), 1/s)$, and when reversals dominate the crossing is the 1-dimensional one for $N/2$ steps. We also treat sticky priors for hidden Markov models and the information in the endpoint.

Keywords: persistent random walk; correlated random walk; Galton board; occupation times of Markov chains; mode; principal Dirichlet eigenvalue; quasi-stationary distribution; local limit theorem; Ising chain; hidden Markov model.

MSC2020: 60J10; 60F05; 82B20; 60F10; 60G50; 62M05.

1 Introduction

Suppose you drop a ball through a Galton board with N rows of pegs. At the first peg the ball goes left or right with probability $\frac{1}{2}$. At every later peg it repeats its previous direction with probability p and switches with probability $q = 1 - p$. Problem 131 of Kagey's Open Problem Collection [1] asks for the law of the bin where the ball lands, and in particular when the middle bin is exactly as likely as an endpoint. An endpoint is reached by a single path, the one that never switches, so it has probability $\frac{1}{2}p^{N-1}$. When p is close to 1 this path wins, and when p is smaller the middle wins. Paper A of this series [2] answers the question at every N . The 2 are equally likely when the expected number of switches $T = Nq$ is close to $\log N$. More precisely, the crossing value z_N of T satisfies $z_N + \frac{1}{2} \log z_N = \log N + \frac{1}{2} \log \frac{\pi}{8} + o(1)$.

Kagey also asks what changes on other boards. In this paper the ball can move in more than 2 directions. Our main example is Kagey's board in 2 dimensions. A ball moves on the square lattice $\mathbb{Z}^2$, one step east, north, west or south at a time. It starts in a uniform direction, and at

each step it keeps its direction, turns left or right, or reverses. Turns and reversals are rare: a turn to each side has probability rT/N and a reversal has probability sT/N . After N steps, which is more likely: the corner $(N, 0)$, which only the path that never turns can reach, or the origin? We show that for every even $N \geq 4$ there is exactly one value T_N^* of T at which the 2 are equally likely, and that

$$\frac{T_N^*}{\log N} \rightarrow \min\left(\frac{2}{2r+s}, \frac{1}{s}\right)$$

(Proposition 6.2 and Theorem 6.3). When reversals dominate ($s > 2r$) the origin is reached mostly by paths that stay on one axis, and then sT_N^* is the crossing $z_{N/2}$ of Kagey's board with $N/2$ steps, up to $O(N^{-\gamma})$ (Theorem 6.4). When turns dominate ($s < 2r$) the origin is reached mostly by paths that use all 4 directions.

1.1 Face selection

The planar result is an instance of a general principle, which is the main subject of this paper. Let the direction be a Markov chain on d states that switches from state i to state j with probability $q_{ij}T/N$ at each step, and count the visits to each state. This count vector, the *composition*, plays the role of the bin. Its support is a set F of states, which we call a *face* (it indexes a face of the simplex of visit fractions). Which face carries the most likely composition, and at which T does the mode move to another face? On Kagey's board there are 2 states, the faces of 1 state are the 2 endpoints and the face of 2 states is the set of interior bins.

Put $T = \tau \log N$ and let λ_F be the rate at which the chain leaves F , that is, the principal Dirichlet eigenvalue on F of $-Q$, where Q is the matrix of the rates q_{ij} . How likely the best composition with support F is depends on 2 effects.

- *The cost of staying.* The chain must stay inside F for all N steps. This has probability about $e^{-T\lambda_F} = N^{-\tau\lambda_F}$. A larger face is harder to leave, so staying on it is cheaper.
- *The cost of spreading.* The mass of F is shared among the compositions with support F , and there are about $N^{|F|-1}$ of them, one factor N for each free coordinate. Up to powers of $\log N$, the best single composition gets a share $N^{-(|F|-1)}$.

So the most likely composition with support F has probability $N^{-\Phi_\tau(F)+o(1)}$, where

$$\Phi_\tau(F) = \tau\lambda_F + |F| - 1,$$

and every mode sits on a face that minimizes Φ_τ (Theorem 3.3). In words, a sticky walk prefers a well-separated community, a set of states that is hard to leave for its size. As τ grows the cost of staying counts for more, and the mode moves to larger faces. On Kagey's board an endpoint has $\Phi_\tau = \tau$ and the interior has $\Phi_\tau = 1$, so the middle takes over at $\tau = 1$, which is the first-order form of the answer of Paper A. Section 3.1 works out 4 states on a cycle by hand, and Theorem 3.4 reads the winners off the convex hull of the points $(\lambda_F, |F| - 1)$.

The planar walk fits this picture. The corner is the face of the single direction east, with exit rate $2r + s$, so it has probability about $N^{-(2r+s)\tau}$. The origin is reached either along an axis, with exit rate $2r$, or with all 4 directions, with exit rate 0. Comparing the 3 exponents gives the limit above (Section 6).

First order fixes the crossing only up to a factor $1 + o(1)$. To locate it up to $o(1)$ we need the constant in front. Say that a face F satisfies (H) if the chain can move between any 2 states of F without leaving F , and the start gives F positive mass. For such faces we prove a local limit

theorem inside the face (Theorem 4.4). It shows (Theorem 4.6) that the best composition with support F , with $k = |F|$ states, has probability

$$M_F = N^{-(k-1)} T^{(k-1)/2} K_F e^{-T\lambda_F} (1 + O(1/T + T^2/N))$$

uniformly for $1 \leq T \leq N^{1/3}$, with an explicit constant K_F . The factor $T^{(k-1)/2}$ appears because the visit fractions fluctuate on the scale $T^{-1/2}$ in each of the $k - 1$ free directions. Now let F and G satisfy (H), and suppose that F has $\Delta \geq 1$ more states than G and an exit rate smaller by $\delta > 0$. Then for large N every $T \in [1, N^{1/3}]$ with $M_F = M_G$ satisfies

$$\delta T + \frac{\Delta}{2} \log T = \Delta \log N - \log \frac{K_F}{K_G} + O\left(\frac{1}{\log N}\right)$$

(Theorem 4.7). So the coefficient $\frac{1}{2}$ of $\log \log N$ that Papers A and B found holds for every such pair, whatever the graph, the rates, the sizes or the start, and without reversibility. For reversible faces K_F involves the square root of a weighted count of spanning trees (Proposition 4.2), and the constants of Papers A and B are short corollaries (Corollaries 4.10 and 4.11). Without (H) the coefficient can change. On a directed 3-cycle it is 1 at one crossing and 0 at the next (Proposition 4.13). Section 5 computes all the constants on the 4-cycle, shows that the start can decide which face wins, and lists the winners on other graphs.

The paper closes with 2 applications, and nothing before them depends on them. Section 7 applies face selection to sticky priors for hidden Markov models, and Section 8 asks how much of the Fisher information about the switching rate is left when we only see the bin of Kagey's board. Section 2 fixes the notation, Section 9 describes the computer checks and open questions, and the longer proofs are in the appendices.

1.2 The series

This is the third of 4 papers on Problem 131, *The middle and the ends I–IV*. We call them Papers A to D, so that Paper A [2] is part I, Paper B [3] is part II, this paper is part III and Paper D [4] is part IV. They share the following notation.

- Kagey's board has N steps (his row n has $N = n - 1$). The first step is left or right with probability $\frac{1}{2}$, and each later step repeats the previous one with probability p . We write $q = 1 - p$ and $u = q/p$, the odds of a switch.
- $P_N(k)$ is the probability of k right steps, that is, that the ball lands in bin k . The endpoints have $P_N(0) = P_N(N) = p^{N-1}/2$.
- $L = \log N$, and $T = Nq$ is the expected number of switches (up to the first step).
- The crossing is the value of p (or T) at which a bin is exactly as likely as an endpoint. For the middle bin it is p_N , and $z_N = N(1 - p_N)$.
- With d states, a face F is a nonempty set of states, and the vertices are the faces of size 1. λ_F is the principal Dirichlet eigenvalue of $-Q$ on F , and K_F is the second-order constant.

Paper A solves Kagey's board: uniform Bessel bounds bracket z_N for every $N \geq 2$ and give $z_N + \frac{1}{2} \log z_N = L + c_0 + 1/(8z_N) + R_N$ with $c_0 = \frac{1}{2} \log(\pi/8)$ and an explicit R_N . Paper B lets the ball switch to each of $d - 1$ other directions at the same rate; there every face ties at first order, and constants a_k break the tie at second order. Paper D asks what memory changes. The present paper treats every rate matrix Q . It explains which face carries the mode and why, computes the constants K_F , recovers c_0 and a_k as special cases, and applies all this to the planar walk. Figure 1 compares the crossings of the 4 papers.

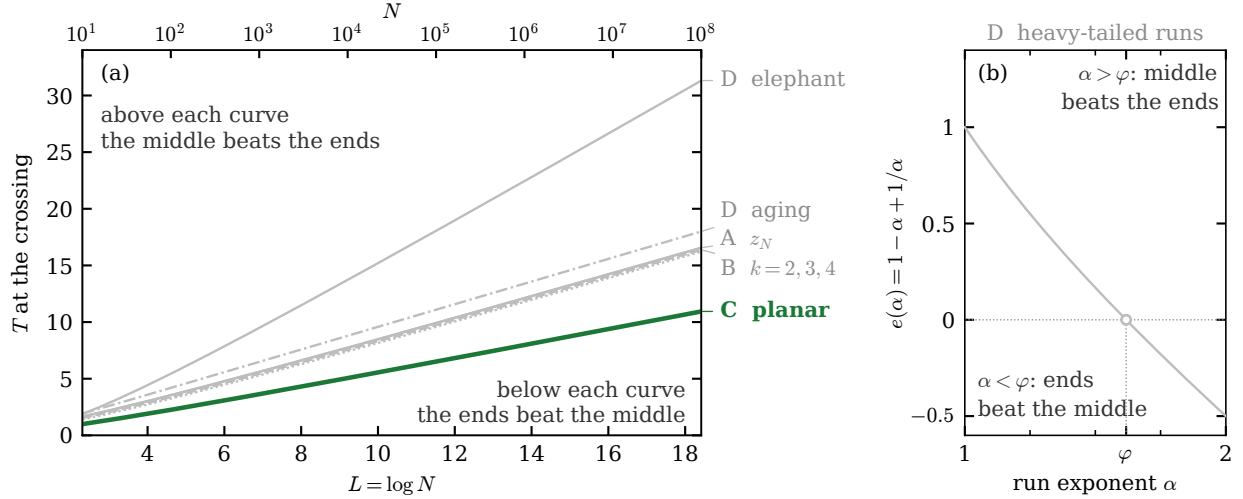


Figure 1: (a) The crossing on each model's switching scale against $L = \log N$ for Papers A to D. The scale is T for Papers A to C, $x = n\varepsilon$ for the elephant walk and the expected number of switches for the aging walk of Paper D. The bold curve solves the first crossing equation of Remark 6.5 for the planar walk with $r = s = 1$, without its $o(1)$ term. The curve A is z_N , and the curves B are the face crossings of Paper B. (b) The exponent $e(\alpha) = 1 - \alpha + 1/\alpha$ of Paper D for runs with heavy tails, which changes sign at the golden ratio φ .

1.3 Related work

The rate I_F of Section 3 is the Donsker–Varadhan rate of the occupation measure of the chain killed on leaving F . Dupuis and Liu give this rate explicitly for reversible jump processes [5, Eq. (3.2)], and on a finite chain their formula agrees with Proposition 3.1(c). Guillin, Nectoux and Wu [6] prove a large deviation principle for the occupation measure of a process conditioned to stay in a set. For reversible processes they show that the exit rate of the set is the least value of the Donsker–Varadhan rate over laws on the set (their Cor. 2). In general they prove an inequality between their rate, the Donsker–Varadhan rate and the exit rate, and they conjecture equality (their (2.15)). For a finite chain with Q_F irreducible, the fact that λ_F is the least value of I_F is the variational formula of Donsker and Varadhan [7], which needs no reversibility; Proposition 3.1(a) gives a short proof in our notation. This does not address the conjecture of Guillin, Nectoux and Wu, which concerns their conditional rate. The quasi-ergodic law goes back to Darroch and Seneta [8, 9], and Colonius and Rasmussen [10] treat the reducible case. Dietz and Sethuraman [11] study chains that switch from i to j at time n with probability $G(i, j)/n$, where every $G(i, j) > 0$. Their chains also make about $\log n$ switches, but spread log-uniformly in time, and their occupation fractions have a limit law with full support on the simplex. In [12] they prove large deviations for chains whose transition matrices converge to a reducible matrix.

Exact laws of transition counts go back to Whittle [13], Dawson and Good [14] and Goodman [15]. Polettini [16] and Carugno, Vivo and Coghi [17] derive them from the BEST theorem and the matrix-tree theorem when every count grows linearly. We group the words by runs instead (Lemma 2.3), which suits a regime with $O(\log N)$ switches. Bertini, Gabrielli and Landim [18] and Landim [19] expand the level-two rate functions of finite chains whose rates depend on a parameter. They describe the time scales of such chains but do not ask where the mode of the composition sits. The sticky HDP-HMM of Fox, Sudderth, Jordan and Willsky [20] adds a self-transition bias

to the hierarchical Dirichlet process hidden Markov model.

On the lattice the correlated random walk goes back to Gillis [21]. Chen and Renshaw [22] treat the Gillis–Domb–Fisher walk, and in [23] they show that the characteristic function of any correlated random walk satisfies a recurrence. So in principle the law of our planar walk is known. Marris and Giuggioli [24] give lattice propagators for walks with forward, backward and lateral steps. In the continuum, Santra, Basu and Sabhapandit [25] find the marginals of a run-and-tumble particle with 4 directions, and Angelani [26] treats switching among 4 orthogonal directions. There the corners carry atoms and the origin only a density, so the comparison of a corner with the origin is a lattice question. Smiga, Radaelli, Binder and Landi [27] compute the Fisher information of the empirical distribution under a Gaussian approximation. Their leading term vanishes for the symmetric walk of Paper A, whose stationary law does not depend on the switching rate. We did not find the face-selection question or the planar crossing of the corner and the origin in the literature.

2 Setting, notation and the exact law

2.1 Setting and notation

The chain. Fix $d \geq 2$ and a generator Q on $[d] = \{1, \dots, d\}$: the rates are $q_{ij} \geq 0$ for $i \neq j$, the exit rates are $q_i = \sum_{j \neq i} q_{ij}$, $Q_{ii} = -q_i$, and $q_{\max} = \max_i q_i$. When we write q for the vector (q_i) this is clear from the context; the scalar $q = 1 - p$ of Kagey’s board only appears in Example 2.5(i) and in Section 8. The start μ is a probability law on $[d]$, and $\mu(F) = \sum_{i \in F} \mu_i$. For $N \geq 2$ and $T > 0$ put

$$h = \frac{T}{N}, \quad L = \log N, \quad \tau = \frac{T}{L}.$$

The chain $X_1, \dots, X_N$ has $X_1 \sim \mu$ and transition matrix $I + hQ$. So at each step it jumps from i to j with probability hq_{ij} , and N steps look like the continuous-time chain with generator Q run for time T . We assume $hq_{\max} \leq \frac{1}{2}$ unless we say otherwise. The *composition* is $n = (n_1, \dots, n_d)$ with $n_i = \#\{t \leq N : X_t = i\}$, we write $P_N(n)$ for its probability and $\alpha = n/N$ for the visit fractions.

Faces. A face is a nonempty set $F \subseteq [d]$, of size $k = k_F = |F|$. Let Δ_F be the set of probability vectors on $[d]$ that vanish off F , and Δ_F° the set of those that are positive on F . We number the states of F as $1, \dots, k$ and use the coordinates $(\alpha_2, \dots, \alpha_k)$ on Δ_F and $(t_2, \dots, t_k)$ on $\{t \in \mathbb{R}^F : t_1 + \dots + t_k = T\}$. We write $x \cdot y = \langle x, y \rangle = \sum_{i \in F} x_i y_i$ and $x \circ y = (x_i y_i)_{i \in F}$. Let $A_F = (q_{ij})_{i, j \in F}$ with zero diagonal, and $Q_F = A_F - \text{diag}(q_i)_{i \in F}$. Since q_i includes the rates of leaving F , Q_F generates the chain killed when it leaves F . The switching graph G_F has an edge $i \rightarrow j$ whenever $i, j \in F$, $i \neq j$ and $q_{ij} > 0$.

Perron roots. A square matrix B is Metzler if its off-diagonal entries are nonnegative, and then its Perron root $\varrho(B)$ is its largest real eigenvalue. If B is irreducible, $\varrho(B)$ is simple and has positive left and right eigenvectors [29].

Definition 2.1 (faces). Let F be a face. We call F *irreducible* if G_F is strongly connected, so every singleton is irreducible. We call F *admissible* if for some N some path $x_1 \cdots x_N$ of positive probability has support exactly F . We say that F satisfies (H) if F is irreducible and $\mu(F) > 0$. The *exit rate* of F is $\lambda_F = -\varrho(Q_F)$, so $\lambda_{\{i\}} = q_i$. If F is irreducible, r_F and l_F are right and left eigenvectors of Q_F for $-\lambda_F$ with positive entries and $\langle l_F, r_F \rangle = 1$, and $\alpha_F^* = l_F \circ r_F$ is the *quasi-ergodic law* of F [8, 9]. The *face maximum* is $M_F = \max\{P_N(n) : \text{supp } n = F\}$, with $\max \emptyset = 0$.

Here is why λ_F is the rate of leaving F . If F is irreducible, then $e^{TQ_F}r_F = e^{-T\lambda_F}r_F$, so the chance that the continuous-time chain started in F stays in F up to time T is $e^{-T\lambda_F}$ up to a bounded factor. If F is reducible, we order the strongly connected components $C_1, \dots, C_r$ of G_F so that every edge between components goes forward. Then Q_F is block triangular with diagonal blocks Q_{C_j} , and $\lambda_F = \min_j \lambda_{C_j}$. In all cases $0 \leq \lambda_F \leq \min_{i \in F} q_i$, since the rows of Q_F have sums at most 0 and a Perron root is at least every diagonal entry. The quasi-ergodic law α_F^* is the long-run law of the visit fractions of the chain conditioned to stay in F ; it is where the best composition on F sits.

Later sections add their own symbols where they are needed: the rate I_F (Section 3), the tilted chain and the constant K_F (Definition 4.1), and the planar notation of Section 6. For Kagey's board we use the following. For $a, b \geq 1$ let $S_{a,b}(u) = \sum_w u^{s(w)}$, the sum over the words w with a right steps and b left steps, where $s(w)$ is the number of switches of w . Then $S_{a,b}(q/p)$ is the ratio $P_N(a)/P_N(0)$ of the probabilities of bin a and bin 0 after $a+b=N$ steps. Let $S_N = S_{\lfloor N/2 \rfloor, \lfloor N/2 \rfloor}$. It has nonnegative coefficients and no constant term, so $S_N(u) = 1$ has exactly 1 positive root u_N , and $p_N = 1/(1+u_N)$ is the crossing of the middle and the endpoints. Put $z_N = N(1-p_N)$ and $c_0 = \frac{1}{2} \log(\pi/8)$. Paper A proves the crossing equation $z_N + \frac{1}{2} \log z_N = L + c_0 + 1/(8z_N) + R_N$ with an explicit remainder $R_N = O(1/L^2)$.

Conventions. Unless we say otherwise, Q , μ and the faces under discussion are fixed, and every constant, every implied constant in $O(\cdot)$ and every threshold N_0 depends only on them and on the compact set $\mathcal{K}$ when one appears. A bound that holds “uniformly for $1 \leq T \leq N^{1/3}$ ” holds for all $N \geq N_0$ and all T in that range with one implied constant. Subscripts as in $O_{\mathcal{K}}$ stress a dependence.

2.2 Admissible faces

Lemma 2.2 (admissible faces). *Let F be a face and $C_1, \dots, C_r$ the strongly connected components of G_F . (a) F is admissible if and only if the components can be numbered so that $\mu(C_1) > 0$ and, for each $j < r$, $q_{xy} > 0$ for some $x \in C_j$ and $y \in C_{j+1}$. (b) An irreducible face is admissible if and only if $\mu(F) > 0$. So (H) holds exactly for the irreducible admissible faces. (c) If Q has symmetric support, that is $q_{ij} > 0$ exactly when $q_{ji} > 0$, then every admissible face is irreducible.*

Proof. The states of the maximal runs of a path of positive probability with support F form a walk in G_F that starts in $\text{supp } \mu$ and visits every state of F , and every such walk arises in this way for large N . (a) Along such a walk a component C cannot be visited again after a state $y \notin C$, since then y and C would be reachable from each other. Hence a path of positive probability visits each strongly connected component of its support in one block of consecutive times, and it passes from one block to the next by a single switch. Conversely, given the numbering and edges $x_j \rightarrow y_{j+1}$, take $y_1 \in C_1$ with $\mu_{y_1} > 0$ and any $x_r \in C_r$, and join walks from y_j to x_j through all of C_j by these edges. (b) is (a) with $r = 1$. (c) Here an edge between 2 components would give an edge back, so no edge joins 2 components, and $r = 1$ by (a). $\square$

So $M_F > 0$ for large N if F is admissible, and $M_F = 0$ otherwise.

2.3 The exact law

For $m \in \mathbb{Z}_{\geq 1}^F$ put $M = \sum_{i \in F} m_i$ and $W(m) = \sum_c \mu_{c_1} \prod_{j < M} q_{c_j c_{j+1}}$, the sum over the words $c = c_1 \cdots c_M$ in F with $c_j \neq c_{j+1}$ in which each $i \in F$ occurs m_i times. Such a word lists the states of the runs of a path, and m_i is the number of runs in state i .

Lemma 2.3 (run expansion). *Let $hq_{\max} \leq 1$ and let n have support F . Then, with $\binom{n_i-1}{m_i-1} = 0$ for $m_i > n_i$ and $0^0 = 1$,*

$$P_N(n) = \sum_{m \in \mathbb{Z}_{\geq 1}^F} W(m) h^{M-1} \prod_{i \in F} \binom{n_i-1}{m_i-1} (1 - hq_i)^{n_i-m_i}.$$

Proof. The states $c_1 \cdots c_M$ of the maximal runs of a path form a word as in $W(m)$. The path has $M-1$ switches, of probabilities $hq_{c_j c_{j+1}}$, and $n_i - m_i$ stays in state i , of probability $1 - hq_i$ each. For a fixed c the lengths of the runs of state i form one of the $\binom{n_i-1}{m_i-1}$ compositions of n_i into m_i positive parts. Summing over c gives the claim. $\square$

Lemma 2.3 is checked in Lean (Section 9). For $F = \{i\}$ the only path is i^N , so uniformly for $1 \leq T \leq N^{1/2}$

$$M_{\{i\}} = \mu_i (1 - hq_i)^{N-1} = \mu_i e^{-q_i T} (1 + O(T^2/N)). \quad (1)$$

For $t \in (0, \infty)^F$ let $H(t) = \sum_m W(m) \prod_{i \in F} t_i^{m_i-1} / (m_i-1)!$, which is finite since $W(m) \leq k^M q_{\max}^{M-1}$. The next lemma says that H is the density of the occupation times of the continuous-time chain. Lemma 4.5 below shows that the discrete law is close to it.

Lemma 2.4 (the continuum density). *Let $(Y_s)_{s \geq 0}$ be the continuous-time chain with generator Q and $Y_0 \sim \mu$, let $\mathcal{O}_i = \int_0^T \mathbf{1}\{Y_s = i\} ds$, and let E_F be the event that Y stays in F during $[0, T]$ and visits every state of F . Then $\mathbb{P}(E_F, \mathcal{O} \in \cdot)$ has the density $e^{-\langle q, t \rangle} H(t)$ in $(t_2, \dots, t_k)$ on $\{t \in (0, \infty)^F : \sum_i t_i = T\}$, and $\mathbb{P}(E_F, \mathcal{O}/T \in \cdot)$ has the density*

$$f_T(\alpha) = T^{k-1} e^{-T \langle q, \alpha \rangle} H(T\alpha)$$

in $(\alpha_2, \dots, \alpha_k)$. For n with support F and $t = hn$, $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q, t \rangle} H(t)$. For $F = \{i\}$, $f_T = \mathbb{P}(E_F) = \mu_i e^{-q_i T}$.

Proof. Up to a null set, E_F is the disjoint union of the events E_c that Y visits exactly $c_1, \dots, c_M$ in this order, over the words c in the sums $W(m)$. Fix c , let $s_1, \dots, s_{M-1}$ be the first $M-1$ holding times and $s_M = T - \sum_{j < M} s_j$. The last holding time only has to exceed s_M . So on $\{s_j > 0 \text{ for } j \leq M\}$ the event E_c has density $\mu_{c_1} \prod_{j < M} q_{c_j c_{j+1}} e^{-\langle q, \mathcal{O} \rangle}$ in $(s_1, \dots, s_{M-1})$. For $i \neq c_M$ we replace the last holding time in state i by $\mathcal{O}_i$, a triangular change of variables with unit diagonal. For fixed $\mathcal{O} = t$ the other $m_i - 1$ holding times in state i range over a simplex of volume $t_i^{m_i-1} / (m_i - 1)!$. Integrating them out and summing over c gives $e^{-\langle q, t \rangle} H(t)$, and any $k-1$ of the coordinates t_i give the same Lebesgue measure. The rest follows from $t = T\alpha$ and $T/N = h$. $\square$

Example 2.5 (the 4 models). (i) For $d = 2$, $q_{12} = q_{21} = 1$ and uniform μ the chain is Kagey's board, the persistent walk of [2], with $p = 1 - h$. So $T = N(1 - p)$, the faces $\{1\}$ and $\{2\}$ are the endpoints, and Lemma 2.3 is the run count of [2]. (ii) For $Q = J - dI$, with J the all-ones matrix, and uniform μ it is the model of [3], which switches to each other state with probability h . (iii) For $Q = \Pi - I$, with Π stochastic with zero diagonal, $\lambda_F = 1 - \varrho(\Pi_F)$. This is the sticky chain of Section 7. (iv) Let $d = 4$, number the directions east, north, west and south as $0, 1, 2, 3$ modulo 4, and put $q_{j,j+1} = r$, $q_{j,j+2} = s$ and μ uniform. Then X_t is the direction of step t of a planar walk that turns left or right with probability rh each and reverses with probability sh (Section 6). For $s = 0$ and $r = 1$ this is the 4-cycle of Section 3.1. In (i), (ii) and (iv), Q has symmetric support and μ full support, so all admissible faces satisfy (H) by Lemma 2.2.

3 First-order face selection

3.1 A worked example: 4 states on a cycle

Before the general theorem we work out one example by hand. Take $d = 4$ states on a cycle, with rate 1 between neighbours (indices modulo 4), and the uniform start. So at each step the chain moves to each of its 2 neighbours with probability $h = T/N$ and stays with probability $1 - 2h$. This is the direction of the planar walk of Section 6 when it turns but never reverses (Example 2.5(iv) with $r = 1$, $s = 0$). Up to rotation and reflection there are 4 kinds of faces: a vertex, an adjacent pair, an arc of 3 states and the whole cycle. (A face such as $\{1, 3\}$ cannot be the support of a path, since the chain cannot pass from 1 to 3 without visiting 2 or 4.) Which kind carries the most likely composition?

A vertex. The composition $(N, 0, 0, 0)$ has a single path, which never switches. So $M_{\{1\}} = \frac{1}{4}(1 - 2h)^{N-1} \approx \frac{1}{4}e^{-2T}$. The exit rate is $\lambda_{\{1\}} = 2$.

An adjacent pair, say $\{1, 2\}$. A path with support $\{1, 2\}$ is a word in the letters 1 and 2. Each stay has probability $1 - 2h$ and each switch probability h , so a word with j switches has probability $\frac{1}{4}(1 - 2h)^{N-1}u^j$ with $u = h/(1 - 2h)$. Summing over the words with a letters 1 gives $P_N(a, N - a, 0, 0) = M_{\{1\}}S_{a, N-a}(u)$, the bin polynomial of Kagey's board. It is largest for the balanced compositions, and the uniform Bessel bound of Paper A, recalled in (8) below, gives $S_N(u) \approx \frac{2}{N}\sqrt{2Nu/\pi}e^{Nu}$ for Nu large. As $Nu \approx T$,

$$M_{\{1,2\}} \approx \frac{1}{4}e^{-2T} \cdot \frac{2}{N}\sqrt{2T/\pi}e^T = \frac{1}{\sqrt{2\pi}}N^{-1}T^{1/2}e^{-T}.$$

This has a simple meaning. The chain leaves $\{1, 2\}$ at rate 1 from each of its states ($1 \rightarrow 4$ and $2 \rightarrow 3$), so it stays in the pair with probability about e^{-T} , and $\lambda_{\{1,2\}} = 1$. Given that it stays, its fraction of time in state 1 is close to $\frac{1}{2}$, with fluctuations of order $T^{-1/2}$. So the mass e^{-T} is shared by about $N/\sqrt{T}$ compositions $(a, N - a, 0, 0)$, and the best one gets about $N^{-1}T^{1/2}e^{-T}$.

An arc of 3 states, say $\{1, 2, 3\}$. Only the end states 1 and 3 can leave, each at rate 1. The chain killed on leaving the arc has the matrix $I + hQ_F$ with $-Q_F = 2I - A$, where A is the adjacency matrix of the path $1 - 2 - 3$. Its largest eigenvalue is $1 - (2 - \sqrt{2})h$, with eigenvector $(1, \sqrt{2}, 1)$, so the chain stays in the arc with probability about $e^{-(2-\sqrt{2})T}$ and $\lambda_{\{1,2,3\}} = 2 - \sqrt{2}$. The visit fractions now fluctuate in 2 directions, so the mass is shared by about $(N/\sqrt{T})^2$ compositions, and $M_{\{1,2,3\}} \approx N^{-2}Te^{-(2-\sqrt{2})T}$.

The whole cycle. The chain never has to leave, so $\lambda = 0$, and the whole mass 1 is shared by about $(N/\sqrt{T})^3$ compositions near $(\frac{N}{4}, \frac{N}{4}, \frac{N}{4}, \frac{N}{4})$. So $M_{[4]} \approx N^{-3}T^{3/2}$.

Comparison. Now put $T = \tau \log N$. Then $e^{-\lambda T} = N^{-\tau\lambda}$, while the powers of T are only powers of $\log N$. So, up to such powers,

$$M_{\{1\}} \approx N^{-2\tau}, \quad M_{\{1,2\}} \approx N^{-(\tau+1)}, \quad M_{\{1,2,3\}} \approx N^{-((2-\sqrt{2})\tau+2)}, \quad M_{[4]} \approx N^{-3}.$$

The mode sits on the face with the smallest exponent. Figure 2(a) plots the 4 exponents against τ . The vertices win for $\tau < 1$, the adjacent pairs for $1 < \tau < 2$ and the whole cycle for $\tau > 2$. The arcs of 3 states never win. They beat the pairs only for $\tau > 1 + \sqrt{2}$, and there the whole cycle beats them, since that needs only $(2 - \sqrt{2})\tau > 1$. Note also that the whole cycle carries almost all the mass for every $\tau > 0$. It carries the mode only for $\tau > 2$, because it spreads its mass over about N^3 compositions.

The general rule. For any face F the same 2 effects give $M_F \approx N^{-(|F|-1)}e^{-T\lambda_F} = N^{-\Phi_\tau(F)}$ up to powers of $\log N$, as announced in Section 1.1. There is a second way to see the cost of spreading.

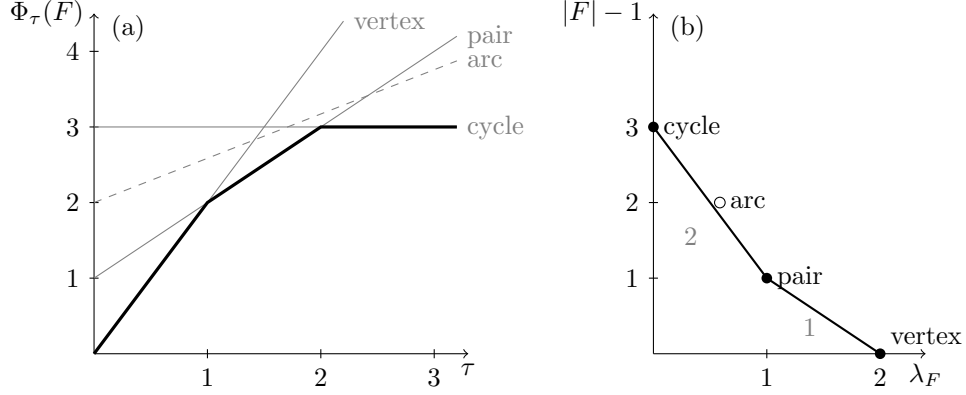


Figure 2: The 4-cycle. (a) The exponents $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$ of the 4 kinds of faces against τ , and their lower envelope (bold). The mode sits on the face whose line is lowest: a vertex for $\tau < 1$, an adjacent pair for $1 < \tau < 2$ and the whole cycle for $\tau > 2$. The dashed line of the arc of 3 states is never lowest. (b) The same faces as points $(\lambda_F, |F| - 1)$, with the lower-left boundary of their convex hull. The gray numbers are minus the slopes of its edges, the values of τ where the winner changes (Theorem 3.4). The arc (open) lies above the boundary.

A path with support F must enter each of the $|F| - 1$ states of F where it does not start, and each first entry is a switch of probability about T/N . Every later switch also costs T/N , but it multiplies the number of ways to place the runs by a factor of order N , so it costs no power of N . Within F the best composition sits near the quasi-ergodic law α_F^* : here $(\frac{1}{2}, \frac{1}{2})$ on a pair, $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ on an arc (the normalized squared Perron vector) and the uniform law on the cycle. Other visit fractions α cost an extra factor $e^{-T(I_F(\alpha) - \lambda_F)}$, with a rate I_F that we now introduce.

We now make this precise. Proposition 3.1 studies the rate inside a face, Theorem 3.3 proves the rule for every Q and every start, and Theorem 3.4 reads the winners off a convex hull as in Figure 2(b). Section 5 returns to the 4-cycle with all constants (Corollary 5.1).

3.2 The rate inside a face

For $\alpha \in \Delta_F$ let

$$I_F(\alpha) = \sup_{u \in (0, \infty)^F} \sum_{i \in F} \alpha_i \left(-\frac{(Q_F u)_i}{u_i} \right) = \sup_{u \in (0, \infty)^F} \sum_{i \in F} \alpha_i \left(q_i - \sum_{j \in F \setminus \{i\}} q_{ij} \frac{u_j}{u_i} \right),$$

and let $q_i^{\text{out}}(F) = \sum_{j \notin F} q_{ij}$. Taking $u = \mathbf{1}$ gives $I_F(\alpha) \geq \sum_i \alpha_i q_i^{\text{out}}(F)$, and dropping the terms with q_{ij} gives $I_F(\alpha) \leq \sum_i \alpha_i q_i$. This is the Donsker–Varadhan rate [7] of the occupation times of the chain killed on leaving F (Section 1.3): the chance that the chain stays in F with visit fractions near α decays like $e^{-TI_F(\alpha)}$. We need its properties without reversibility, and the proofs are short.

Proposition 3.1 (the rate on a face). *Let F be irreducible with $k = |F| \geq 2$.*

- (a) $I_F \geq \lambda_F$ on Δ_F , with equality exactly at $\alpha_F^* = l_F \circ r_F$.
- (b) I_F is convex on Δ_F and strictly convex on Δ_F° . For $\alpha \in \Delta_F^\circ$ the supremum defining $I_F(\alpha)$ is attained at a vector $u = r_\alpha$, which is unique up to a positive factor, and α is the stationary law of the generator on F with off-diagonal entries $q_{ij}r_{\alpha,j}/r_{\alpha,i}$.

(c) If there is $\nu \in (0, \infty)^F$ with $\nu_i q_{ij} = \nu_j q_{ji}$ for all $i, j \in F$, then for every $\alpha \in \Delta_F$

$$I_F(\alpha) = \sum_{\{i,j\} \subseteq F, i \neq j} (\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}})^2 + \sum_{i \in F} \alpha_i q_i^{\text{out}}(F).$$

(d) For every nonempty face $F' \subsetneq F$ and every $\alpha \in \Delta_{F'}$, $I_F(\alpha) = I_{F'}(\alpha) \geq \lambda_{F'} > \lambda_F$.

Proof. Write $u = e^\psi$, so that $I_F(\alpha) = \sup_\psi \Psi_\alpha(\psi)$ with $\Psi_\alpha(\psi) = \sum_i \alpha_i q_i - \sum_{i \neq j} \alpha_i q_{ij} e^{\psi_j - \psi_i}$, where i, j run over F . It is concave in ψ and affine in α . Let G^ψ be the generator on F with off-diagonal entries $q_{ij} e^{\psi_j - \psi_i}$. Then $\nabla_\psi \Psi_\alpha = -\alpha^\top G^\psi$, so ψ is a maximizer exactly when α is stationary for G^ψ , which is irreducible.

The lower bound in (a), for every face F' . Write $Q_{F'} = B - cI$ with $B \geq 0$ and let $s > \varrho(Q_{F'})$. Then $u = (sI - Q_{F'})^{-1} \mathbf{1} = \sum_{n \geq 0} (s + c)^{-n-1} B^n \mathbf{1}$ is positive and $-(Q_{F'} u)_i / u_i = -s + 1/u_i > -s$. So $I_{F'} > -s$, and $s \downarrow \varrho(Q_{F'})$ gives the claim.

The value at α_F^* . With $\psi = \log r_F$ and $R = \text{diag}(r_F)$ we get $G^\psi = R^{-1}(Q_F + \lambda_F I)R$ and $(\alpha_F^*)^\top G^\psi = l_F^\top (Q_F + \lambda_F I)R = 0$. So $I_F(\alpha_F^*) = \Psi_{\alpha_F^*}(\log r_F) = \lambda_F$.

(b) I_F is convex as a supremum of affine functions. Let $\alpha \in \Delta_F^\circ$, let a be the least positive rate q_{ij} in F , normalize $\min_i \psi_i = 0$ and let $M = \max_i \psi_i$. Some edge $i \rightarrow j$ of a shortest path in G_F from a minimum of ψ to a maximum has $\psi_j - \psi_i \geq M/(k-1)$. So $\Psi_\alpha(\psi) \leq \sum_i \alpha_i q_i - a(\min_i \alpha_i) e^{M/(k-1)}$, and the supremum is attained. If ψ and $\psi + \delta$ both attain it, concavity makes each term $\alpha_i q_{ij} e^{\psi_j - \psi_i + t(\delta_j - \delta_i)}$ affine in $t \in [0, 1]$. So $\delta_j = \delta_i$ on every edge, and δ is constant. For strict convexity let $\alpha^0 \neq \alpha^1$ lie in Δ_F° , let $\alpha^t = (1-t)\alpha^0 + t\alpha^1$ with $0 < t < 1$, and let ψ attain $I_F(\alpha^t)$. Then $I_F(\alpha^t) = (1-t)\Psi_{\alpha^0}(\psi) + t\Psi_{\alpha^1}(\psi) \leq (1-t)I_F(\alpha^0) + tI_F(\alpha^1)$, and equality would make both α^0 and α^1 stationary for G^ψ .

(d) For $\alpha \in \Delta_{F'}$ only the rows in F' enter, and the terms with $j \in F \setminus F'$ are subtracted. Dropping them gives $I_F(\alpha) \leq I_{F'}(\alpha)$, and $u_j = \varepsilon \rightarrow 0$ for $j \notin F'$ gives equality. Then $I_{F'} \geq \lambda_{F'}$ by the lower bound, and $\lambda_{F'} > \lambda_F$ because the Perron root of an irreducible nonnegative matrix, here $Q_F + cI$, is larger than that of each proper principal submatrix [29, Ch. 2].

(a) If $\alpha \neq \alpha_F^*$ lies on the boundary of Δ_F , (d) gives $I_F(\alpha) > \lambda_F$. If $\alpha \in \Delta_F^\circ$ and $I_F(\alpha) = \lambda_F$, strict convexity gives $I_F(\frac{1}{2}(\alpha + \alpha_F^*)) < \lambda_F$, against the lower bound.

(c) Since $q_i = \sum_{j \in F \setminus \{i\}} q_{ij} + q_i^{\text{out}}(F)$, the objective is $\sum_i \alpha_i q_i^{\text{out}}(F)$ plus, for each pair $\{i, j\}$, the term $\alpha_i q_{ij}(1 - u_j/u_i) + \alpha_j q_{ji}(1 - u_i/u_j)$. By the AM-GM inequality this term is at most $(\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}})^2$. The choice $u_i = \sqrt{\alpha_i/\nu_i}$, with $u_i = \varepsilon \rightarrow 0$ when $\alpha_i = 0$, attains the bound in the limit by reversibility. $\square$

In words, the cheapest way to stay on F is to spend the fractions α_F^* of the time in its states, and any other fractions cost more. Part (d) says that a composition on the boundary of F , which misses some states of F , is really a composition of a smaller face, and is more expensive. The lower bound in (a), the value at α_F^* and the interior form of (c) are checked in Lean. For symmetric Q_F , $l_F = r_F$ and α_F^* is the entrywise square of the Perron vector normalized in ℓ^2 .

3.3 Face selection

We also need an upper bound that holds on a whole face, including near its boundary, where the local law of Section 4 does not apply. For an irreducible face F put $V_F = \{\theta \in \mathbb{R}^F : \varrho(A_F - \text{diag } \theta) = 0\}$. Note that $q - \lambda_F \mathbf{1} \in V_F$, since $A_F - \text{diag}(q - \lambda_F \mathbf{1}) = Q_F + \lambda_F I$. Each $\theta \in V_F$ gives an exponential bound with the linear rate $\alpha \mapsto \langle q - \theta, \alpha \rangle$, and I_F is the supremum of these linear rates.

Lemma 3.2 (a uniform upper bound). *Let F be irreducible with $k = |F| \geq 2$, and let $\theta \in V_F$. There is C_θ , depending on θ , Q and μ , such that for every $n \in \mathbb{Z}_{\geq 0}^F$ with $\text{supp } n = F$ and every h with $h q_{\max} \leq \frac{1}{2}$, the probability that $X_1, \dots, X_{|n|}$ have composition n is at most $C_\theta h^{k-1} \exp(-h\langle n, q - \theta \rangle + C_\theta h^2 |n|)$.*

The proof is in Appendix A. With $\theta = q - \lambda_F \mathbf{1}$ it gives $P_N(n) \leq C h^{k-1} e^{-T\lambda_F + CT^2/N}$ for every n with support F . This is the rule $M_F \lesssim N^{-(k-1)} e^{-T\lambda_F}$ of Section 3.1, up to a power of T . The bound does not depend on N , so it also applies to the blocks of a path.

Theorem 3.3 (first-order face selection). *Fix Q and μ .*

- (i) *Let F satisfy (H), let $k = |F|$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. There is N_0 such that, uniformly for $N \geq N_0$, $1 \leq T \leq N^{1/3}$ and $\text{supp } n = F$ with $n/N \in \mathcal{K}$,*

$$\log P_N(n) = -(k-1)L - T I_F(n/N) + O_{\mathcal{K}}(1 + \log T).$$

- (ii) *Let F be admissible with components $C_1, \dots, C_r$, and let k^* be the least size of a component C_j with $\lambda_{C_j} = \lambda_F$. There are κ , T_0 and N_0 such that for $N \geq N_0$ and $T_0 \leq T \leq N^{1/3}$*

$$\begin{aligned} \log M_F &\geq -(k_F - 1)L + (k_F - k^*) \log T - T\lambda_F - \kappa, \\ \log M_F &\leq -(k_F - 1)L + (k_F - 1) \log T - T\lambda_F + \kappa. \end{aligned}$$

If F is not admissible, $M_F = 0$.

- (iii) *Let $T = \tau L$ with $\tau > 0$ fixed, and let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ . Then $\log \max_n P_N(n) = -L \min_F \Phi_\tau(F) + O(\log L)$, with the minimum over admissible faces, and for large N every global mode $\hat{n}$ has $\text{supp } \hat{n} \in \mathcal{F}_\tau$. This holds for every start μ .*
- (iv) *If in (iii) every face in $\mathcal{F}_\tau$ is irreducible, then every global mode $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^* + O(1/L)$ with $F = \text{supp } \hat{n}$. This is the case when Q has symmetric support or μ has full support (Theorem 3.4(c)).*

In words: (i) a composition with fractions α inside F has probability $N^{-(k-1)} e^{-T I_F(\alpha)}$ up to powers of T ; (ii) the best composition on an admissible face has probability $N^{-(k_F-1)} e^{-T\lambda_F}$ up to a power of T between 0 and $k_F - 1$; (iii) so with $T = \tau L$ the exponent of N is $-\Phi_\tau(F)$, and the mode lies on a face that minimizes it, whatever the start; (iv) inside the winning face the mode sits near the quasi-ergodic law.

Idea of the proof. For faces with (H), part (i) and the face maxima come from the local law of Section 4, which we prove independently of this theorem. A reducible face is visited in blocks, one for each strongly connected component. For the upper bound in (ii) we apply Lemma 3.2 to each block, and for the lower bound we write down explicit paths that spend almost all their time in a component of least exit rate. Part (iii) compares the faces, and (iv) follows from Theorem 4.6.

Proof. (i) For $k = 1$ this is (1), since $I_{\{i\}} = q_i$. For $k \geq 2$, Lemma 4.5 gives $P_N(n) = N^{-(k-1)} f_T(n/N)(1 + O(T^2/N))$ with $T^2/N \leq N^{-1/3}$. By Theorem 4.4, and since f_T is positive and continuous for bounded T , $\log f_T(\alpha) = \frac{k-1}{2} \log T + \log C_F(\alpha) - T I_F(\alpha) + O_{\mathcal{K}}(1)$ on $\mathcal{K}$, and $\log C_F$ is bounded there. This gives (i).

(ii) If F is not admissible, $M_F = 0$. If F is admissible and irreducible, it satisfies (H) by Lemma 2.2(b), and $k^* = k_F$. Then Theorem 4.6 gives both bounds, since $0 \leq \frac{k_F-1}{2} \log T \leq (k_F - 1) \log T$. For a reducible face, a path visits each component in one block of consecutive times

(proof of Lemma 2.2). For the upper bound, Lemma 3.2 (or a direct count if $|C| = 1$) bounds a block of length N_C in the component C by $O(h^{|C|-1}e^{-h\lambda_C N_C})$, and each of the $r - 1$ switches between blocks gives a factor h . As $\lambda_C \geq \lambda_F$, this gives $M_F = O(h^{k_F-1}e^{-T\lambda_F})$. For the lower bound the path spends about N/T visits in each component except one of least exit rate, which loses a bounded factor per component, and we divide by the number of compositions. The details are in Appendix A.6.

(iii) For large N we have $T_0 \leq T = \tau L \leq N^{1/3}$, and by (ii) $\log M_F = -L \Phi_\tau(F) + O(\log L)$ for every admissible face, while $M_F = 0$ for the others. Since $\max_n P_N(n) = \max_F M_F$, the first claim follows. Let $m_\tau = \min_F \Phi_\tau(F)$. An admissible $F \notin \mathcal{F}_\tau$ has $\Phi_\tau(F) \geq m_\tau + \varepsilon_0$ for a fixed $\varepsilon_0 > 0$, so $M_F \leq N^{-m_\tau - \varepsilon_0 + o(1)}$, which is smaller than $\max_n P_N(n)$ for large N . A global mode has positive probability, so its support is admissible and lies in $\mathcal{F}_\tau$.

(iv) By (iii), for large N a global mode $\hat{n}$ maximizes P_N on its face $F \in \mathcal{F}_\tau$, which is irreducible and admissible and so satisfies (H) by Lemma 2.2(b). So Theorem 4.6 with $T = \tau L \leq N^{1/3}$ gives $\hat{n}/N = \alpha_F^* + O(1/L)$. $\square$

For a reducible face we only have the bounds in (ii), and Proposition 4.13 shows that the power of T can differ from $(k_F - 1)/2$.

3.4 Reading the winners off a convex hull

For an admissible face F put $P_F = (\lambda_F, |F| - 1)$, and let S be the set of these points. Then $\Phi_\tau(F) = \tau x + y$ at $(x, y) = P_F$. Picture a line of slope $-\tau$ moving up from below the plane: the first points of S it touches are the winners at τ . For each $\tau > 0$ the points of $\text{conv } S$ where $\tau x + y$ is least form a vertex or an edge of $\text{conv } S$, and we call the union of these over all $\tau > 0$ the lower-left boundary of $\text{conv } S$. Figure 2(b) shows it for the 4-cycle.

Theorem 3.4 (geometry of the winners). *Let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ .*

- (a) *The lower-left boundary of $\text{conv } S$ is a path whose vertices lie in S . Read from right to left, they are $v_i = (\lambda^{(i)}, y^{(i)})$, $0 \leq i \leq m$, with $\lambda^{(0)} > \dots > \lambda^{(m)}$ and $y^{(0)} < \dots < y^{(m)}$. Put $\tau_i = (y^{(i)} - y^{(i-1)})/(\lambda^{(i-1)} - \lambda^{(i)})$ for $1 \leq i \leq m$, $\tau_0 = 0$ and $\tau_{m+1} = \infty$. Then $\tau_1 < \dots < \tau_m$. For $\tau_i < \tau < \tau_{i+1}$, $\mathcal{F}_\tau$ is the set of faces F with $P_F = v_i$. At $\tau = \tau_i$ it is the set of faces whose point lies on the closed edge $[v_{i-1}, v_i]$.*
- (b) *If $0 < \tau < \tau'$, $F \in \mathcal{F}_\tau$ and $G \in \mathcal{F}_{\tau'}$, then $\lambda_F \geq \lambda_G$ and $|F| \leq |G|$. Both inequalities are strict when $\mathcal{F}_\tau \cap \mathcal{F}_{\tau'} = \emptyset$.*
- (c) *If Q has symmetric support, or μ has full support, then every face in $\mathcal{F}_\tau$ is irreducible, for every $\tau > 0$.*

In words: as τ grows the winners walk up the lower-left boundary of the hull, from small faces with large exit rates to large faces with small exit rates, and they change at the values τ_i , minus the slopes of its edges.

Proof. (a) Let $g(\tau) = \min_{(x,y) \in S} (\tau x + y)$. It is a minimum of finitely many affine functions, so it is concave and piecewise affine. Let $\tau_1 < \dots < \tau_m$ be the points where its slope changes. A point of S that attains g inside (τ_i, τ_{i+1}) attains it on the whole interval, so exactly one point v_i does. The slope decreases, so $\lambda^{(0)} > \dots > \lambda^{(m)}$, and continuity at τ_i gives the formula for τ_i and $y^{(i)} > y^{(i-1)}$. A point of S that attains $g(\tau_i)$ lies on the line through v_{i-1} and v_i , and comparing with g slightly

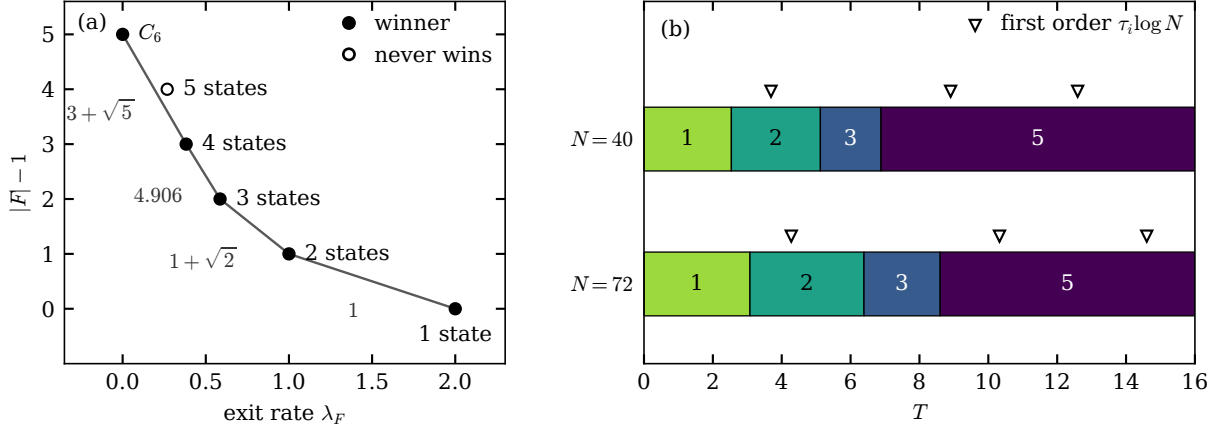


Figure 3: (a) The points $(\lambda_F, |F| - 1)$ of the arcs of the 6-cycle and of C_6 , with the lower-left boundary of their hull. The numbers beside the edges are the switch values τ_i of Theorem 3.4. The winners (filled) are the arcs of 1 to 4 states and C_6 , and the arc of 5 states (open) never wins. (b) The support size of the exact mode on C_5 with the uniform start, against T , at $N = 40$ and $N = 72$. The mode skips the arc of 4 states. The triangles mark the first-order switch points $\tau_i \log N$ with $\tau_i = 1, 1 + \sqrt{2}$ and $2 + \sqrt{2}$ (Table 1).

below and above τ_i puts it on the closed edge. The minimizers on $\text{conv } S$ are the hull of those on S , so the lower-left boundary is the path $v_0 \cdots v_m$.

(b) Adding $\Phi_\tau(F) \leq \Phi_\tau(G)$ and $\Phi_{\tau'}(G) \leq \Phi_{\tau'}(F)$ gives $(\tau' - \tau)(\lambda_F - \lambda_G) \geq 0$, and then $|F| - |G| \leq \tau(\lambda_G - \lambda_F) \leq 0$. If $\lambda_F = \lambda_G$, then $|F| = |G|$ and $F \in \mathcal{F}_{\tau'}$. So if the 2 sets are disjoint, $\lambda_F > \lambda_G$ and then $|F| < |G|$.

(c) With symmetric support use Lemma 2.2(c). If μ has full support and F is admissible and reducible, a component C with $\lambda_C = \lambda_F$ is admissible by Lemma 2.2(b), and $\Phi_\tau(C) \leq \Phi_\tau(F) - 1$. $\square$

The finite form of Theorem 3.4 is checked in Lean. A face with small λ_F for its size is a set of states that the chain rarely leaves, a well-separated community. As τ grows the exit rate counts for more, and the mode moves to larger communities. It need not grow around its first vertex, and it can move to a disjoint community (Table 1). For symmetric Q , λ_F is the least value of $\sum_{i < j} q_{ij}(f_i - f_j)^2$, the sum over all pairs in $[d]$, over unit vectors f supported in F (Rayleigh–Ritz), and it decreases strictly as a connected face grows, by Proposition 3.1(d). So only the least exit rate of each size matters. Figure 3(a) shows S for the 6-cycle.

4 The second-order theory

First order locates a crossing only up to a factor $1 + o(1)$ in T . To locate it up to $o(1)$ we need the constant in front of $N^{-(k-1)}e^{-T\lambda_F}$ for every face that satisfies (H), together with the power of T . The plan is as follows. Section 4.1 defines the constant K_F . Section 4.2 proves a local limit theorem inside a face and deduces M_F up to a factor $1 + O(1/T + T^2/N)$. Section 4.3 compares 2 faces, Section 4.4 recovers Papers A and B, and Section 4.5 shows what can change without (H). In this section gradients and Hessians in α are taken in the coordinates $\alpha' = (\alpha_2, \dots, \alpha_k)$, $|\cdot|$ is the Euclidean norm, and $q = (q_i)_{i \in F}$.

4.1 The tilted chain and the constant

To find the probability of a composition with fractions α we change the rates so that α becomes typical. The optimal vector r_α in the definition of $I_F(\alpha)$ gives a new generator $\widehat{Q}_\alpha$ on F , with the same switching graph, whose stationary law is exactly α . Under it the visit fractions satisfy a central limit theorem with covariance Γ_α/T , and the price of the change of rates is $e^{-T I_F(\alpha)}$ times boundary factors for the start and the end of the path. The constant $C_F(\alpha)$ collects these factors and the Gaussian normalization.

Definition 4.1 (tilt and constants). Let F be irreducible with $k \geq 2$. For $\alpha \in \Delta_F^\circ$ let r_α attain the supremum defining $I_F(\alpha)$ (Proposition 3.1(b)), and put $\theta(\alpha) = A_F r_\alpha / r_\alpha$ and $l_\alpha = \alpha / r_\alpha$ (entrywise). The *tilted generator* is $\widehat{Q}_\alpha = R^{-1}(A_F - \text{diag } \theta(\alpha))R$ with $R = \text{diag } r_\alpha$. With $D = \text{diag } \alpha$ let

$$Z_\alpha = (\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)^{-1} - \mathbf{1}\alpha^\top, \quad \Gamma_\alpha = DZ_\alpha + Z_\alpha^\top D,$$

and let Σ_α be Γ_α with the row and column of state 1 deleted. Finally let $K_{\{i\}} = \mu_i$ for a single state, and

$$C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(2\pi)^{(k-1)/2} \sqrt{\det \Sigma_\alpha}}, \quad K_F = C_F(\alpha_F^*).$$

Nothing in Definition 4.1 changes when r_α is scaled. Note that $l_\alpha \cdot r_\alpha = 1$ and $I_F(\alpha) = \langle q - \theta(\alpha), \alpha \rangle$. Since r_α minimizes $u \mapsto \sum_{i \neq j} \alpha_i q_{ij} u_j / u_i$, its derivative in u_j , which is $\sum_i l_{\alpha,i} q_{ij} - l_{\alpha,j} \theta_j(\alpha)$, vanishes at r_α . Hence

$$l_\alpha^\top (A_F - \text{diag } \theta(\alpha)) = 0. \quad (2)$$

Also $(A_F - \text{diag } \theta(\alpha))r_\alpha = 0$ with $r_\alpha > 0$, so $\theta(\alpha) \in V_F$ [29]. So $\widehat{Q}_\alpha$ is an irreducible generator with the pattern of A_F and, by (2), stationary law α . If $(\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)v = 0$, multiplying by $\alpha^\top$ gives $\alpha^\top v = 0$, so $\widehat{Q}_\alpha v = 0$ and $v = 0$. Thus Z_α exists, and

$$Z_\alpha \mathbf{1} = 0, \quad \alpha^\top Z_\alpha = 0, \quad -\widehat{Q}_\alpha Z_\alpha = I - \mathbf{1}\alpha^\top. \quad (3)$$

So $\Gamma_\alpha \mathbf{1} = 0$. As Γ_α is symmetric, its adjugate is then a multiple of $\mathbf{1}\mathbf{1}^\top$, so deleting another state instead of state 1 gives the same $\det \Sigma_\alpha$. At α_F^* the vector $u = r_F$ gives the value $\lambda_F = I_F(\alpha_F^*)$, so $r_{\alpha_F^*}$ is a multiple of r_F , $\theta(\alpha_F^*) = q - \lambda_F \mathbf{1}$, and $l_{\alpha_F^*}$ is the matching multiple of l_F . In the local law below, $\mu \cdot r_\alpha$ accounts for the start of the path, $l_\alpha \cdot \mathbf{1}$ for its free end, and Σ_α for the Gaussian spread of the occupation fractions.

For conductances $c_{ij} = c_{ji}$ on F let $\text{tree}(c) = \sum_{\mathcal{T}} \prod_{\{i,j\} \in \mathcal{T}} c_{ij}$, the sum over the spanning trees $\mathcal{T}$ of the complete graph on F .

Proposition 4.2 (the constant). *Let F be irreducible with $k \geq 2$.*

- (a) *The maps $\alpha \mapsto \theta(\alpha)$, $I_F(\alpha)$, Σ_α , r_α and l_α (with $\sum_i r_{\alpha,i} = 1$) are real-analytic on Δ_F° , Σ_α is positive definite, and $\nabla^2 I_F(\alpha) = \Sigma_\alpha^{-1}$. Hence $\det \Sigma_\alpha \cdot \det \nabla^2 I_F(\alpha) = 1$ and $C_F(\alpha) = (\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) \sqrt{\det \nabla^2 I_F(\alpha) / (2\pi)^{k-1}}$.*
- (b) *Suppose that $\nu_i q_{ij} = \nu_j q_{ji}$ for $i, j \in F$ and some $\nu \in (0, \infty)^F$. Then r_α is a multiple of $\sqrt{\alpha/\nu}$ and l_α of $\sqrt{\alpha\nu}$, and $\widehat{Q}_\alpha$ is reversible with conductances $c_{ij}(\alpha) = \sqrt{\alpha_i \alpha_j} q_{ij} \sqrt{\nu_i / \nu_j}$. Also $I_F(\alpha) = \sum_i q_i \alpha_i - \sum_{i \neq j} c_{ij}(\alpha)$, and*

$$\det \Sigma_\alpha = \frac{2^{k-1} (\prod_i \alpha_i)^2}{\text{tree}(c(\alpha))}, \quad C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(4\pi)^{(k-1)/2}} \frac{\sqrt{\text{tree}(c(\alpha))}}{\prod_i \alpha_i}.$$

Proof. The proof of (a) is in Appendix A and uses Lemma A.1.

(b) Put $u = \sqrt{\alpha/\nu}$ and $l = \alpha/u = \sqrt{\alpha\nu}$. Then $\sum_i l_i q_{ij} = \sum_i \sqrt{\alpha_i \nu_i} q_{ij}$ and $l_j (A_F u)_j / u_j = \nu_j \sum_i q_{ji} \sqrt{\alpha_i / \nu_i}$ agree because $\nu_j q_{ji} = \nu_i q_{ij}$. So u is a critical point of $\sum_{i \neq j} \alpha_i q_{ij} u_j / u_i$, which is convex in $\log u$, and r_α is a multiple of u . Then $\alpha_i (\hat{Q}_\alpha)_{ij} = l_i q_{ij} u_j = c_{ij}(\alpha)$, which is symmetric by reversibility, and $\alpha_i \theta_i(\alpha) = \sum_j c_{ij}(\alpha)$ gives I_F .

For the determinant let $\mathcal{L} = -D\hat{Q}_\alpha$, the Laplacian of the conductances. Fix η with $\eta_1 = 0$, and let $u = Z_\alpha \eta$ and $\zeta = D\eta - (\alpha \cdot \eta)\alpha$. By (3), $\mathcal{L}u = \zeta$ and $\alpha \cdot u = 0$, so $\eta^\top \Gamma_\alpha \eta = 2(D\eta)^\top u = 2\zeta^\top u$. Let $v = u - u_1 \mathbf{1}$. As $\zeta \cdot \mathbf{1} = 0$ and $\mathcal{L}\mathbf{1} = 0$, we get $\zeta^\top u = \zeta^\top v$ and $\mathcal{L}v = \zeta$, and since $v_1 = 0$ the rows $2, \dots, k$ give $\mathcal{L}_{(1)} v' = \zeta'$. Here $\mathcal{L}_{(1)}$ is $\mathcal{L}$ without row and column 1, primes denote the coordinates $2, \dots, k$, and $\zeta' = (D' - \alpha' \alpha'^\top) \eta'$ with $D' = \text{diag } \alpha'$. Hence $\Sigma_\alpha = 2(D' - \alpha' \alpha'^\top) \mathcal{L}_{(1)}^{-1} (D' - \alpha' \alpha'^\top)$. The matrix determinant lemma gives $\det(D' - \alpha' \alpha'^\top) = \prod_{i \geq 2} \alpha_i (1 - \sum_{i \geq 2} \alpha_i) = \prod_i \alpha_i$, and the matrix-tree theorem [30] gives $\det \mathcal{L}_{(1)} = \text{tree}(c(\alpha))$. $\square$

Corollary 4.3 (symmetric faces). *Let F satisfy (H) with $k \geq 2$, and let Q_F be symmetric with constant row sums. Then $\lambda_F = q_i - \sum_{j \in F \setminus \{i\}} q_{ij}$ for every $i \in F$, α_F^* is uniform on F , and*

$$K_F = \mu(F) k^{(k+1)/2} \sqrt{\text{tree}(q_F)} (4\pi)^{-(k-1)/2},$$

where $\text{tree}(q_F)$ is $\text{tree}(c)$ for the conductances $c_{ij} = q_{ij}$, $i, j \in F$.

Proof. The positive vector $\mathbf{1}$ is a right and left eigenvector of Q_F , so it belongs to the Perron root. So λ_F is minus the row sum, r_F and l_F are multiples of $\mathbf{1}$, and $\alpha_F^* = \mathbf{1}/k$. At $\alpha = \mathbf{1}/k$, Proposition 4.2(b) with $\nu = \mathbf{1}$ gives $r_\alpha = \mathbf{1}$, $l_\alpha = \alpha$ and $(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) = \mu(F)$. Also $c_{ij}(\alpha) = q_{ij}/k$, so $\text{tree}(c(\alpha)) = k^{-(k-1)} \text{tree}(q_F)$, and $\prod_i \alpha_i = k^{-k}$. $\square$

4.2 The local law and the face maxima

Inside the face the occupation fractions fluctuate on the scale $T^{-1/2}$ in $k-1$ directions. So a single composition gets the factor $T^{(k-1)/2}$ times the volume $N^{-(k-1)}$ of a lattice cell.

Theorem 4.4 (local law). *Let F satisfy (H) with $k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,*

$$f_T(\alpha) = T^{(k-1)/2} C_F(\alpha) e^{-T I_F(\alpha)} (1 + O_{\mathcal{K}}(1/T)).$$

For $F = \{i\}$, $f_T = \mu_i e^{-q_i T}$.

In words, the density of the visit fractions on the event that the chain stays in F is a Gaussian factor $T^{(k-1)/2}$ times the large-deviation factor $e^{-T I_F(\alpha)}$ times the constant $C_F(\alpha)$, with a relative error $O(1/T)$ that is uniform away from the boundary of the face.

Idea of the proof. The proof is in Appendix A. The sum over words that defines H is a matrix resolvent, so Cauchy's formula writes H as an integral over a torus of radii $\theta(\alpha)$ (Lemma A.2). All the coefficients are nonnegative, so the integrand is largest at the real point, and a Laplace lemma with a parameter (Lemma B.1) gives the expansion. The Hessian at the real point is computed from the Perron root of a perturbed generator (Lemma A.1), and this is where Σ_α comes from. The hypothesis $\mu(F) > 0$ is needed, since $f_T = C_F = 0$ when $\mu(F) = 0$. Also $f_T(\alpha)$ is continuous and positive in (T, α) , as H is a convergent power series with some $W(m) > 0$ by (H).

Lemma 4.5 (discrete and continuum). *Let F satisfy (H) with $k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly for $\text{supp } n = F$ with $n/N \in \mathcal{K}$ and $1 \leq T \leq N^{1/2}$, $P_N(n) = N^{-(k-1)} f_T(n/N) (1 + O_{\mathcal{K}}(T^2/N))$.*

The proof is in Appendix A. With Theorem 4.4 it sharpens the error in Theorem 3.3(i) to $\frac{k-1}{2} \log T + \log C_F(n/N) + O(1/T + T^2/N)$.

Theorem 4.6 (face maxima). *Let F satisfy (H). If $k \geq 2$, then uniformly for $1 \leq T \leq N^{1/3}$,*

$$M_F = N^{-(k-1)} T^{(k-1)/2} K_F e^{-T\lambda_F} (1 + O(1/T + T^2/N)),$$

and every maximizer $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^ + O(1/T)$. For $k = 1$, $M_{\{i\}}$ is given by (1). In both cases, uniformly for $1 \leq T \leq N^{1/3}$,*

$$\log M_F = -(k-1)L + \frac{k-1}{2} \log T + \log K_F - T\lambda_F + O(1/T + T^2/N). \quad (4)$$

In words, the best composition on F sits within $O(N/T)$ of $N\alpha_F^*$, and its probability is the Gaussian peak of the local law at α_F^* . The factor $N^{-(k-1)}$ is the volume of a lattice cell, and $T^{(k-1)/2}$ is the height of a Gaussian density whose spread in each free direction is of order $T^{-1/2}$.

Idea of the proof. Near α_F^* we use the local law and Lemma 4.5, and maximize a Gaussian with a small linear perturbation. Away from α_F^* the rate exceeds λ_F by a fixed amount, and Lemma 3.2, applied with finitely many vectors θ , shows that these compositions are exponentially less likely in T . For bounded T positivity and continuity suffice.

Proof. For $k = 1$ this is (1). Let $k \geq 2$, put $X = N^{-(k-1)} T^{(k-1)/2} K_F e^{-T\lambda_F}$ and $x = \alpha' - \alpha_F^*$. By Propositions 3.1(a) and 4.2(a), I_F and $\log C_F$ are real-analytic near α_F^* , I_F has its minimum λ_F there and $\nabla^2 I_F > 0$. So there is a closed ball $U = \{|x| \leq \rho_0\} \subset \Delta_F^\circ$ on which $c_U |x|^2 \leq I_F - \lambda_F \leq C_U |x|^2$, $|\log(C_F/K_F) - \gamma \cdot x| \leq C_1 |x|^2$ with $\gamma = \nabla \log C_F(\alpha_F^*)$, and $\log C_F$ is C_3 -Lipschitz.

Outside U . As I_F is lower semicontinuous, Proposition 3.1(a) gives $I_F \geq \lambda_F + 3\eta$ on $\Delta_F \setminus U^\circ$ for some $\eta > 0$. Since $I_F(\beta) = \sup_{u>0} \langle q - A_F u/u, \beta \rangle$ and $A_F u/u \in V_F$, compactness gives finitely many $\theta_j \in V_F$ with $\max_j \langle q - \theta_j, \beta \rangle > \lambda_F + 2\eta$ on $\Delta_F \setminus U^\circ$. Lemma 3.2 then gives $P_N(n) \leq C(T/N)^{k-1} e^{-T(\lambda_F + 2\eta)} = X \cdot O(T^{(k-1)/2} e^{-2\eta T})$ for $\text{supp } n = F$ and $n/N \notin U^\circ$.

Inside U . By Theorem 4.4 and Lemma 4.5 with $\mathcal{K} = U$, for $\alpha = n/N \in U$,

$$P_N(n) = X \frac{C_F(\alpha)}{K_F} e^{-T(I_F(\alpha) - \lambda_F)} (1 + E), \quad |E| \leq C_2(1/T + T^2/N). \quad (5)$$

For $T \geq 2C_1/c_U$ the middle factors are at most $\exp(\sup_x [\gamma \cdot x - \frac{c_U T}{2} |x|^2]) = e^{|\gamma|^2/(2c_U T)}$, so $M_F \leq X(1 + O(1/T + T^2/N))$. For the lower bound let n^0 have support F and $|n_i^0 - N\alpha_{F,i}^*| < 1$ for all i , which exists for $N \geq N_0$ since every $\alpha_{F,i}^* > 0$. Then $C_F(n^0/N) \geq K_F e^{-kC_3/N}$ and $T(I_F(n^0/N) - \lambda_F) \leq C_U k^2 T/N^2$, so (5) gives $P_N(n^0) \geq X(1 - O(1/T + T^2/N))$. Fix T_2 so large that the 2 bounds above give $M_F/X \in [\frac{1}{2}, \frac{3}{2}]$ for $T \geq T_2$. For $1 \leq T \leq T_2$, f_T is positive and continuous on $[1, T_2] \times U$, so Lemma 4.5 gives $M_F/X \in [c, C]$. This gives the first display and (4).

The maximizer. For $T \geq T_3$ with T_3 large, the bound outside U is below $X/2 \leq P_N(n^0)$, so $\hat{\alpha} = \hat{n}/N \in U$ and $|E| \leq \frac{1}{2}$. Taking logarithms in $P_N(\hat{n}) \geq P_N(n^0)$ gives $c_U T |\hat{x}|^2 \leq C_3 |\hat{x}| + C_4(1/T + T^2/N) \leq C_3 |\hat{x}| + 2C_4/T$ with $\hat{x} = \hat{\alpha}' - \alpha_F^*$, because $T \leq N^{1/3}$ gives $T^2/N \leq 1/T$. This step uses the exponent $\frac{1}{3}$, and it gives $|\hat{x}| = O(1/T)$. For $1 \leq T < T_3$ the claim holds since $|\hat{x}|$ is bounded. $\square$

4.3 Crossings and tie windows

We now compare 2 faces. Taking logarithms in Theorem 4.6, the difference $\log M_F - \log M_G$ is an explicit increasing function of T up to $O(1/T + T^2/N)$, and its zero is the crossing.

Theorem 4.7 (the crossing equation). *Let F and G satisfy (H), with $\Delta = k_F - k_G \geq 1$ and $\delta = \lambda_G - \lambda_F > 0$, and let $D_N(T) = \log M_F - \log M_G$.*

(i) *For large N , D_N has a zero in $[\frac{\Delta}{2\delta}L, \frac{2\Delta}{\delta}L]$.*

(ii) *For large N , every zero $T_N \in [1, N^{1/3}]$ of D_N satisfies*

$$\delta T_N + \frac{\Delta}{2} \log T_N = \Delta L - \log \frac{K_F}{K_G} + O\left(\frac{1}{L}\right). \quad (6)$$

(iii) *All such zeros lie in one interval of length $O(1/L)$, and*

$$T_N = \frac{\Delta}{\delta} \left[L - \frac{1}{2} \log L + b_{FG} \right] + O\left(\frac{\log L}{L}\right), \quad b_{FG} = -\frac{1}{2} \log \frac{\Delta}{\delta} - \frac{1}{\Delta} \log \frac{K_F}{K_G}.$$

Proof. By (4), uniformly on $[1, N^{1/3}]$, $D_N(T) = -\Delta L + \frac{\Delta}{2} \log T + \delta T + \log \frac{K_F}{K_G} + O(1/T + T^2/N)$.

(i) At $T = \frac{\Delta}{2\delta}L$ this is $-\frac{\Delta}{2}L + O(\log L) < 0$, and at $T = \frac{2\Delta}{\delta}L$ it is $\Delta L + O(\log L) > 0$. For fixed N , M_F is a maximum of finitely many polynomials in h , and it is positive because F is admissible. So D_N is continuous and has a zero in between.

(ii) Let $\psi(T) = \delta T + \frac{\Delta}{2} \log T$, which is increasing. At a zero $T_N \in [1, N^{1/3}]$, $\psi(T_N) = \Delta L - \log(K_F/K_G) + O(1)$. This is larger than $\psi(\frac{\Delta}{2\delta}L) = \frac{\Delta}{2}L + O(\log L)$, and $\delta T_N \leq \psi(T_N)$. So $\frac{\Delta}{2\delta}L < T_N \leq \frac{2\Delta}{\delta}L$, and then $O(1/T_N + T_N^2/N) = O(1/L)$.

(iii) Let $\psi(T^\circ) = \Delta L - \log(K_F/K_G)$. As $\psi' \geq \delta$, every zero is within $O(1/L)$ of T° . Also $T^\circ = \frac{\Delta}{\delta}L + O(\log L)$, so $\log T^\circ = \log L + \log \frac{\Delta}{\delta} + O(\log L/L)$, which gives the explicit form. $\square$

In words: the larger face F pays $N^{-\Delta}$ for spreading and gains $e^{\delta T}$ for staying, and its Gaussian peak gives an extra factor $T^{\Delta/2}$; the crossing balances the 3. The explicit form in (iii) converges slowly, so computations should solve (6) (see Corollary 5.1). We do not prove that the zero of D_N is unique in general.

Sometimes several faces tie at first order: at some τ^* more than 2 faces minimize Φ_{τ^*} , as all faces do for the model of Paper B. Then the competition plays out in a window $T = \tau^*(L - \frac{1}{2} \log L + t)$ with t of order 1. In this window each tied face has a line $\ell_F(t)$, and the mode follows the highest line.

Theorem 4.8 (tie windows). *Fix $\tau^* > 0$. Let m^* be the least value of Φ_{τ^*} over the admissible faces and $\mathcal{F}^*$ the set of minimizers. Assume that every face in $\mathcal{F}^*$ is irreducible. By Theorem 3.4(c) this holds if Q has symmetric support or μ has full support. Put $T = \tau^*(L - \frac{1}{2} \log L + t)$, and for $F \in \mathcal{F}^*$ let $\ell_F(t) = \log K_F + \frac{k_F-1}{2} \log \tau^* - \tau^* \lambda_F t$ and $E(t) = \max_{G \in \mathcal{F}^*} \ell_G(t)$.*

(i) *Uniformly for t in compact sets, for $F \in \mathcal{F}^*$, $\log M_F = -m^*L + \frac{m^*}{2} \log L + \ell_F(t) + O(\log L/L)$.*

(ii) *There are $\varepsilon_1, \varepsilon_0 > 0$ such that for large N and $|T/L - \tau^*| \leq \varepsilon_1$, every admissible $F \notin \mathcal{F}^*$ has $M_F \leq N^{-\varepsilon_0} \max_{G \in \mathcal{F}^*} M_G$.*

(iii) *There are $\varepsilon, t_0 > 0$ such that for large N and $t_0 \leq |t| \leq \varepsilon L$, every global mode lies on a face $F \in \mathcal{F}^*$ with $\ell_F = E$ on $[t_0, \infty)$ if $t > 0$, or on $(-\infty, -t_0]$ if $t < 0$. Such a face has the least λ_F in $\mathcal{F}^*$ if $t > 0$ and the largest if $t < 0$.*

Consequently, for every $\eta > 0$ and large N , if $|t| \leq \varepsilon L$ is at distance at least η from every kink of E , then every global mode lies on a face whose line equals E near t . So the line of the face that carries the global mode changes only at $T = \tau^*(L - \frac{1}{2} \log L + t_j) + o(1)$, where t_j runs over the kinks of E , and it changes near each kink. Faces with identical lines (the same k_F , λ_F and K_F) tie at this order and give no crossing.

In words: inside the window the logarithm of every tied face maximum is the same leading term plus the line $\ell_F(t)$, faces that do not tie are smaller by a power of N , and far out in the window the face with the least (for $t > 0$) or largest (for $t < 0$) exit rate wins.

Proof. Every $F \in \mathcal{F}^*$ is admissible and irreducible, so it satisfies (H) by Lemma 2.2(b). Write $T = \tau^*L(1+y)$ with $y = (t - \frac{1}{2}\log L)/L$, so that $|y| \leq \frac{1}{2}$ for $|t| \leq L/4$ and large N . By (4) and $\tau^*\lambda_F + k_F - 1 = m^*$, uniformly for $|t| \leq L/4$,

$$\log M_F = -m^*L + \frac{m^*}{2}\log L + \ell_F(t) + \frac{k_F-1}{2}\log(1+y) + O(1/L) \quad (F \in \mathcal{F}^*). \quad (7)$$

For t in a compact set, $\log(1+y) = O(\log L/L)$, which gives (i).

(ii) Let $g > 0$ be the least value of $\Phi_{\tau^*}(F) - m^*$ over the admissible $F \notin \mathcal{F}^*$. As $0 \leq \lambda_F \leq q_{\max}$, for $|\tau - \tau^*| \leq \varepsilon_1 = g/(4q_{\max} + 4)$ we have $\Phi_{\tau}(F) - \Phi_{\tau}(G) \geq g/2$ for such F and all $G \in \mathcal{F}^*$. With $\tau = T/L$, Theorem 3.3(ii) gives $\log M_F - \log M_G \leq -\frac{g}{2}L + O(\log L) \leq -\frac{g}{4}L$.

(iii) Take $t_0 \geq 1$ larger than every $|t_j|$ and $\varepsilon \leq \min(\frac{1}{4}, \varepsilon_1/(2\tau^*))$, so that (ii) applies. Let $t \geq t_0$, let $\ell_F = E$ on $[t_0, \infty)$, and let $G \in \mathcal{F}^*$ with $\ell_G \neq \ell_F$, so that $\lambda_F \leq \lambda_G$. As both faces are tied, $\Delta = k_F - k_G = \tau^*(\lambda_G - \lambda_F) \geq 0$. If $\Delta = 0$, (7) gives $\log(M_F/M_G) = \log(K_F/K_G) + O(1/L)$ with $K_F > K_G$. If $\Delta > 0$, $\log(M_F/M_G) = \Delta t + \ell_F(0) - \ell_G(0) + \frac{\Delta}{2}\log(1+y) + O(1/L)$, where $|\log(1+y)| \leq 2|y| \leq t/2$ for large N , so it is positive if t_0 is large. The case $t \leq -t_0$ is the same.

For the consequence, fix $\eta > 0$ and let $S_\eta = [-t_0, t_0] \setminus \bigcup_j (t_j - \eta, t_j + \eta)$. For $G \in \mathcal{F}^*$, $E - \ell_G$ is convex, piecewise linear and nonnegative with kinks only at kinks of E . So its zero set is empty, a single kink of E , or an interval whose finite endpoints are kinks of E , and on each component of S_η either $\ell_G = E$ or $E - \ell_G \geq g_\eta > 0$. Then (i) and (ii) give the claim on S_η , and (iii) for $t_0 \leq |t| \leq \varepsilon L$. $\square$

4.4 Papers A and B

We use the polynomials $S_{a,b}$ and S_N , the root u_N , the crossing p_N , $z_N = N(1 - p_N)$ and $c_0 = \frac{1}{2}\log(\pi/8)$ of Section 2.1. The first result shows that a single state against a symmetric pair is Kagey's board at every N .

Proposition 4.9 (an exact pair identity). *Let $i \neq j$ with $q_{ij} = q_{ji} = a$, $q_i = q_j = q$ and $\mu_i = \mu_j > 0$, and let $F = \{i, j\}$. For every $N \geq 2$, $1 \leq m \leq N - 1$ and $h > 0$ such that $I + hQ$ is stochastic and $qh < 1$,*

$$\frac{P_N(me_i + (N - m)e_j)}{P_N(Ne_i)} = S_{m, N-m}(u), \quad u = \frac{ah}{1 - qh}.$$

So the balanced compositions, with $m = \lfloor N/2 \rfloor$ or $m = \lceil N/2 \rceil$, are the most likely ones with support F , $M_F = M_{\{i\}}S_N(u)$, and $M_F > M_{\{i\}}$ exactly when $u > u_N$.

Proof. A path with this composition is a word in i, j with m letters i . Each stay has probability $1 - qh$ and each switch ah , so a word with s switches has probability $\mu_i(1 - qh)^{N-1-s}(ah)^s = \mu_i(1 - qh)^{N-1}u^s$, while $P_N(Ne_i) = \mu_i(1 - qh)^{N-1}$. Sum over the words. The bin polynomials of [2] grow coefficientwise as the bin moves toward the middle, and $S_{m, N-m} = S_{N-m, m}$. So the balanced m give the largest $S_{m, N-m}(u)$. Finally S_N increases strictly and equals 1 only at u_N (Section 2.1). $\square$

The identity in Proposition 4.9 is checked in Lean.

Corollary 4.10 (Paper A). *Let $d = 2$, $q_{12} = q_{21} = 1$ and $\mu = (\frac{1}{2}, \frac{1}{2})$, the walk of [2] with $T = N(1 - p)$. Then $K_{\{1\}} = K_{\{2\}} = \frac{1}{2}$ and $K_{\{1,2\}} = \sqrt{2/\pi}$, so $b_{FG} = c_0$ for $F = \{1, 2\}$ and $G = \{1\}$. For every $N \geq 2$, $M_{\{1,2\}} = M_{\{1\}}$ holds exactly at $T = z_N$, and $z_N + \frac{1}{2} \log z_N = L + c_0 + O(1/L)$.*

Proof. Corollary 4.3 with $k = 2$, $\text{tree}(q_F) = 1$ and $\mu(F) = 1$ gives $K_{\{1,2\}}$, and then $b_{FG} = -\log(K_{\{1,2\}}/K_{\{1\}}) = c_0$. Proposition 4.9 with $a = q = 1$ gives $M_{\{1,2\}} = M_{\{1\}} S_N(h/(1-h))$, and $h/(1-h) = u_N$ exactly when $p = p_N$. So for every N the only zero of D_N is z_N , and by Theorem 4.7(i) it lies in $[L/2, 2L] \subset [1, N^{1/3}]$ for large N . Then Theorem 4.7(ii) with $\Delta = \delta = 1$ gives the equation. $\square$

This is the crossing equation of [2] without its terms $1/(8z_N)$ and R_N . The term $1/(8z_N)$ is $(c_{\{1\}} - c_{\{1,2\}})/z_N$ in the notation of Remark 4.14, whose proof we omit, with $c_{\{1\}} = 0$ and $c_{\{1,2\}} = -\frac{1}{8}$.

Corollary 4.11 (Paper B). *Let $Q = J - dI$, where J is the all-ones matrix, and let μ be uniform, the model of [3]. A face of k states has $\lambda_F = d - k$, α_F^* uniform and $K_F = \frac{1}{d} k^{k+1/2} (4\pi)^{-(k-1)/2}$. So $\Phi_1(F) = d - 1$ for every face, and all faces tie at $\tau = 1$. The crossing of a face of $k \geq 2$ states with a vertex has $\Delta = \delta = k - 1$ and $b_{FG} = a_k = \frac{1}{2} \log(4\pi) - \frac{k+1/2}{k-1} \log k$, the constant of the vertex crossing of [3].*

Proof. Each Q_F is symmetric with row sums $-(d - k)$, and the complete graph on F has k^{k-2} spanning trees by Cayley's formula. So Corollary 4.3 with $\mu(F) = k/d$ gives λ_F , α_F^* and K_F . A vertex has $K = \frac{1}{d}$ and $\lambda = d - 1$, so $b_{FG} = -\frac{1}{k-1} \log(dK_F) = a_k$. The arithmetic that gives b_{FG} from K_F here and in Corollary 4.10 is checked in Lean. At $\tau^* = 1$, Theorem 4.8 describes the tie window, and [3] resolves this window in full. $\square$

4.5 Reducible faces

Remark 4.12 (scope of the $\frac{1}{2}$). The coefficient $\frac{1}{2}$ of $\log \log N$ in Theorems 4.7 and 4.8 is due to the power $T^{(k-1)/2}$ in Theorem 4.6, so it needs only (H). At a kink of E where the line of G gives way to that of F as t increases, $\Delta = \tau^* \delta$ and $t_j = b_{FG}$. So every change in a tie window is a crossing of Theorem 4.7. The best composition of a reducible face can visit a state only once, with no Gaussian window in that direction. Then the $\frac{1}{2}$ can fail, and in the next example it is 1 and then 0.

Proposition 4.13 (a directed 3-cycle). *Let $d = 3$, $q_{12} = b > 2$, $q_{23} = q_{31} = 1$, all other rates 0, and $\mu_1 = 1$.*

- (i) *The admissible faces are $\{1\}$, $\{1, 2\}$ (reducible, $\lambda = 1$) and $\{1, 2, 3\}$ (irreducible, $\lambda = 0$). Exactly $M_{\{1\}} = (1 - bh)^{N-1}$ and $M_{\{1,2\}} = bh(1 - h)^{N-2}$, and the latter is attained only at $n = (1, N - 1, 0)$. For the full face, $\alpha^* = (1, b, b)/(1 + 2b)$, $\det \nabla^2 I = (1 + 2b)^4/(3b^2)$ and $K_{\{1,2,3\}} = (1 + 2b)^2/(2\sqrt{3}\pi b)$.*
- (ii) *$M_{\{1\}} = M_{\{1,2\}}$ has exactly one root $T_N^{(1)}$ with $bh < 1$. It satisfies $(b - 1)T + \log T = L - \log b + O(T^2/N)$, so $T_N^{(1)} = \frac{1}{b-1} [L - \log L + \log \frac{b-1}{b}] + O(\log L/L)$, and the coefficient of $\log \log N$ in the bracket is 1.*
- (iii) *Every root $T \in [L/2, 2L]$ of $M_{\{1,2\}} = M_{\{1,2,3\}}$ satisfies $T = L + \log \frac{2\sqrt{3}\pi b^2}{(1+2b)^2} + O(1/L)$, so the coefficient of $\log \log N$ is 0. These roots lie in an interval of length $O(1/L)$.*

(iv) Both are changes of the global mode. For large N and $T \leq N^{1/3}$, the mode is $(N, 0, 0)$ for $T < T_N^{(1)}$, it is $(1, N-1, 0)$ between $T_N^{(1)}$ and the smallest root in (iii), and it lies on the full face above the largest root in (iii).

In words: the face $\{1, 2\}$ is reducible, and its best composition $(1, N-1, 0)$ visits state 1 once and then stays in state 2. Its maximum is about $b(T/N)e^{-T}$, with the power T^1 where an irreducible pair would have $T^{1/2}$. This changes the coefficient of $\log \log N$ at both crossings.

Proof. (i) The chain moves only along $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ from 1, which gives the supports and the components $\{1\}, \{2\}$ of $\{1, 2\}$. A path with support $\{1, 2\}$ is $1^{n_1}2^{n_2}$, of probability $(1-bh)^{n_1-1}bh(1-h)^{n_2-1}$, and raising n_1 by one multiplies it by $(1-bh)/(1-h) < 1$. For the full face $\lambda = 0$, $r = 1$, and $l = \alpha^* = (1, b, b)/(1+2b)$ is the stationary law, so $(\mu \cdot r)(l \cdot 1) = 1$. The ratios u_2/u_1 , u_3/u_2 and u_1/u_3 have product 1, so the AM-GM inequality gives $I(\alpha) = \sum_i q_i \alpha_i - 3(b\alpha_1\alpha_2\alpha_3)^{1/3}$. Let $g = (\alpha_1\alpha_2\alpha_3)^{1/3}$, $w = 1/\alpha$ and $s = 1+2b$. The Hessian of g in all 3 coordinates is $-gM$ with $M = \frac{1}{3} \text{diag}(w)^2 - \frac{1}{9}ww^\top$, so $\nabla^2 I = 3b^{1/3}gJ^\top MJ$, where J is the Jacobian of $(\alpha_2, \alpha_3) \mapsto (1-\alpha_2-\alpha_3, \alpha_2, \alpha_3)$. At α^* , $3b^{1/3}g = 3b/s$ and $w = s(1, y, y)$ with $y = 1/b$. So $J^\top MJ$ has diagonal entries $\frac{2}{9}s^2(1+y+y^2)$ and off-diagonal entries $\frac{1}{9}s^2(2+2y-y^2)$, and $\det J^\top MJ = \frac{s^4}{81} \cdot 3y^2(2+y)^2 = \frac{s^6}{27b^4}$. Hence $\det \nabla^2 I = s^4/(3b^2)$, and Proposition 4.2(a) gives $K_{\{1,2,3\}}$. The exact parts of (i) are checked in Lean.

(ii) Let $D_1 = \log M_{\{1,2\}} - \log M_{\{1\}} = \log(bh) + (N-2)\log(1-h) - (N-1)\log(1-bh)$. Then $D'_1(h) = \frac{1}{h} - \frac{N-2}{1-h} + \frac{b(N-1)}{1-bh} > 0$ for $bh < 1$, while $D_1 \rightarrow -\infty$ as $h \rightarrow 0$ and $D_1 \rightarrow \infty$ as $bh \rightarrow 1$. So there is exactly one root (this is checked in Lean). Expanding, $D_1 = \log b + \log T - L + (b-1)T + O(T^2/N)$, which is positive at $T = L/(b-1)$ and negative at $T = L/(2(b-1))$. So the root lies between them, and $\log T = \log L - \log(b-1) + O(\log L/L)$ gives the expansion.

(iii) Here $\log M_{\{1,2\}} = \log b + \log T - L - T + O(T^2/N)$, and Theorem 4.6 for the full face ($k=3$, $\lambda=0$, $\mu(F)=1$) gives $\log M_{\{1,2,3\}} = -2L + \log T + \log K_{\{1,2,3\}} + O(1/T + T^2/N)$. So on $[L/2, 2L]$ the difference is $L - T + \log b - \log K_{\{1,2,3\}} + O(1/L)$, which gives (iii).

(iv) Lemma 3.2 with $\theta = q$ gives $M_{\{1,2,3\}} \leq C(T/N)^2$. For $T < T_N^{(1)}$, $M_{\{1\}} > M_{\{1,2\}}$ by (ii), and $M_{\{1\}} \geq N^{-b/(b-1)-o(1)}$ since $T_N^{(1)} \leq L/(b-1)$. This beats $C(T/N)^2$ because $b/(b-1) < 2$. For $T > T_N^{(1)}$, $M_{\{1,2\}} > M_{\{1\}}$. For $T \leq L/2$, $\log M_{\{1,2\}} - \log M_{\{1,2,3\}} \geq L - T - \log T - C > 0$. On $[L/2, 2L]$ the sign is that of the formula in (iii), which is positive at $L/2$ and negative at $2L$. For $2L \leq T \leq N^{1/3}$ the same formula is at most $-L + C < 0$. $\square$

Remark 4.14 (the next term, proof omitted). Carrying the Laplace argument behind Theorem 4.4 one term further gives the factor $1 + c(\alpha)/T + O(T^{-3/2})$ there. Here $c(\alpha)$ is the coefficient of $1/T$ in the next step of Lemma B.1 for the integral (16), divided by $\chi(\alpha, 0)$. It is a Gaussian moment of the φ -derivatives of Ψ of orders 3 and 4 and of χ of orders 1 and 2 at $\varphi = 0$. Then (4) gains the term c_F/T , with $c_F = c(\alpha_F^*) + \frac{1}{2}\gamma^\top \Sigma_{\alpha_F^*} \gamma$ and $\gamma = \nabla \log C_F(\alpha_F^*)$, and the error becomes $O(T^{-3/2} + T^2/N)$. So at every crossing

$$T_N \left(\Delta L - \frac{\Delta}{2} \log T_N - \delta T_N - \log \frac{K_F}{K_G} \right) = c_F - c_G + O(T_N^{-1/2} + T_N^3/N).$$

Here $c_{\{i\}} = 0$. Comparing with the terms of order $1/z$ in the expansions of [2] and [3] gives $c_F = -1/(8q_{ij})$ for a pair with $q_{ij} = q_{ji}$, $q_i = q_j$ and $\mu_i = \mu_j$, and $c_F = -(k-1)/8$ for a face of k states when $Q = J - dI$ and μ is constant on F . We omit the details.

5 The 4-cycle and other graphs

In this section Q is symmetric. So the admissible faces are the connected faces with positive start mass, and they satisfy (H) (Lemma 2.2). By Theorem 3.4(c), Theorem 4.8 applies at every tie. We call the faces in $\mathcal{F}_\tau$ the *winners* at τ . We first complete the worked example of Section 3.1 with all its constants, then show on K_3 that the start can decide the winner, and then list the outcomes for other graphs.

Corollary 5.1 (the 4-cycle cascade). *Let Q be the 4-cycle C_4 with $q_{i,i\pm 1} = 1$ (indices mod 4), and let μ be uniform.*

- (a) *The admissible faces are the vertices, the adjacent pairs, the arcs of 3 states and C_4 , with exit rates 2, 1, $2 - \sqrt{2}$ and 0. The winners are the vertices for $\tau < 1$, the adjacent pairs for $1 < \tau < 2$ and C_4 for $\tau > 2$. The arcs of 3 states never win.*
- (b) *$K_{\text{vertex}} = \frac{1}{4}$, $K_{\text{pair}} = 1/\sqrt{2\pi}$, $K_{\text{arc3}} = (4 + 3\sqrt{2})/(4\pi)$ and $K_{C_4} = 8/\pi^{3/2}$, and $\det \Sigma = 1/512$ on C_4 .*
- (c) *For every $N \geq 2$ and $0 < h < \frac{1}{2}$, $M_{\text{pair}} = M_{\text{vertex}} S_N(u)$ with $u = h/(1 - 2h)$. So the face maxima of a vertex and an adjacent pair cross exactly at $u = u_N$, the crossing of [2]. For large N this crossing satisfies $T + \frac{1}{2} \log T = L + c_0 + O(1/L)$.*
- (d) *For large N , every crossing $T_N \in [1, N^{1/3}]$ of the face maxima of an adjacent pair and C_4 satisfies $T_N + \log T_N = 2L - \log(8\sqrt{2}/\pi) + O(1/L)$.*
- (e) *Put $b = -\frac{1}{2} \log 2 - \frac{1}{2} \log(8\sqrt{2}/\pi)$. In the tie window at $\tau = 1$ the face that carries the global mode changes from the vertices to the adjacent pairs only at $T = L - \frac{1}{2} \log L + c_0 + o(1)$. In the tie window at $\tau = 2$ it changes from the adjacent pairs to C_4 only at $T = 2(L - \frac{1}{2} \log L + b) + o(1)$.*

In words: the mode moves from a vertex to an adjacent pair at the crossing of Kagey's board, exactly at every N , and from the pairs to the whole cycle at a crossing whose equation has the coefficient $1 = \frac{\Delta}{2}$ of $\log T$, since the cycle has 2 more states than a pair.

Proof. (a) A proper face that is not an arc is disconnected, so it is not admissible. For an adjacent pair $-Q_F$ has rows $(2, -1)$ and $(-1, 2)$, with least eigenvalue 1. For an arc of 3 states $-Q_F = 2I - A$, where A is the adjacency matrix of the path on 3 states, and $r = (1/\sqrt{2}, 1, 1/\sqrt{2})$ is a positive eigenvector with eigenvalue $2 - \sqrt{2}$. Also $Q\mathbf{1} = 0$. So Φ_τ equals $2\tau, \tau + 1, (2 - \sqrt{2})\tau + 2$ and 3, and comparing the first, second and fourth gives the winners. Since $2 - \sqrt{2} > \frac{1}{2}$, the 3-arc has $\Phi_\tau > \frac{1}{2}(\tau + 1) + \frac{1}{2} \cdot 3$, which is at least the smaller of the exponents of the pair and of C_4 . So it never wins.

(b) A vertex has $K = \mu_i = \frac{1}{4}$. The pair and C_4 have symmetric Q_F with constant row sums, so Corollary 4.3 applies, with $\mu(F) = \frac{1}{2}$ and one spanning tree for the pair, and $\mu(F) = 1$ and 4 spanning trees for C_4 . On C_4 the law α^* is uniform and every edge has conductance $\frac{1}{4}$, so Proposition 4.2(b) gives $\det \Sigma = 2^3 \cdot 4^{-8}/(4 \cdot 4^{-3}) = 1/512$. The 3-arc has unequal row sums, and we use Proposition 4.2(b) with $\nu = \mathbf{1}$. Here $\alpha^* = r \circ r/|r|^2 = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ and $r_\alpha = l_\alpha = \sqrt{\alpha^*}$, so $(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) = \frac{1}{4}(1 + 1/\sqrt{2})^2 = (3 + 2\sqrt{2})/8$. The only spanning tree is the path, with tree(c) = $c_{12}c_{23} = \frac{1}{8}$, and $\prod_i \alpha_i^* = \frac{1}{32}$. So $K_{\text{arc3}} = \frac{3+2\sqrt{2}}{8} \cdot \frac{32}{2\sqrt{2}} \cdot \frac{1}{4\pi} = \frac{4+3\sqrt{2}}{4\pi}$.

(c) This is Proposition 4.9 with $a = 1$ and $q = 2$. As $K_{\text{pair}}/K_{\text{vertex}} = 2\sqrt{2/\pi} = e^{-c_0}$, Theorem 4.7 with $\Delta = \delta = 1$ gives the equation.

(d) This is Theorem 4.7 with $F = C_4$, G an adjacent pair, $\Delta = 2$, $\delta = 1$ and $K_F/K_G = 8\sqrt{2\pi}/\pi^{3/2} = 8\sqrt{2}/\pi$.

(e) At $\tau = 1$ the tied faces are the vertices and the adjacent pairs, and at $\tau = 2$ they are the adjacent pairs and C_4 . Faces of one type have the same line. By Remark 4.12 the kink of E is at $t = b_{FG}$, which is c_0 at $\tau = 1$ by (c) and b at $\tau = 2$ by (d). So Theorem 4.8 gives (e). $\square$

The finite parts of (a) are checked in Lean. Lean also checks that a closed form for the constants of all arcs, which we do not use, gives the values of K_{pair} and K_{arc3} above. Remark 4.14 gives $c_{\text{pair}} = -\frac{1}{8}$ without proof, and an exact computer evaluation of the next Laplace term gives $c_{\text{arc3}} = \frac{5}{4}\sqrt{2} - 2$ and $c_{C_4} = -\frac{9}{16}$. So if Remark 4.14 holds, then at the crossings in (d) the quantity $T_N(2L - \log T_N - T_N - \log(8\sqrt{2}/\pi))$ tends to $-\frac{9}{16} + \frac{1}{8} = -\frac{7}{16}$ (computer-assisted, proof omitted).

Theorem 4.7 also applies to a face that never carries the mode, and this is a good test of the implicit equation (6). Take the 3-arc F against a vertex G , so that $\Delta = 2$, $\delta = \sqrt{2}$ and $\log(K_F/K_G) = \log((4 + 3\sqrt{2})/\pi) = 0.964591$. At $N = 10^{14}$ a numerical computation of the face maxima, over the symmetric compositions $(a, N - 2a, a)$, puts the crossing at $T_N = 42.2633$, where $\Delta L - \frac{\Delta}{2} \log T_N - \delta T_N = 0.959070$. The root of the implicit equation is within 0.004 of T_N , while the explicit form in Theorem 4.7 gives 42.2059. Also $T_N(0.959070 - 0.964591) = -0.2333$, close to the value $c_{\text{arc3}} - c_{\text{vertex}} = -0.2322$ that Remark 4.14 predicts.

By Corollary 4.3 the start enters K_F only through $\mu(F)$, and this can change which face wins inside a tie window.

Corollary 5.2 (the start on K_3). *Let $d = 3$ and $Q = J - 3I$, put $T = L - \frac{1}{2} \log L + t$, and let $w = \log \frac{16}{9\sqrt{3}} = 0.026$.*

(a) *Let $\mu = (\frac{1}{2}, \frac{1}{2}, 0)$. For every $\eta > 0$ and large N , if $c_0 + \eta \leq t \leq c_0 + w - \eta$, then every global mode lies on the face $\{1, 2\}$.*

(b) *Let μ be uniform. For every compact set of t and large N , no global mode lies on a face of 2 states.*

Proof. All faces tie at $\tau^* = 1$ (Corollary 4.11), and Theorem 4.8 applies since Q is symmetric. By Corollary 4.3 the lines of a vertex i , an edge $\{i, j\}$ and $[3]$ are $\log \mu_i - 2t$, $\log((\mu_i + \mu_j)\sqrt{2}/\pi) - t$ and $\log K_{[3]}$ with $K_{[3]} = 3^{5/2}/(4\pi)$. In (a) the line of $\{1, 2\}$ lies strictly above the lines of the other admissible faces exactly when $c_0 < t < \frac{1}{2} \log \frac{2}{\pi} - \log K_{[3]} = c_0 + w$. So E has no kink in this interval, and the consequence in Theorem 4.8 gives (a). In (b) the gap between $\max(\log \frac{1}{3} - 2t, \log K_{[3]})$ and an edge line is least where the first 2 lines meet, and there it is $\frac{1}{2} \log(3^{7/2}/32) > 0.18$. So Theorem 4.8(i) gives (b). $\square$

The left end c_0 is the crossing of Paper A. Indeed Proposition 4.9 with $a = 1$ and $q = 2$ shows that for every N the face maxima of $\{1, 2\}$ and of a vertex cross exactly at $u = u_N$. The comparison of the lines in (a) and the value of w are checked in Lean. With the uniform start the mode jumps from the vertices straight to $[3]$, as [3] shows for every complete graph. Paper B [3] also compares the window in (a) with exact computations at finite N , where it approaches its width w slowly.

Remark 5.3 (the outcomes table). Table 1 lists the outcomes for other graphs. Each row comes from Theorem 3.4 applied to the exit rates of the connected faces. On C_d a proper face that is not an arc is disconnected, hence not admissible, and the growth of its switch values follows from $2 - 2\cos x = x^2 + O(x^4)$. The rows marked “proof omitted” use the same hull argument, and we

graph	winners in order, with the value of τ where each one starts	status
K_d	$\lambda_F = d - F $. Vertices for $\tau < 1$ and $[d]$ for $\tau > 1$. Every face ties at $\tau = 1$, and [3] resolves the tie at second order.	Lean
symmetric Q	$\lambda_{[d] \setminus \{x\}} \leq q_x / (d - 1)$, with equality exactly when $q_{yx} = q_x / (d - 1)$ for all $y \neq x$. With unit rates on a connected graph, the winners go from vertices straight to $[d]$ exactly when the graph is complete.	Lean
$C_d, d \geq 4$	Arcs of $k = 1, \dots, k_d$ states with $k_d = \lfloor 2d/3 \rfloor$, then C_d . The arc of $k \geq 2$ states wins from $1/(\lambda_{k-1} - \lambda_k)$ with $\lambda_k = 2 - 2 \cos \frac{\pi}{k+1}$, and C_d from $\lfloor d/3 \rfloor / \lambda_{k_d}$. These values are 1, $1 + \sqrt{2}$, 4.906 and 8.771 for $k = 2, \dots, 5$, and they grow like $k^3 / (2\pi^2)$. Arcs of $k \geq 3$ states win exactly when $d \geq \lceil 3k/2 \rceil$.	Lean, asymptotics by hand
$P_d, d \geq 3$	End-arcs of $1, \dots, \lfloor 2d/3 \rfloor$ states, then P_d . The end-arc of k states has exit rate $2 - 2 \cos \frac{\pi}{2k+1}$, and the first switch is at the golden ratio $(1 + \sqrt{5})/2$.	Lean in part, proof omitted
P_4 at finite N	The end-arc of 3 states never wins, and its exit rate is only 0.0071 too large to win. Yet it is the exact mode for $T \in (9.381, 12.502)$ at $N = 40$, and on a window at every N tested up to 300.	numerical
Θ_ε	Triangles $\{1, 2, 3\}$ and $\{4, 5, 6\}$ joined by a link 3–4 of rate $\varepsilon < \varepsilon_1 = (64 + \sqrt{46})/60$. The states 1, 2, 5, 6, then $\{1, 2\}$ and $\{5, 6\}$ from 1, the triangles from $1 + \varepsilon/3 + O(\varepsilon^2)$, and [6] from $9/\varepsilon + 2 + O(\varepsilon)$.	Lean in part, 3 exact resultants, proof omitted
paw	A triangle $\{1, 2, 3\}$ with a leaf 4 at 1. $\{4\}$, then $\{1, 4\}$ from $1 + \sqrt{2}$, $\{1, 2, 3\}$ from $\sqrt{2} + \sqrt{3}$, and all 4 states from $2 + \sqrt{3}$.	Lean
house	The square 1, 2, 3, 4 with a roof state 5 joined to 1, 2. $\{3\}, \{4\}, \{5\}$, then $\{3, 4\}$ from 1, $\{1, 2, 5\}$ from $1 + \sqrt{2}$, and all 5 states from $2 + \sqrt{2}$.	Lean
K_4 plus a vertex	A state 5 joined to 1, 2 of K_4 . $\{5\}$, then $\{1, 2, 3, 4\}$ from $3(\sqrt{17} + 1)/8 = 1.921$, and all 5 states from $(5 + \sqrt{17})/4 = 2.281$.	Lean

Table 1: First-order winners on some graphs with unit rates and a start of full support (the second row allows any symmetric Q). The last column says how each row is established (Remark 5.3).

omit the details. A well-separated community wins for a long range of τ , such as a triangle of Θ_ε from $\tau \approx 1$ to $\tau \approx 9/\varepsilon$. The mode need not grow around its first vertex. It leaves the leaf of the paw, and on the house and on K_4 plus a vertex it jumps to a disjoint community. The second row explains the last one. The state 5 feeds $\{1, 2, 3, 4\}$ unequally, so the point of $\{1, 2, 3, 4\}$ lies strictly left of the chord from $\{5\}$ to $[5]$, and the winners cannot jump from $\{5\}$ straight to $[5]$. At feasible N the switch values are approached slowly (Figure 3(b)), and on P_4 even the order of the phases differs.

6 The planar persistent walk

We now return to the question of the introduction, Kagey’s board in 2 dimensions. A ball moves on $\mathbb{Z}^2$ in the directions east, north, west and south, numbered 0, 1, 2, 3 modulo 4, with unit vectors $e_0 = (1, 0)$, $e_1 = (0, 1)$, $e_2 = -e_0$ and $e_3 = -e_1$. Let $r, s \geq 0$ with $\beta = 2r + s > 0$, and let $0 < h < 1/\beta$. The first direction is uniform. After that the ball keeps its direction with probability $p_0 = 1 - \beta h$, turns left or right with probability rh each, and reverses with probability sh . This

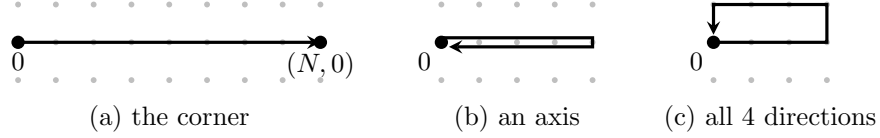


Figure 4: Paths of $N = 8$ steps. (a) The only path to the corner $(N, 0)$ never switches. A path to the origin either (b) stays on one axis and only reverses, or (c) turns and then uses all 4 directions.

is the chain of Example 2.5(iv), with $q_{j,j\pm 1} = r$ and $q_{j,j+2} = s$, and its position after N steps is $Y_N = (n_0 - n_2, n_1 - n_3)$. So the bin is now a linear function of the composition. In this section N is even, since otherwise the origin cannot be reached, and we allow every $h < 1/\beta$. Only the path that always goes east reaches the corner Ne_0 , so

$$c_N = \mathbb{P}(Y_N = Ne_0) = \frac{1}{4}p_0^{N-1}.$$

We compare it with $o_N = \mathbb{P}(Y_N = 0)$. Figure 4 shows the 3 kinds of paths that matter.

We write $u = sh/p_0$, $v = rh/p_0$ and $w = \beta h/p_0 = 2v + u$, so $1 + w = 1/p_0$. These are the odds of a reversal, of a turn to a given side and of any switch, as in the shared notation $u = q/p$. We use S_N , u_N , p_N , $z_N = N(1 - p_N)$ and c_0 from Section 2.1. As N is even, $S_N(u)$ is the sum of u^j over the words with $N/2$ letters of each of 2 kinds, where j is the number of switches. It has nonnegative coefficients and no constant term, so it increases strictly on $[0, \infty)$.

The answer at first order. Face selection predicts the answer, up to powers of $\log N$. Let $h = \tau \log N/N$. The corner is the vertex $\{0\}$, which has exit rate β , so c_N is about $N^{-\beta\tau}$. A vertex, a face with 3 directions, or a face with 2 adjacent ones misses the opposite of one of its directions, so it cannot reach the origin. On the axis $\{0, 2\}$, which has exit rate $2r$, the origin is the single composition $(N/2, 0, N/2, 0)$, of probability about $N^{-(2r\tau+1)}$. On the full face a single composition has probability about N^{-3} , but about $N/2$ compositions (a, b, a, b) end at the origin, and together they give about N^{-2} . So the origin should beat the corner when $\beta\tau > \min(2r\tau + 1, 2)$, that is, when $\tau > \tau^* = \min(2/\beta, 1/s)$. At τ^* the axes win if $s > 2r$ and the full face wins if $s < 2r$. This is only a heuristic, since it applies Theorem 3.3 to a linear function of the composition, which we have not proved in general. Theorem 6.3 proves the conclusion directly, and Theorem 6.4 finds the crossing up to $O(N^{-\gamma})$ when reversals dominate. The tool is a decomposition of a path into 3 independent levels.

Lemma 6.1 (3 levels). *Write a sequence of directions as its number m of switches, its skeleton $Z = (Z_0, \dots, Z_m)$, which lists the directions of the runs so that $Z_t \neq Z_{t-1}$, and its run lengths $\ell = (\ell_0, \dots, \ell_m)$, which form a composition of N into $m + 1$ positive parts. This is a bijection, and $Y_N = \sum_{t=0}^m \ell_t e_{Z_t}$. Under the walk, (a) m has the law $\text{Bin}(N - 1, \beta h)$. Given m , (b) Z and ℓ are independent, (c) Z is the random walk on $\mathbb{Z}_4$ with uniform Z_0 whose independent increments are ± 1 with probability r/β each and 2 with probability s/β , and (d) ℓ is uniform on the $\binom{N-1}{m}$ compositions.*

Proof. Put $q(\pm 1) = r$ and $q(2) = s$. The sequence with data (m, Z, ℓ) has probability

$$\frac{1}{4}p_0^{N-1-m} \prod_{t=1}^m h q(Z_t - Z_{t-1}) = \binom{N-1}{m} (\beta h)^m p_0^{N-1-m} \cdot \frac{1}{4} \prod_{t=1}^m \frac{q(Z_t - Z_{t-1})}{\beta} \cdot \binom{N-1}{m}^{-1}.$$

For fixed m the middle factor is the probability of Z under the walk in (c), so it sums to 1 over the skeletons, and the last factor sums to 1 over the compositions. So the 3 factors are the laws in (a), (c) and (d), and the product form gives (b). $\square$

Proposition 6.2 (exact decomposition and a unique crossing). *Let $N \geq 2$ be even and $0 < h < 1/\beta$. Let $o_N^{(0)}$ and $o_N^{(4)}$ be the probabilities that $Y_N = 0$ and the walk never turns, and that $Y_N = 0$ and the walk uses all 4 directions. Then $o_N = o_N^{(0)} + o_N^{(4)}$,*

$$o_N^{(0)} = 2S_N(u) c_N, \quad 0 \leq o_N^{(4)} \leq \frac{w^2}{2p_0} (1 - p_0^N).$$

Equivalently, $o_N/c_N = 2S_N(u) + E_N$ with $0 \leq E_N \leq 2w^2((1+w)^N - 1)$. Also, o_N/c_N is a polynomial in v and u with nonnegative coefficients and no constant term. If $s > 0$, or if $s = 0$ and $N \geq 4$, it increases strictly in h from 0 at $h = 0$ to ∞ as $h \uparrow 1/\beta$. So there is exactly one $h_N^ \in (0, 1/\beta)$ with $o_N = c_N$, and the corner is more likely than the origin exactly when $h < h_N^*$.*

In words: the origin mass splits exactly into the axis part, which is twice the middle bin of Kagey's board measured against its endpoint, and a part from paths with all 4 directions, which is small when w is small. The ratio o_N/c_N grows with the switching rate, so the corner and the origin cross exactly once, at every even $N \geq 4$.

Proof. An origin path with no turn stays on the axis of its first step. An origin path that turns uses both axes, and since $n_0 = n_2$ and $n_1 = n_3$ at the origin, it uses all 4 directions. So every origin path is of exactly one of the 2 kinds. The origin paths on the axis $\{0, 2\}$ are the paths with $n_0 = n_2 = N/2$, and the path that always goes east has probability c_N . So Proposition 4.9 with $a = s$ and $q = \beta$, for the pairs $\{0, 2\}$ and $\{1, 3\}$, gives $o_N^{(0)} = 2S_N(u)c_N$. The bound on $o_N^{(4)}$ follows from Lemma 6.1 and a count of the run lengths that bring a skeleton back to the origin. We prove it in Appendix C. Dividing by c_N gives the form with E_N .

A path with a turns and b reversals has probability $\frac{1}{4}p_0^{N-1-a-b}(rh)^a(sh)^b = c_N v^a u^b$. So o_N/c_N is the sum of $v^a u^b$ over the origin paths, and $a + b \geq 1$ for each of them. Both u and v are nondecreasing in h , and each of them with a positive rate increases strictly to ∞ as $h \uparrow 1/\beta$. If $s > 0$, the path that goes east $N/2$ times and then west gives the term u . If $s = 0$ and $N \geq 4$, then $r > 0$, and a path whose 4 runs go east, north, west and south, with lengths $a, b, a, b \geq 1$, gives the term v^3 . This proves the claims on monotonicity and the limits. For $s = 0$ and $N = 2$ the origin cannot be reached. $\square$

The identity and the unique crossing for $s > 0$ are checked in Lean (Section 9). We write $T_N^* = Nh_N^*$ and $u_N^* = sh_N^*/(1 - \beta h_N^*)$. At the crossing $o_N = c_N$, so $2S_N(u_N^*) = o_N^{(0)}/o_N$ there. We call it the zero-turn share of the origin mass.

We compare S_N with Bessel functions by the uniform Bessel bound of [2]. Put $\psi(z) = z(I_0(z) + I_1(z))$. In our notation that bound says that for every $N \geq 2$ and $z > 0$

$$\left(1 - \frac{z(z+2)}{2N}\right)\psi(z) \leq \frac{N}{2} S_N\left(\frac{z}{N}\right) \leq \psi(z). \quad (8)$$

Since $I'_0 = I_1$ and $(zI_1)' = zI_0$, we have $\psi' = \psi + I_0$, so $(\log \psi)' \geq 1$.

Theorem 6.3 (first order). *Fix $r > 0$ and $s \geq 0$, put $\tau^* = \min(2/\beta, 1/s)$ with $1/0 = \infty$, and let $N \rightarrow \infty$ through even integers.*

- (a) *Fix $\tau > 0$ and put $h = \tau \log N/N$. If $\tau < \tau^*$, then $o_N/c_N \leq N^{-\gamma}$ for some $\gamma > 0$ and all large N . If $\tau > \tau^*$, then $o_N/c_N \geq N^\gamma$ for some $\gamma > 0$ and all large N .*
- (b) *$T_N^*/\log N \rightarrow \tau^*$.*

(c) If $s > 2r$, then $2S_N(u_N^*) \rightarrow 1$, so the origin mass at the crossing comes from the 2 axes, half from each. If $s < 2r$, then $2S_N(u_N^*) \leq N^{-\gamma}$ for some $\gamma > 0$ and all large N , so it comes from the paths that use all 4 directions.

In words, the heuristic above is right: the crossing sits at $T \approx \tau^* \log N$, and at the crossing the origin is reached through the axes when reversals dominate and through all 4 directions when turns dominate.

Idea of the proof. The proof is in Appendix C. The upper bound in (a) follows from Proposition 6.2 and (8). For the lower bound when $\beta\tau > 2$ we count balanced skeletons with Lemma 6.1, and they give $o_N^{(4)}$ of order at least $N^{-2}T^{-1/2}$. Part (b) follows from (a) and the monotonicity in Proposition 6.2. For $r = 0$ the same limit $1/s$ follows from Theorem 6.4(c).

When reversals dominate, the crossing is the crossing of [2] for $N/2$ steps. We write $z_{N/2} = \frac{N}{2}(1 - p_{N/2})$ for it, and put

$$\mathcal{A}(\Lambda) = \Lambda - \frac{1}{2} \log \Lambda + c_0 + \frac{\frac{1}{4} \log \Lambda - \frac{1}{2} c_0 + \frac{1}{8}}{\Lambda},$$

so that the third-order expansion of z_N in [2] reads $z_N = \mathcal{A}(L) + O((\log L)^2/L^2)$.

Theorem 6.4 (reversals dominate). (a) Let $s > 0$. For all even $N \geq 2$ and $0 < h < 1/\beta$,

$$\frac{o_N}{c_N} = 2S_N(u)(1 + \varepsilon_N(h)), \quad 0 \leq \varepsilon_N(h) \leq \frac{w^2(1+w)^N}{S_N(u)}.$$

(b) Let $r \geq 0$ and $s > 2r$, let $N \rightarrow \infty$ through even integers, and let $0 < \gamma < \min(1, (s - 2r)/s)$. Then

$$\varepsilon_N(h_N^*) = O((\log N)^{3/2-r/s} N^{-(s-2r)/s}), \quad Nu_N^* = z_{N/2} + O(N^{-\gamma}).$$

(c) Under the hypotheses of (b), $sT_N^* = z_{N/2} + O(N^{-\gamma})$. Hence

$$sT_N^* = \mathcal{A}(L - \log 2) + O\left(\frac{(\log L)^2}{L^2}\right), \quad sT_N^* - z_N = -\log 2 + \frac{\log 2}{2L} + O\left(\frac{(\log L)^2}{L^2}\right).$$

In words: when reversals dominate, the planar walk at its crossing behaves like Kagey's board with half as many steps, and its crossing is smaller than Kagey's by $\log 2$ in the limit. Here is why. At the crossing $o_N \approx o_N^{(0)} = 2S_N(u)c_N$, so the 2 axes double the mass at the origin. By (8), applied with N and with $N/2$ steps, the equations $2S_N(z/N) = 1$ and $S_{N/2}(z/(N/2)) = 1$ both say that $\psi(z) \approx N/4$. So doubling S_N has the same effect as halving N .

Proof. (a) Since $s > 0$, $S_N(u) \geq 2u > 0$. Divide Proposition 6.2 by $2S_N(u)$. Then $\varepsilon_N(h) = E_N/(2S_N(u))$ and $E_N \leq 2w^2(1+w)^N$.

(b) Write $u^* = u_N^*$, $z^* = Nu^*$ and $E^* = E_N(h_N^*)$. At the crossing Proposition 6.2 gives $2S_N(u^*) + E^* = 1$. So $S_N(u^*) \leq \frac{1}{2} < S_N(u_N)$, and $u^* < u_N$. Put $w_N = \beta u_N/s$. Since $w \mapsto 2w^2(1+w)^N$ is increasing, $E^* \leq \eta_N := 2w_N^2(1+w_N)^N$. By the second-order expansion of z_N in [2], $Nu_N = z_N/p_N = L - \frac{1}{2} \log L + O(1)$. So $(1+w_N)^N \leq e^{Nw_N} = O(N^{\beta/s} L^{-\beta/(2s)})$ and $w_N^2 = O(L^2/N^2)$. As $\beta/s - 2 = -(s - 2r)/s$ and $2 - \beta/(2s) = \frac{3}{2} - \frac{r}{s}$, this gives $\eta_N = O(L^{3/2-r/s} N^{-(s-2r)/s})$, which tends to 0. Then $\varepsilon_N(h_N^*) = E^*/(1 - E^*) \leq \eta_N/(1 - \eta_N)$.

Next let $z' = \frac{N}{2}u_{N/2}$, so that $S_{N/2}(2z'/N) = 1$. By (8) with $N/2$ steps, $\frac{N}{4} \leq \psi(z') \leq \frac{N}{4}(1 - \delta'_N)^{-1}$ with $\delta'_N = z'(z' + 2)/N$. By (8) with N steps and $\frac{1}{2}(1 - \eta_N) \leq S_N(u^*) \leq \frac{1}{2}$, we get

$\frac{N}{4}(1 - \eta_N) \leq \psi(z^*) \leq \frac{N}{4}(1 - \delta_N)^{-1}$ with $\delta_N = z^*(z^* + 2)/(2N)$. Both z' and $z^* < Nu_N$ are $O(L)$, so $\delta_N, \delta'_N = O(L^2/N)$. As $(\log \psi)' \geq 1$, $|z^* - z'| \leq |\log \psi(z^*) - \log \psi(z')| = O(\eta_N + L^2/N)$. Finally $z_{N/2} = z'/(1 + u_{N/2}) = z' + O(L^2/N)$. So $z^* - z_{N/2} = O(\eta_N + L^2/N)$, which is $O(N^{-\gamma})$.

(c) Since $h_N^* = u^*/(s + \beta u^*)$, we have $sT_N^* = z^*/(1 + \beta u^*/s) = z^* + O(L^2/N)$, as $u^* < u_N = O(L/N)$. With (b) this gives $sT_N^* = z_{N/2} + O(N^{-\gamma})$. The third-order expansion of z_N in [2], for $N/2$ steps and for N steps, gives $z_{N/2} = \mathcal{A}(L - \log 2) + O((\log L)^2/L^2)$ and $z_N = \mathcal{A}(L) + O((\log L)^2/L^2)$. The last term of $\mathcal{A}$ has derivative $O(\log L/L^2)$ near L , so

$$\mathcal{A}(L - \log 2) - \mathcal{A}(L) = -\log 2 - \frac{1}{2} \log \left(1 - \frac{\log 2}{L}\right) + O\left(\frac{\log L}{L^2}\right) = -\log 2 + \frac{\log 2}{2L} + O\left(\frac{\log L}{L^2}\right). \quad \square$$

For $(r, s) = (0.1, 1)$ the difference $sT_N^* - z_N$ is -0.6317 at $N = 640$ and -0.6493 at $N = 10240$, while the first 2 terms in (c) give -0.6395 and -0.6556 (Figure 5(a)). When turns dominate, the origin mass is carried by the paths with all 4 directions, and we need their local limit.

Remark 6.5 (when turns matter, proof omitted). Let $r > 0$ and $s \geq 0$, let $N \rightarrow \infty$ through even integers, and let $T = Nh \rightarrow \infty$ with $T \leq N^{1/8}$. Then $N^2 o_N^{(4)} = \frac{2(r+s)}{\pi} T(1 + o(1))$. This is twice the density at 0 of a centered Gaussian vector with covariance $\frac{N^2}{2(r+s)T} I_2$, and the factor 2 appears because Y_N lies on a sublattice of index 2. If $s < 2r$, it gives

$$\beta T_N^* + \log T_N^* = 2L + \log \frac{\pi}{8(r+s)} + o(1), \quad \text{or} \quad \beta T_N^* = 2z_N + \log \frac{\beta}{2(r+s)} + o(1).$$

For $r = s = 1$ the bold curve of Figure 1 solves the first equation without its $o(1)$ term. The coefficient of $\log T_N^*$ is 1 and not $\frac{1}{2}$, because the origin is a point of a 2-dimensional local limit. Still $T_N^* = \tau^*[L - \frac{1}{2} \log L + O(1)]$ for every $s \geq 0$. At the switch $s = 2r$ the zero-turn share tends to $2\sqrt{2}/(\sqrt{2} + \sqrt{5}) = 0.7749$, and the term $\log((\sqrt{5} - \sqrt{2})/(\sqrt{5} + \sqrt{2}))$ is added to the right side of the first equation. The approach is slow. At $N = 10240$ the zero-turn share is 0.7650 for $(r, s) = (0.5, 1)$ and still 0.1051 for $(1, 1)$ (Figure 5(b)). With turns only ($s = 0$), we rotate the lattice by 45° . If $Y_N = (Y^{(1)}, Y^{(2)})$, then $Y^{(1)} + Y^{(2)}$ and $Y^{(1)} - Y^{(2)}$ move by ± 1 at each step, and their sign words are 2 copies of the walk of [2] that never switch at the same step. So $Y_N = 0$ exactly when both words are balanced, which is the lattice form of the factorization in [26, Eq. (36)]. A union bound over the pairs of balanced words that switch at a common step then gives $rT_N^* = z_N + O((\log N)^4/N)$. The other proofs use Lemma 6.1, a local limit for the run lengths given the skeleton and a martingale central limit theorem [28] for the skeleton. We omit the details.

7 Application: sticky priors for hidden Markov models

This section and the next are applications. Nothing earlier depends on them, and a reader who is interested only in the general theory and the planar walk can skip to Section 9. Here we ask which hidden states the most likely count vector of a sticky prior uses. In words, the answer is face selection with $\lambda_F = 1 - \varrho(\Pi_F)$: the prior mode uses 1 state, then a proper set of states, then all of them, and only the uniform kernel jumps from 1 state straight to all.

Fox et al. [20] add a self-transition bias $\kappa \geq 0$ to the HDP hidden Markov model, so that the hidden chain stays in each state longer. In their weak-limit approximation with base weights $\bar{\beta}$ and concentration ω , row j of the transition matrix is Dirichlet with parameters $\omega \bar{\beta}_1, \dots, \omega \bar{\beta}_j + \kappa, \dots, \omega \bar{\beta}_d$. By neutrality, row j off the diagonal, divided by $1 - \pi_{jj}$, is Dirichlet with parameters

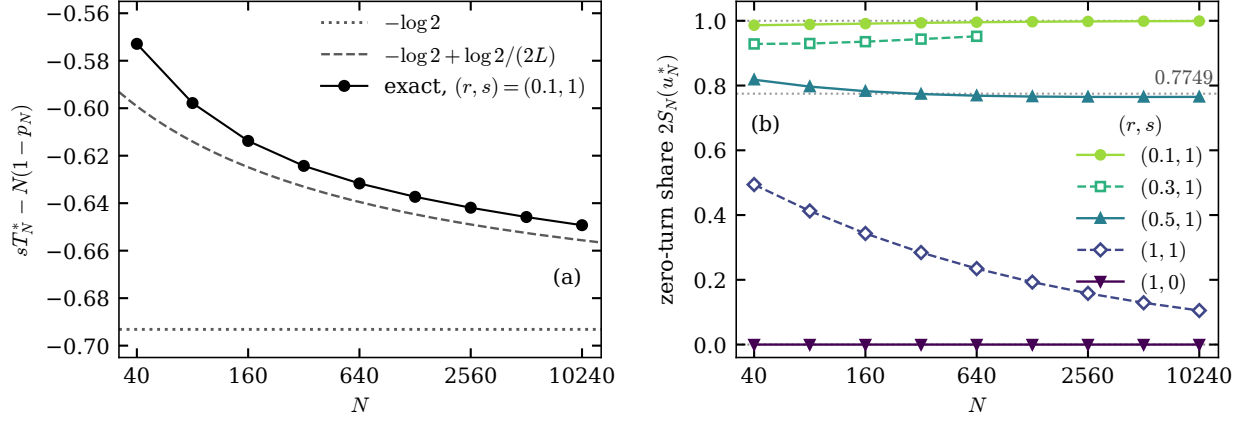


Figure 5: (a) The exact crossing of the planar walk with $(r, s) = (0.1, 1)$, written as $sT_N^* - z_N$ with $z_N = N(1 - p_N)$, for $N = 40, 80, \dots, 10240$. The dashed curve is $-\log 2 + \log 2/(2L)$ from Theorem 6.4(c), and the dotted line is its limit $-\log 2$. (b) The zero-turn share $2S_N(u_N^*)$ of the origin mass at the crossing. It tends to 1 for $s > 2r$ and to 0 for $s < 2r$ (Theorem 6.3), and by Remark 6.5 (proof omitted) to $2\sqrt{2}/(\sqrt{2} + \sqrt{5}) = 0.7749$ at the switch $s = 2r$. For $(r, s) = (1, 0)$ there are no reversals and the share is 0 for every N . The pair $(0.3, 1)$ was computed up to $N = 640$.

$(\omega\bar{\beta}_l)_{l \neq j}$ and independent of π_{jj} . So κ only sets how sticky the chain is. We fix every exit probability at $h = \tau L/N$, the scale of Kagey's crossing, although the prior mixes all scales of h .

So let $\Pi = (\pi_{ij})$ be a stochastic matrix on $[d]$ with zero diagonal, $\tau > 0$ and $h = \tau L/N < 1$. Then $(1 - h)I + h\Pi = I + hQ$ with $Q = \Pi - I$, so $q_i = 1$ and $T = \tau L$ (Example 2.5(iii)). We assume that $\pi_{ij} > 0$ for all $i \neq j$, which we call (A+), and that μ has full support. Under the prior (A+) holds almost surely. Let $\rho_F = \varrho(\Pi_F)$ for the principal submatrix Π_F on F . Then $\lambda_F = 1 - \rho_F$, and Q_F and Π_F have the same Perron vectors. A vertex has $\rho_F = 0$, and $\rho_{[d]} = 1$. If $2 \leq |F| \leq d - 1$, then Π_F is irreducible by (A+), and $0 < \rho_F < 1$ since ρ_F is at least the least row sum of Π_F and the Perron root of the irreducible Π exceeds that of each proper principal submatrix [29]. So $\Phi_\tau(F) = \tau(1 - \rho_F) + |F| - 1$, and we put

$$\tau_- = \min_{|F| \geq 2} \frac{|F| - 1}{\rho_F}, \quad \tau_+ = \max_{F \neq [d]} \frac{d - |F|}{1 - \rho_F}.$$

We call a global mode of P_N , the most likely value of the counts under the prior, a *prior mode*.

Corollary 7.1 (support and location of the prior mode). *Assume (A+), let μ have full support, and fix $\tau > 0$. Let $\mathcal{F}_\tau$ be the set of faces that minimize Φ_τ . There is $N_0(\tau)$ such that for $N \geq N_0(\tau)$ every prior mode n^* satisfies $\text{supp } n^* \in \mathcal{F}_\tau$ and $n^*/N = \alpha_F^* + O(1/L)$ with $F = \text{supp } n^*$. If $\mathcal{F}_\tau = \{F^*\}$, then $n^*/N \rightarrow \alpha_{F^*}^* = l \circ r$, where l and r are the left and right Perron vectors of Π_{F^*} with $l \cdot r = 1$. No reversibility is needed.*

Proof. By (A+) every face is irreducible and, as μ has full support, admissible. So Theorem 3.3(iii) and (iv) give the claims, with $\alpha_F^* = l \circ r$. $\square$

At $\tau = \tau_-$ the vertices tie with the faces that attain τ_- , and Theorem 4.8(iii) with $\tau^* = \tau_-$ and a fixed large $t = t_0$ shows that for large N the prior mode is not a single state at $\tau = \tau_-(1 - \frac{\log L - 2t_0}{2L}) < \tau_-$. So $N_0(\tau) \rightarrow \infty$ as $\tau \uparrow \tau_-$.

Proposition 7.2 (thresholds). *Assume (A+) and $d \geq 3$.*

(a) *The minimizers of Φ_τ are exactly the vertices if and only if $\tau < \tau_-$, and no vertex minimizes Φ_τ if $\tau > \tau_-$. The face $[d]$ is the unique minimizer if and only if $\tau > \tau_+$, and $[d]$ does not minimize Φ_τ if $\tau < \tau_+$.*

(b) $1 < \tau_- \leq d - 1 \leq \tau_+$.

(c) $\tau_- = d - 1$ if and only if $\tau_+ = d - 1$, and both hold if and only if

$$\rho_F \leq \frac{|F| - 1}{d - 1} \quad \text{for every face } F. \quad (9)$$

If (9) fails, then $\tau_- < d - 1 < \tau_+$, and for $\tau \in (\tau_-, \tau_+)$ every minimizer F of Φ_τ has $2 \leq |F| \leq d - 1$.

Consequently, fix τ and let $N \geq N_0(\tau)$. If $\tau < \tau_-$, the prior modes are the Ne_i with $\mu_i = \max_l \mu_l$. If $\tau > \tau_+$, every prior mode has full support. If $\tau_- < \tau < \tau_+$, every prior mode has a support F with $2 \leq |F| \leq d - 1$, and $F = F^$ if F^* is the unique minimizer of Φ_τ .*

Proof. (a) A vertex has $\Phi_\tau = \tau$. For $|F| \geq 2$, $\Phi_\tau(F) - \tau = \rho_F((|F| - 1)/\rho_F - \tau)$ with $\rho_F > 0$, and for $F \neq [d]$, $\Phi_\tau(F) - (d - 1) = (1 - \rho_F)(\tau - (d - |F|)/(1 - \rho_F))$ with $1 - \rho_F > 0$, while $\Phi_\tau([d]) = d - 1$. (b) For $2 \leq |F| \leq d - 1$, $\rho_F < 1$ gives $(|F| - 1)/\rho_F > 1$, and $[d]$ gives $d - 1 \geq 2$, so $\tau_- > 1$. Also $[d]$ gives $\tau_- \leq d - 1$, and a vertex gives $\tau_+ \geq d - 1$. (c) By (b), $\tau_- = d - 1$ exactly when $(|F| - 1)/\rho_F \geq d - 1$ for all $|F| \geq 2$, and $\tau_+ = d - 1$ exactly when $(d - |F|)/(1 - \rho_F) \leq d - 1$ for all $F \neq [d]$. Both say $(d - 1)\rho_F \leq |F| - 1$, which holds for the vertices and $[d]$. If (9) fails, we get $\tau_- < d - 1 < \tau_+$, and (a) excludes the vertices and $[d]$ on (τ_-, τ_+) . The consequence follows from Corollary 7.1 and $P_N(Ne_i) = \mu_i(1 - h)^{N-1}$. Parts (a) to (c) are checked in Lean. $\square$

We say that Π has a *direct jump* when (9) holds. Then at first order the prior mode goes from one state straight to all d states at $\tau = d - 1$. The uniform kernel $\pi_{ij} = 1/(d - 1)$ has constant row sums on F , so $\rho_F = (|F| - 1)/(d - 1)$. So (9) says that no set of states is more closed than under the uniform kernel. That kernel has $Q = (J - dI)/(d - 1)$, and all its faces tie at $\tau = d - 1$, the degeneracy of [3] (Corollary 4.11).

Corollary 7.3 (only the uniform kernel). *Let $d \geq 3$ and assume (A+). (a) If Π is symmetric, then Π has a direct jump if and only if $\pi_{ij} = 1/(d - 1)$ for all $i \neq j$. (b) If $d = 3$, then Π has a direct jump if and only if $\pi_{ij}\pi_{ji} \leq \frac{1}{4}$ for all 3 pairs. In particular every cyclic kernel $\pi_{i,i+1} = a$, $\pi_{i,i-1} = 1 - a$ (indices mod 3) has a direct jump.*

Proof. A pair has $\rho_{\{i,j\}} = \sqrt{\pi_{ij}\pi_{ji}}$. (a) Here $\rho_{\{i,j\}} = \pi_{ij}$, and since the rows sum to 1 the average of π_{ij} over the pairs is $1/(d - 1)$. So (9) fails unless all π_{ij} are equal, and the uniform kernel satisfies it with equality. (b) For $d = 3$, (9) is automatic for the vertices and [3], and for a pair it reads $\pi_{ij}\pi_{ji} \leq \frac{1}{4}$. The cyclic kernel has $a(1 - a) \leq \frac{1}{4}$. Both parts are checked in Lean. $\square$

Proposition 7.4 (the direct jump under the prior). *Let $d = 3$ and let the base weights be uniform. In the weak-limit sticky HDP-HMM the rows of Π are then independent, and the first off-diagonal entry of each row has the law $\text{Beta}(\omega/3, \omega/3)$. Write $\mathbb{P}_\omega$ for this law of Π and D for the event that Π has a direct jump. Then (a) $\mathbb{P}_\omega(D) \rightarrow \frac{1}{4}$ as $\omega \rightarrow 0$, and (b) $\mathbb{P}_\omega(D) = \frac{3\sqrt{3}}{8\pi\omega}(1 + o(1))$ as $\omega \rightarrow \infty$.*

Proof. Let B_1, B_2, B_3 be independent with law $\text{Beta}(\omega/3, \omega/3)$, and put $\pi_{12} = B_1$, $\pi_{21} = B_2$ and $\pi_{31} = B_3$, so that the other off-diagonal entries are $1 - B_j$. By Corollary 7.3(b), D is the event that $B_1 B_2$, $(1 - B_1) B_3$ and $(1 - B_2)(1 - B_3)$ are all at most $\frac{1}{4}$. (a) As $\omega \rightarrow 0$ each B_j tends in law to a fair coin on $\{0, 1\}$, and the 3 products then take only the values 0 and 1. So ∂D has limit probability 0, and $\mathbb{P}_\omega(D)$ tends to the limit probability of D . Only $(1, 0, 1)$ and $(0, 1, 0)$ make all 3 products 0 (checked in Lean), so the limit is $\frac{1}{4}$. Part (b) is proved in Appendix D. $\square$

For (b), for large ω the set D is a thin tube around the cyclic kernels, with cross-section of area about $\frac{9}{8}t^4$ at the logit coordinate t along the cyclic kernels (Appendix D), which gives the order $1/\omega$. So the uniform tie is not typical. At $\omega = 1$, Monte Carlo with 10^6 draws gives $\mathbb{P}_\omega(D) = 0.115$, so about 0.88 of random kernels have a window of τ in which the prior mode sits on a pair (Proposition 7.2(c)).

Corollary 7.5 (an exact link to Kagey's crossing). *Let $i \neq j$ satisfy $\pi_{ij} = \pi_{ji} = \pi_0 > 0$ and $\mu_i = \mu_j > 0$. We do not assume (A+). Then for every $N \geq 2$ the balanced compositions are the most likely ones with support $\{i, j\}$, and they are more likely than Ne_i if and only if $\tau > \tau^{\text{pair}}(N) := Nu_N/((\pi_0 + u_N)L)$. If also $\mu_i = \mu_j = \max_l \mu_l$, then for every τ with $\tau^{\text{pair}}(N) < \tau < N/L$ no prior mode is a single state. Moreover,*

$$\tau^{\text{pair}}(N) = \frac{1}{\pi_0} \left[1 - \frac{\frac{1}{2} \log L - c_0}{L} + O\left(\frac{\log L}{L^2}\right) \right],$$

which tends to the first-order value $1/\rho_{\{i,j\}} = 1/\pi_0$.

Proof. Every exit rate is 1, so this is Proposition 4.9 with $a = \pi_0$ and $q = 1$, where $u = \pi_0 h/(1-h) > u_N$ exactly when $h > u_N/(\pi_0 + u_N)$. For the prior modes, $P_N(Ne_l) = \mu_l(1-h)^{N-1} \leq P_N(Ne_i)$ for every l , and the range of τ is not empty as $u_N/(\pi_0 + u_N) < 1$. Finally $Nu_N = z_N/p_N$ and $u_N = O(L/N)$, so $\tau^{\text{pair}}(N) = \frac{z_N}{\pi_0 L}(1 + O(L/N))$, and the second-order expansion of z_N in [2] gives the claim. $\square$

Proposition 7.6 (face masses). *Assume (A+) and $0 < h < 1$. For every face F with $|F| \geq 2$ and $\Lambda_F = 1 - h\lambda_F$,*

$$\frac{\min r_F}{\max r_F} \mu(F) \Lambda_F^{N-1} \leq \mathbb{P}(\text{supp } n \subseteq F) \leq \frac{\max r_F}{\min r_F} \mu(F) \Lambda_F^{N-1},$$

and for fixed τ , $\Lambda_F^{N-1} = N^{-\tau\lambda_F} e^{O(L^2/N)}$. Also $\mathbb{P}(\text{one state}) = (1-h)^{N-1} \sim N^{-\tau}$ and $\mathbb{P}(\text{supp } n = [d]) \rightarrow 1$.

Proof. Let $U_F = (1-h)I + h\Pi_F \geq 0$. Then $\mathbb{P}(\text{supp } n \subseteq F) = \mu_F^\top U_F^{N-1} \mathbf{1}$ and $U_F r_F = \Lambda_F r_F$. As $r_F/\max r_F \leq \mathbf{1} \leq r_F/\min r_F$ entrywise, this gives the bounds, and $(N-1)\log(1-h\lambda_F) = -\tau\lambda_F L + O(L^2/N)$. A vertex composition is a single path, and every $F \neq [d]$ has $\lambda_F > 0$, which gives the last 2 claims. The finite bounds are checked in Lean. $\square$

Remark 7.7 (prior mode versus mass). Let $d \geq 3$. For fixed $\tau < \tau_+$ and large N every prior mode has a proper support, while $\mathbb{P}(\text{supp } n = [d]) \rightarrow 1$. For $\tau < \tau_-$ the prior mode uses one state, an event of probability about $N^{-\tau}$. A face of size k holds about N^{k-1} compositions, so it can carry most of the mass while each of them is less likely than a vertex. Table 2 shows this. At $\tau = 0.8$ the prior mode is a single state, yet $\mathbb{P}(\text{one state}) = 0.0122$. At $\tau = 8$ first order puts the prior mode on $\{1, 2\}$, but at $N = 240$ it has full support. So the prior mode is a poor summary of a sticky prior, and face selection says when it misleads.

τ	prior mode	one state	$\{1, 2\}$	[3]
0.5	one state	0.0643	0.465	0.379
0.8	one state	0.0122	0.423	0.531
3	(120, 120, 0)	4.3×10^{-8}	0.129	0.871
8	(111, 111, 18)	1.2×10^{-21}	0.0081	0.992

Table 2: The kernel Π with rows $(0, 0.9, 0.1)$, $(0.9, 0, 0.1)$, $(0.5, 0.5, 0)$, the uniform start and $N = 240$, where $\tau_- = 10/9$ and $\tau_+ = 10$. The prior mode (from exact dynamic programming, with a 3-way tie at $\tau = 0.5$ and 0.8) and the prior probabilities of the supports.

8 Application: the information in the endpoint

This second application returns to Kagey's board and asks a statistical question. If we only see the bin where the ball lands, how much do we learn about the switching probability? The answer is that the bin keeps about a fraction $1/(2T)$ of the Fisher information of the whole path, so about $1/(2 \log N)$ at Kagey's crossing. We use the shared notation of Section 1.2: the first step is left or right with probability $\frac{1}{2}$ each, and each later step repeats the one before it with probability $p = 1 - q$, where $0 < q < 1$. This is Example 2.5(i) with $h = q$, so $T = Nq$. Put $\rho = 1 - 2q$, $u = q/p$ and $y = Nu = T/p$. Let J be the number of switches, K the number of right steps, $X = 2K - N$ the endpoint and $W = X/N$. A path with J switches has probability $\frac{1}{2}p^{N-1-J}q^J$. So $J \sim \text{Bin}(N-1, q)$, and the path carries the Fisher information $(N-1)/(pq)$ about q . The endpoint carries $I_K(q) = \sum_{k=0}^N (\partial_q \mathbb{P}(K = k))^2 / \mathbb{P}(K = k)$, and we call $e_N(q) = I_K(q) pq / (N-1)$ the *efficiency* of the endpoint. It is the fraction of the information left when we see only the endpoint.

The endpoint sees the switches only through how far the walk spreads. A normal X with mean 0 and variance Np/q , close to $\text{Var } X$ for large T , would carry the information $\frac{1}{2}(\partial_q \log(Np/q))^2 = 1/(2p^2q^2)$, which is the fraction $1/(2(N-1)pq) \approx 1/(2T)$. Theorem 8.3 confirms this.

Proposition 8.1 (a correlation ratio). *For $N \geq 2$ and $0 < q < 1$,*

$$e_N(q) = \frac{\text{Var}(\mathbb{E}[J \mid K])}{\text{Var } J}.$$

For $0 < k < N$ we have $\mathbb{E}[J \mid K = k] = \frac{d}{dt} \log S_{k, N-k}(e^t)$ at $t = \log(q/p)$, where $S_{a,b}$ is the polynomial of Section 4.4. For $k \in \{0, N\}$ we have $J = 0$.

Proof. The score of a path is $(J - (N-1)q)/(pq)$. Since $\mathbb{P}(K = k)$ is a sum of path probabilities, $\partial_q \log \mathbb{P}(K = k) = \mathbb{E}[(J - (N-1)q)/(pq) \mid K = k]$. As $\mathbb{E}J = (N-1)q$, this gives $I_K(q) = \text{Var}(\mathbb{E}[J \mid K])/(pq)^2$, and we divide by $(N-1)/(pq) = \text{Var } J/(pq)^2$. By the run count of [2], for $0 < k < N$ the probability $\mathbb{P}(K = k, J = j)$ is $\frac{1}{2}p^{N-1}u^j$ times the number of words with k right steps, $N-k$ left steps and j switches. Summing over j gives $\mathbb{P}(K = k) = \frac{1}{2}p^{N-1}S_{k, N-k}(u)$, so $\mathbb{E}[J \mid K = k] = uS'_{k, N-k}(u)/S_{k, N-k}(u)$. Only the 2 constant paths have $K \in \{0, N\}$. $\square$

This identity, checked in Lean, says that e_N is the correlation ratio of J on K . Since $\mathbb{E}[J \mid K]$ is the orthogonal projection of J onto the functions of K in L^2 , the Cauchy-Schwarz inequality gives

$$e_N(q) \geq \frac{\text{Cov}(J, g(K))^2}{\text{Var } J \text{Var } g(K)} \quad \text{whenever } \text{Var } g(K) > 0. \quad (10)$$

Remark 8.2 (few switches, proof omitted). For fixed N we have $e_N(q) = 1 - \frac{N-2}{2}q + O_N(q^2)$ as $q \rightarrow 0$. If $q \rightarrow 0$ and $T = Nq$ stays bounded, then $e_N(q) \rightarrow 1$ if and only if $T \rightarrow 0$. The reason is that 1 switch and 2 switches both leave the endpoint uniform on the interior bins. The proof applies (10) to the indicator of $K \in \{0, N\}$ and compares the 2 words with 1 switch with the $N - 2$ words with 2 switches in each interior bin. We omit the details.

Theorem 8.3 (many switches). *Let $R_N^2(q) = \text{Cov}(J, X^2)^2 / (\text{Var } J \text{ Var } X^2)$ be the squared correlation of J and X^2 . For $N \geq 2$ and $0 < q < 1$,*

$$R_N^2(q) = \frac{1}{2(N-1)pq} \cdot \frac{[1 + \rho^N - p(1 - \rho^N)/T]^2}{V},$$

where

$$V = 1 - \frac{1 + 8\rho + \rho^2 + 8\rho^{N+1}}{4pT} + \frac{\rho(1 - \rho^N)[2(1 + 2\rho)(2 + \rho) - \rho(1 - \rho^N)]}{8p^2T^2}.$$

There are absolute constants C and N_0 such that for $N \geq N_0$, $0 < q \leq \frac{1}{2}$ and $1 \leq T \leq N^{1/4}$,

$$0 \leq e_N(q) - R_N^2(q) \leq C \left(\frac{1}{T^3} + \frac{T^3}{N} \right).$$

Consequently, in the same range and with an absolute implied constant,

$$e_N(q) = \frac{1}{2T} + \frac{1}{4T^2} + O \left(\frac{1}{T^3} + \frac{T^3}{N} \right).$$

The lower bound $e_N \geq R_N^2$ is (10) with $g(K) = X^2$, for all $N \geq 2$ and $0 < q < 1$. For the upper bound, Proposition 8.1 and the uniform Bessel estimate for every bin in [2] make $\mathbb{E}[J \mid K]$ close to $y\sqrt{1 - W^2} + \frac{1}{2}$. As W is of order $T^{-1/2}$, this is close to $y(1 - W^2/2) + \frac{1}{2}$, which lies in the span of 1 and X^2 . Appendix E proves the closed form and the bound.

Proof of the expansion in Theorem 8.3. If $T \leq 18$, then $|e_N - \frac{1}{2T} - \frac{1}{4T^2}| \leq 1 \leq 18^3 T^{-3}$, since e_N and $\frac{1}{2T} + \frac{1}{4T^2}$ lie in $[0, 1]$. Let $T \geq 18$. As $q \leq \frac{1}{2}$, we have $p \geq \frac{1}{2}$ and $0 \leq \rho < 1$, and $\rho^N \leq e^{-2Nq} = e^{-2T} \leq T^{-2}$. The middle term of V lies in $[0, 9/T]$, and since $1 + 8\rho + \rho^2 = 10 - 20q + 4q^2$ it equals $\frac{5}{2T} + O(q/T + T^{-2})$. The last term lies in $[0, 9/T^2]$. So $V \geq \frac{1}{2}$, $1/V = 1 + \frac{5}{2T} + O(q/T + T^{-2})$, the squared bracket is $1 - \frac{2}{T} + O(q/T + T^{-2})$, and $1/(2(N-1)pq) = (1 + O(q + 1/N))/(2T)$. Multiplying, $R_N^2 = \frac{1}{2T} + \frac{1}{4T^2} + O(T^{-3} + q/T + 1/(NT))$ with absolute constants. Since $q/T = 1/N \leq T^3/N$, the bound on $e_N - R_N^2$ gives the expansion. $\square$

Corollary 8.4 (at Kagey's crossing). *Let p_N be the value of p at which the middle bin and the endpoints are equally likely, and let $q_N = 1 - p_N$ and $z_N = Nq_N$ (Section 4.4). Then*

$$e_N(q_N) = \frac{1}{2L} + \frac{\log L + 1 + \log(8/\pi)}{4L^2} + O \left(\frac{(\log L)^2}{L^3} \right).$$

Proof. Since $S_N(u) \geq 2u$, we have $u_N \leq \frac{1}{2}$, so $p_N \geq \frac{2}{3}$ and $q_N \leq \frac{1}{3}$. By the second-order expansion of z_N in [2], $z_N = L(1 - \varepsilon)$ with $\varepsilon = (\frac{1}{2} \log L - c_0)/L + O(\log L/L^2)$. So Theorem 8.3 applies with $T = z_N$ for large N and gives $e_N(q_N) = \frac{1}{2z_N} + \frac{1}{4z_N^2} + O(L^{-3})$. Here $\frac{1}{2z_N} = \frac{1}{2L} + \frac{\log L - 2c_0}{4L^2} + O((\log L)^2/L^3)$, $\frac{1}{4z_N^2} = \frac{1}{4L^2} + O(\log L/L^3)$ and $-2c_0 = \log(8/\pi)$. $\square$

So $e_N \sim 1/(2 \log N)$ there, but the relative correction is of order $(\log \log N)/\log N$ and is large at every N we computed. The product $2Le_N$ is 1.74 at $N = 100$ and 1.35 at $N = 3200$ (Table 3).

N	z_N	$e_N(q_N)$	$2z_N e_N$	$2Le_N$	$R_N^2(q_N)$	$\frac{1}{2z_N} + \frac{1}{4z_N^2}$
50	2.8998	0.24117	1.3987	1.8869	0.20873	0.20216
100	3.5034	0.18869	1.3221	1.7379	0.16642	0.16309
400	4.7501	0.12681	1.2047	1.5196	0.11715	0.11634
1600	6.0246	0.09433	1.1366	1.3919	0.09007	0.08988
3200	6.6684	0.08361	1.1151	1.3496	0.08070	0.08060

Table 3: The efficiency of the endpoint at Kagey’s crossing $q_N = 1 - p_N$, computed from the exact law, with $z_N = Nq_N$ and $L = \log N$. The last 2 columns are the squared correlation of Theorem 8.3 and its 2-term expansion.

9 Verification and open questions

No proof in this paper rests on a floating-point computation. Exact computer algebra enters in 3 places. (1) In Lemma E.1 each moment lies in a space of sequences of dimension at most 10. The script `verify_sticky_fisher.py` computes the 3 moments exactly for $N \leq 12$ by a dynamic program in rational arithmetic with q as a symbol, and checks the identity for $\text{Var } X^2$ in sympy. (2) The row Θ_ε of Table 1 rests on 3 exact resultants of characteristic polynomials, computed in sympy by `verify_examples.py` (this row is marked “proof omitted”). (3) The values of c_F on the 4-cycle after Corollary 5.1 are exact evaluations, by `verify_second_order.py`, of the next Laplace term of Remark 4.14, whose proof we omit. Every other number in the paper is a closed form or a computation at finite size, done in 2 independent ways where possible (for example by the run expansion and by a torus integral, or by a dynamic program and an exact formula). All 4 papers on Problem 131 share one verification supplement, `SUPPLEMENT.md` in the Problem 131 folder of <https://github.com/apemm/Kagey-Problems>. It lists each check and the 2 methods behind it, and python `verify_all.py --paper c` in that folder runs the checks for this paper. Before most computations we wrote a prediction and a condition that would refute it in a ledger, and the outcome is recorded below it. Many finite and algebraic steps, and the rows of Table 1 marked Lean, are proved in Lean 4 in the folder `lean/PaperC`, and its file `MAP.md` matches each statement of the paper to its Lean theorems. The asymptotic results are not in Lean.

Several questions remain open. Theorem 4.4 holds on compact subsets of the open face, and a local law up to the boundary is open. The same formula cannot hold there. Take a star with center d and $\ell = d - 1 \geq 3$ leaves, and a start of full support. Then $P_N(n) = 0$ whenever $\text{supp } n = [d]$ and $n_d < \ell - 1$, since a path visits the center between any 2 runs at the leaves, although n/N lies in the open face. This zero set is checked in Lean. For a reducible face, Theorem 3.3(ii) places the power of T in M_F between $k_F - k^*$ and $k_F - 1$, and heuristically it is $(k^* - 1)/2 + k_F - k^*$. Theorem 4.7 puts all crossings of 2 faces in an interval of length $O(1/L)$, but uniqueness is proved only in special cases, such as Papers A and B, the pairs of Proposition 4.9 and the planar walk. We have no tree formula for K_F on non-reversible faces like Proposition 4.2(b), and no closed form for c_F . At finite N the order of the phases can differ from first order when a face lies close to the hull, as on the path P_4 (Table 1). For the planar walk the next term of the crossing equation in Remark 6.5 and the crossover near $s = 2r$ are open. A heuristic suggests that the next term comes from replacing T by $T + \frac{1}{4r} - \frac{1}{2(r+s)}$. For sticky priors we do not treat random exit probabilities as in [20], or the probability of a direct jump for $d \geq 4$. Finally, we do not compute the constant in Theorem 8.3, and the limits of e_N at fixed q and at fixed Nq are open.

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A Proofs of the local law and of the first-order bounds

This appendix proves Proposition 4.2(a), Lemma 3.2, Theorem 4.4, Lemma 4.5 and Theorem 3.3(ii) for reducible faces. Except in Section A.6, F is a face with $k \geq 2$ states, numbered $1, \dots, k$. For $x \in \mathbb{C}^F$ we write $x' = (x_2, \dots, x_k)$. We write i for the imaginary unit, since i denotes a state, and $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$. We use 2 standard facts [29]. First, let $B \geq 0$ be square and S diagonal with positive diagonal. Then $\varrho(B - S) < 0$ if and only if $\varrho(S^{-1}B) < 1$, and also if and only if $(S - B)v > 0$ for some $v > 0$. In that case $(S - B)^{-1} = \sum_{j \geq 0} (S^{-1}B)^j S^{-1} \geq 0$. Second, if M is an irreducible Metzler matrix, then $\varrho(M)$ is larger than the Perron root of each proper principal submatrix of M , and every nonnegative eigenvector of M is positive and belongs to $\varrho(M)$.

A.1 The Perron root under a diagonal perturbation

An irreducible generator on F is a Metzler matrix G with $G\mathbf{1} = 0$ whose graph is strongly connected. It has a unique stationary law $\pi > 0$. The argument after Definition 4.1, with (G, π) in place of $(\hat{Q}_\alpha, \alpha)$, shows that $Z = (\mathbf{1}\pi^\top - G)^{-1} - \mathbf{1}\pi^\top$ exists and satisfies (3). Put $D = \text{diag } \pi$ and $\Gamma = DZ + Z^\top D$.

Lemma A.1 (the Perron root). *Let G , π , Z and Γ be as above. Then $\kappa(\eta) = \varrho(G + \text{diag } \eta)$ is real-analytic near 0, with $\kappa(0) = 0$, $\nabla \kappa(0) = \pi$ and $\nabla^2 \kappa(0) = \Gamma$. For $\eta \in \mathbb{R}^F$ and $u = Z\eta$,*

$$\eta^\top \Gamma \eta = \sum_{i \neq j} \pi_i G_{ij} (u_i - u_j)^2. \quad (11)$$

So Γ is positive semidefinite with kernel $\mathbb{R}\mathbf{1}$, and Σ , which is Γ without the row and column of state 1, is positive definite.

Proof. Since $G + \text{diag } \eta$ is irreducible, its Perron root is a simple eigenvalue. So by the implicit function theorem, κ and the right eigenvector $v(\eta)$ with $\pi^\top v = 1$ are real-analytic near 0, with $\kappa(0) = 0$ and $v(0) = \mathbf{1}$. With subscripts for derivatives at $\eta = 0$, differentiating $(G + \text{diag } \eta)v = \kappa v$ in η_j gives $Gv_j + e_j = \kappa_j \mathbf{1}$, and multiplying by $\pi^\top$ gives $\kappa_j = \pi_j$. Then $v_j = Ze_j$, because by (3) both vectors solve $Gx = \pi_j \mathbf{1} - e_j$ with $\pi^\top x = 0$, and the kernel of G is $\mathbb{R}\mathbf{1}$. Differentiating again, in η_i , gives $Gv_{ij} + \text{diag}(e_i)v_j + \text{diag}(e_j)v_i = \kappa_{ij} \mathbf{1} + \kappa_i v_j + \kappa_j v_i$, and multiplying by $\pi^\top$ gives $\kappa_{ij} = \pi_i Z_{ij} + \pi_j Z_{ji} = \Gamma_{ij}$.

For (11) let $f = \eta - (\pi \cdot \eta)\mathbf{1}$. By (3), $-Gu = f$ and $\pi \cdot u = 0$, so $\eta^\top \Gamma \eta = 2 \sum_i \pi_i \eta_i u_i = 2 \sum_i \pi_i f_i u_i = -2 \sum_i \pi_i u_i (Gu)_i$. Expanding the squares in (11) and using $G\mathbf{1} = 0$ and $\pi^\top G = 0$ gives the same expression. Also $\Gamma\mathbf{1} = 0$ by (3). If $\eta^\top \Gamma \eta = 0$, then u is constant along every edge of the strongly connected graph of G , so u is constant. Then $u = 0$ because $\pi \cdot u = 0$, so $f = 0$ and $\eta \in \mathbb{R}\mathbf{1}$. A nonzero vector that vanishes at state 1 is not in $\mathbb{R}\mathbf{1}$, which gives the claim about Σ . $\square$

Proof of Proposition 4.2(a). For $\vartheta \in \mathbb{R}^F$ let $\Lambda(\vartheta) = \varrho(Q_F + \text{diag } \vartheta)$, so that $\Lambda(\vartheta + c\mathbf{1}) = \Lambda(\vartheta) + c$. It is a simple eigenvalue with positive right and left eigenvectors $\hat{r}, \hat{l}$, normalized by $\sum_i \hat{r}_i = 1$ and $\hat{l} \cdot \hat{r} = 1$, and by the implicit function theorem Λ , $\hat{r}$ and $\hat{l}$ are real-analytic in ϑ . Put $\pi = \hat{l} \circ \hat{r}$ and $G = \hat{R}^{-1}(Q_F + \text{diag } \vartheta - \Lambda(\vartheta)I)\hat{R}$ with $\hat{R} = \text{diag } \hat{r}$. Then G is an irreducible generator with stationary law π . Since $\hat{R}$ commutes with diagonal matrices, $\Lambda(\vartheta + \eta) = \Lambda(\vartheta) + \varrho(G + \text{diag } \eta)$. So by Lemma A.1, $\nabla \Lambda(\vartheta) = \pi$ and $\nabla^2 \Lambda(\vartheta) = \Gamma$, with Γ built from (G, π) .

Let $\Lambda_0(\beta) = \Lambda(0, \beta)$ for $\beta \in \mathbb{R}^{k-1}$. Its Hessian is Γ without the row and column of state 1, which is positive definite by Lemma A.1. So $\nabla \Lambda_0$ is injective, and by the inverse function theorem it has a real-analytic inverse on its open image. Fix $\alpha \in \Delta_F^\circ$ and put $\vartheta(\alpha) = q - \theta(\alpha)$. Then $Q_F + \text{diag } \vartheta(\alpha) = A_F - \text{diag } \theta(\alpha)$ has Perron root 0 and, by (2), Perron vectors r_α and l_α . So at $\vartheta(\alpha)$ we have $\pi = \alpha$, $G = \widehat{Q}_\alpha$ and $\Gamma = \Gamma_\alpha$. Let $\beta(\alpha)$ be the last $k-1$ coordinates of $\vartheta(\alpha) - \vartheta_1(\alpha)\mathbf{1}$. By the shift property, $\nabla \Lambda_0(\beta(\alpha)) = \alpha'$ and $\nabla^2 \Lambda_0(\beta(\alpha)) = \Sigma_\alpha$. Hence $\beta(\alpha) = (\nabla \Lambda_0)^{-1}(\alpha')$ is real-analytic, and so are $\vartheta_1(\alpha) = -\Lambda_0(\beta(\alpha))$, $\theta(\alpha)$, its Perron vectors and Σ_α . Moreover $I_F(\alpha) = \langle \vartheta(\alpha), \alpha \rangle = \langle \beta(\alpha), \alpha' \rangle - \Lambda_0(\beta(\alpha))$. Differentiating and using $\nabla \Lambda_0(\beta) = \alpha'$ gives $\nabla I_F = \beta$. So $\nabla^2 I_F$ is the Jacobian of $(\nabla \Lambda_0)^{-1}$ at α' , which is Σ_α^{-1} . The formula for C_F follows from Definition 4.1. $\square$

A.2 The torus representation

We split

$$A_F = \begin{pmatrix} 0 & a_1^\top \\ b_1 & A' \end{pmatrix}, \quad \mu_F = (\mu_1, \mu'),$$

so $a_1 = (q_{1j})_{j \geq 2}$, $b_1 = (q_{i1})_{i \geq 2}$ and $A' = (q_{ij})_{i,j \geq 2}$. For $z' \in \mathbb{C}^{k-1}$ let $\mathcal{Z}' = \text{diag}(z')$. Where $\mathcal{Z}' - A'$ is invertible we put $X(z') = (\mathcal{Z}' - A')^{-1}$, $g(z') = a_1^\top X(z')b_1$ and

$$b(z') = (\mu_1 + \mu'^\top X(z')b_1)(1 + a_1^\top X(z')\mathbf{1}).$$

Lemma A.2 (torus representation). *Let $s' \in (0, \infty)^{k-1}$ satisfy $\varrho(A' - \text{diag } s') < 0$. On the set of $z' \in \mathbb{C}^{k-1}$ with $|z_i| \geq s_i$ for every $i \geq 2$, the matrix $\mathcal{Z}' - A'$ is invertible, and X , g and b are given by absolutely convergent power series in $1/z_2, \dots, 1/z_k$ with nonnegative coefficients. In particular $|g(z')| \leq g(s')$ and $|b(z')| \leq b(s')$ there. For every $t \in (0, \infty)^F$,*

$$H(t) = \frac{1}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} \exp\left(t_1 g(z') + \sum_{i \geq 2} t_i z_i\right) b(z') \prod_{i \geq 2} z_i d\varphi, \quad z_i = s_i e^{i\varphi_i}.$$

Proof. By the first fact, $\varrho(\text{diag}(s')^{-1}A') < 1$. So the series $X(z') = \sum_{j \geq 0} (\mathcal{Z}'^{-1}A')^j \mathcal{Z}'^{-1}$ converges absolutely for $|z_i| \geq s_i$, since its terms are bounded entrywise in modulus by those at $z' = s'$. As a_1 , b_1 and μ are nonnegative, this gives the claims about X , g and b .

Choose $s_1 > g(s')$ and put $S = \text{diag}(s_1, s')$. We show that $\varrho(S^{-1}A_F) < 1$. For $\varepsilon > 0$ let $v = (1, v')$ with $v' = X(s')(b_1 + \varepsilon\mathbf{1}) > 0$. Then $((S - A_F)v)_i = \varepsilon$ for $i \geq 2$, and $((S - A_F)v)_1 = s_1 - g(s') - \varepsilon a_1^\top X(s')\mathbf{1}$. For small ε both are positive, and the first fact gives the claim.

Now let $\mathcal{Z} = \text{diag}(z)$ with $|z_i| = s_i$ for every $i \in F$. Since $\varrho(A_F S^{-1}) < 1$, expanding $(\mathcal{Z} - A_F)^{-1} = \mathcal{Z}^{-1}(I - A_F \mathcal{Z}^{-1})^{-1}$ gives

$$\mu_F^\top (\mathcal{Z} - A_F)^{-1} \mathbf{1} = \sum_c \mu_{c_1} \prod_{j < M} q_{c_j c_{j+1}} \prod_{i \in F} z_i^{-m_i(c)},$$

absolutely and uniformly on the torus. Here $c = c_1 \cdots c_M$ runs over the words in F with $M \geq 1$ and $c_j \neq c_{j+1}$, and $m_i(c)$ is the number of letters i in c . For an integer m and $t_i > 0$, Cauchy's formula gives $\frac{1}{2\pi} \int_0^{2\pi} e^{t_i z_i} z_i^{1-m} d\varphi_i = t_i^{m-1}/(m-1)!$ if $m \geq 1$ and 0 if $m \leq 0$. So integrating term by term removes the words that miss a state of F , and grouping the others by $m(c)$ gives

$$H(t) = \frac{1}{(2\pi)^k} \int_{\mathbb{T}^k} e^{(t, z)} \mu_F^\top (\mathcal{Z} - A_F)^{-1} \mathbf{1} \prod_{i \in F} z_i d\varphi. \quad (12)$$

The Schur complement $z_1 - g(z')$ of $\mathcal{Z}' - A'$ in $\mathcal{Z} - A_F$ is not 0, since $|g(z')| \leq g(s') < s_1$. Block inversion gives

$$\mu_F^\top (\mathcal{Z} - A_F)^{-1} \mathbf{1} = \frac{b(z')}{z_1 - g(z')} + \mu'^\top X(z') \mathbf{1}.$$

The second term does not depend on z_1 . So by Cauchy's formula the integral over φ_1 in (12), divided by 2π , is $e^{t_1 g(z')} b(z')$ times the factors without z_1 . This proves the formula. $\square$

All coefficients are nonnegative, so the integrand has its largest modulus at $\varphi = 0$. This gives Lemma 3.2, and a Laplace expansion around $\varphi = 0$ gives Theorem 4.4.

A.3 Proof of Lemma 3.2

Let F be irreducible and $\theta \in V_F$. Then $A_F - \text{diag } \theta$ is an irreducible Metzler matrix with Perron root 0. So it has a right eigenvector $u > 0$ for 0, and $\theta_i = (A_F u)_i / u_i$. Each row of A_F has a positive entry, since G_F is strongly connected and $k \geq 2$, so $\theta > 0$. By the second fact, $\varrho(A' - \text{diag } \theta') < 0$. The rows $i \geq 2$ of $(A_F - \text{diag } \theta)u = 0$ give $u' = u_1 X(\theta') b_1$, and then row 1 gives $\theta_1 = g(\theta')$. Hence

$$\theta > 0, \quad \varrho(A' - \text{diag } \theta') < 0, \quad g(\theta') = \theta_1 \quad \text{for every } \theta \in V_F. \quad (13)$$

Now fix n with support F and h with $h q_{\max} \leq \frac{1}{2}$, and put $|n| = \sum_i n_i$, $t = hn$ and $\sigma = e^{h q_{\max}}$. Lemma 2.3 with $|n|$ in place of N gives the probability $P_{|n|}(n)$ in the lemma. Only terms with $m_i \leq n_i$ for every i are nonzero, and in them $h^{m_i-1} \binom{n_i-1}{m_i-1} \leq t_i^{m_i-1} / (m_i-1)!$ and $(1 - h q_i)^{n_i-m_i} \leq e^{-q_i t_i} \sigma^{m_i}$. Note that $h^{M-1} = h^{k-1} \prod_i h^{m_i-1}$ and $\sigma^M W(m) = \sigma W_\sigma(m)$, where W_σ is W for σA_F in place of A_F . So

$$P_{|n|}(n) \leq \sigma h^{k-1} e^{-\langle q, t \rangle} H_\sigma(t), \quad (14)$$

where H_σ is H for σA_F . We bound H_σ by Lemma A.2 for σA_F with the radii $\sigma \theta'$, which is allowed since $\varrho(\sigma A' - \text{diag } \sigma \theta') = \sigma \varrho(A' - \text{diag } \theta') < 0$. The functions of Lemma A.2 for σA_F satisfy $X_\sigma(\sigma z') = \sigma^{-1} X(z')$, so $g_\sigma(\sigma z') = \sigma g(z')$ and $b_\sigma(\sigma z') = b(z')$. By (13) the integrand has modulus at most $e^{\sigma \langle t, \theta \rangle} b(\theta') \sigma^{k-1} \prod_{i \geq 2} \theta_i$, so for every $\sigma > 0$

$$H_\sigma(t) \leq \sigma^{k-1} b(\theta') \prod_{i \geq 2} \theta_i e^{\sigma \langle t, \theta \rangle}. \quad (15)$$

With (14) this gives $P_{|n|}(n) \leq \sigma^k b(\theta') \prod_{i \geq 2} \theta_i h^{k-1} \exp(-h \langle n, q - \theta \rangle + (\sigma - 1) h \langle n, \theta \rangle)$. Finally $\sigma \leq e^{1/2}$ and $\sigma - 1 \leq 2 h q_{\max}$, so the lemma holds with $C_\theta = \max(e^{k/2} b(\theta') \prod_{i \geq 2} \theta_i, 2 q_{\max} \max_i \theta_i)$. The bound after the lemma is the case $\theta = q - \lambda_F \mathbf{1}$. $\square$

A.4 Proof of Theorem 4.4

For $k = 1$ the formula is part of Lemma 2.4. Let $k \geq 2$ and $\alpha \in \mathcal{K}$, and write $\theta = \theta(\alpha)$, $r = r_\alpha$, $l = l_\alpha$ and $\Theta' = \text{diag}(\theta')$. By (13) we may use Lemma A.2 with the radii θ' and $t = T\alpha$. With Lemma 2.4 this gives

$$f_T(\alpha) = \frac{T^{k-1} e^{-T \langle q, \alpha \rangle}}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} e^{T \Psi(\alpha, \varphi)} \chi(\alpha, \varphi) d\varphi, \quad (16)$$

where $z_i = \theta_i e^{i\varphi_i}$, $\Psi(\alpha, \varphi) = \alpha_1 g(z') + \sum_{i \geq 2} \alpha_i z_i$ and $\chi(\alpha, \varphi) = b(z') \prod_{i \geq 2} z_i$. We check the hypotheses of Lemma B.1 with $m = k - 1$.

Hypothesis (iii). By (13), $\Psi(\alpha, 0) = \langle \alpha, \theta \rangle$ is real. By Lemma A.2, $|g(z')| \leq g(\theta') = \theta_1$ on the torus, so

$$\operatorname{Re} \Psi(\alpha, \varphi) \leq \Psi(\alpha, 0) - \sum_{i \geq 2} \alpha_i \theta_i (1 - \cos \varphi_i).$$

All α_i and θ_i are positive, so the sum is positive unless $\varphi = 0$.

Hypotheses (i) and (ii). For $z \in \mathbb{R}^F$ near θ let $\lambda(z) = \varrho(A_F - \operatorname{diag} z)$. With $R = \operatorname{diag} r$ we have $A_F - \operatorname{diag}(\theta - \eta) = R(\widehat{Q}_\alpha + \operatorname{diag} \eta)R^{-1}$, and $\widehat{Q}_\alpha$ is an irreducible generator with stationary law α . So Lemma A.1 gives $\nabla \lambda(\theta) = -\alpha$ and $\nabla^2 \lambda(\theta) = \Gamma_\alpha$. For real z' near θ' we have $\varrho(A' - \operatorname{diag} z') < 0$, so $(1, X(z')b_1) \geq 0$ by the first fact. By the definition of g this vector lies in the kernel of $A_F - \operatorname{diag}(g(z'), z')$, so $\lambda(g(z'), z') = 0$ by the second fact. Write g_j, g_{ij} for the derivatives of g at θ' and put $w_j = e_j - (\alpha_j/\alpha_1)e_1 \in \mathbb{R}^F$ for $j \geq 2$. Differentiating $\lambda(g(z'), z') = 0$ once and twice at θ' gives

$$g_j = -\frac{\alpha_j}{\alpha_1}, \quad \alpha_1 g_{ij} = w_i^\top \Gamma_\alpha w_j.$$

So $\partial \Psi / \partial z_j = \alpha_1 g_j + \alpha_j = 0$ at θ' , and the Hessian $\mathcal{H}$ of Ψ in z' at θ' is $V^\top \Gamma_\alpha V$, where V has columns $w_2, \dots, w_k$. Since $\partial / \partial \varphi_j = i z_j \partial / \partial z_j$ and the first derivatives vanish, $\nabla_\varphi \Psi(\alpha, 0) = 0$ and $-\nabla_\varphi^2 \Psi(\alpha, 0) = \Theta^\top \mathcal{H} \Theta$. Let $B = I + \mathbf{1} \alpha'^\top / \alpha_1$. Then $Vy + (\alpha'^\top y / \alpha_1) \mathbf{1} = (0, By)$ for $y \in \mathbb{R}^{k-1}$, and $\Gamma_\alpha \mathbf{1} = 0$, so $\mathcal{H} = B^\top \Sigma_\alpha B$. By Lemma A.1, Σ_α is positive definite, and hence so is $\mathcal{H}$. The matrix determinant lemma gives $\det B = 1/\alpha_1$, so $\det \mathcal{H} = \det \Sigma_\alpha / \alpha_1^2$.

Regularity. By Proposition 4.2(a), $\theta(\alpha)$ is real-analytic. Fix $\alpha_0 \in \mathcal{K}$. By (13) and continuity there is $\delta \in (0, \min_i \theta_i(\alpha_0))$ with $\varrho(A' - \operatorname{diag}(\theta'(\alpha_0) - \delta \mathbf{1})) < 0$. By Lemma A.2, X, g and b are holomorphic where $|z_i| > \theta_i(\alpha_0) - \delta$ for every $i \geq 2$, and this holds on the torus of radii $\theta'(\alpha)$ for α near α_0 . So Ψ and χ are real-analytic near $\{\alpha_0\} \times \mathbb{T}^{k-1}$. Covering $\mathcal{K}$ by finitely many such sets shows that their φ -derivatives are jointly continuous on $\mathcal{K} \times \mathbb{T}^{k-1}$.

The constant. Lemma B.1 now gives, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$\int_{\mathbb{T}^{k-1}} e^{T\Psi} \chi d\varphi = e^{T\langle \alpha, \theta \rangle} \left(\frac{2\pi}{T} \right)^{\frac{k-1}{2}} \frac{\alpha_1}{\prod_{i \geq 2} \theta_i \sqrt{\det \Sigma_\alpha}} \left(\chi(\alpha, 0) + O\left(\frac{1}{T} \right) \right).$$

Here $\chi(\alpha, 0) = b(\theta') \prod_{i \geq 2} \theta_i$. As in the derivation of (13), $r' = r_1 X(\theta') b_1$. In the same way the columns $j \geq 2$ of $l'^\top (A_F - \operatorname{diag} \theta) = 0$, which is (2), give $l' = l_1 X(\theta')^\top a_1$. Hence $\mu \cdot r = r_1(\mu_1 + \mu'^\top X(\theta') b_1)$ and $l \cdot \mathbf{1} = l_1(1 + a_1^\top X(\theta') \mathbf{1})$. Since $l_1 r_1 = \alpha_1$,

$$b(\theta') = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{\alpha_1}.$$

As $r_\alpha, l_\alpha > 0$ and $\mu(F) > 0$ by (H), $\chi(\alpha, 0)$ is positive and continuous, hence bounded below on $\mathcal{K}$. So the last bracket is $\chi(\alpha, 0)(1 + O(1/T))$. Putting this into (16) and using $I_F(\alpha) = \langle q - \theta, \alpha \rangle$ gives $f_T(\alpha) = T^{(k-1)/2} C_F(\alpha) e^{-T I_F(\alpha)} (1 + O(1/T))$, uniformly on $\mathcal{K}$. $\square$

A.5 Proof of Lemma 4.5

Under the continuum weights a path has of order T runs, and each run after the first in state i changes the number of placements by a factor $1 - O(m_i/n_i)$, which suggests a relative error of order T^2/N . Let $t = hn$ and $\alpha = n/N \in \mathcal{K}$. By Lemma 2.4, $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q, t \rangle} H(t) = \sum_m v_m$ with

$$v_m = W(m) h^{k-1} \prod_{i \in F} e^{-q_i t_i} \frac{t_i^{m_i-1}}{(m_i-1)!}, \quad m \in \mathbb{Z}_{\geq 1}^F.$$

Since $h^{m_i-1} \binom{n_i-1}{m_i-1} = \frac{t_i^{m_i-1}}{(m_i-1)!} \prod_{1 \leq j < m_i} (1 - j/n_i)$, the m -th term of Lemma 2.3 is $\kappa_m v_m$ with

$$\kappa_m = \prod_{i \in F} \left[\prod_{1 \leq j < m_i} \left(1 - \frac{j}{n_i}\right) \right] (1 - hq_i)^{n_i - m_i} e^{hq_i n_i}.$$

Let $n_{\min} = \min_i n_i$. If all $m_i \leq n_i$, then $(1 - hq_i)^{n_i - m_i} \leq e^{-hq_i(n_i - m_i)}$ gives $\kappa_m \leq e^{hq_{\max} M}$. Also $\log(1 - x) \geq -x - x^2$ for $0 \leq x \leq \frac{1}{2}$ gives $(1 - hq_i)^{n_i - m_i} e^{hq_i n_i} \geq e^{-n_i h^2 q_i^2}$, and the product of these over i is at least $e^{-q_{\max}^2 T^2 / N}$. The factors $1 - j/n_i$ lie in $[0, 1]$, so their product is at least $1 - \sum_i m_i^2 / (2n_i) \geq 1 - M^2 / (2n_{\min})$. If some $m_i > n_i$, then $\kappa_m = 0$, while $M \geq n_{\min} + 1$ gives $M^2 / (2n_{\min}) \geq 2$. In all cases

$$\left(1 - \frac{M^2}{2n_{\min}}\right) e^{-q_{\max}^2 T^2 / N} \leq \kappa_m \leq e^{hq_{\max} M}. \quad (17)$$

Let $w_m = v_m / \sum_{m'} v_{m'}$, a probability since $H(t) > 0$ under (H). By Lemma 2.3 the ratio in the lemma is $\mathbb{E}_w \kappa_m$, so by (17) we need $\mathbb{E}_w e^{hq_{\max} M}$ and $\mathbb{E}_w M^2$. Since W for σA_F is $\sigma^{M-1} W(m)$, the function $\psi(c) = \log \mathbb{E}_w e^{c(M-1)} = \log H_{e^c}(t) - \log H(t)$ is convex with $\psi(0) = 0$. So $\psi(c) \leq c\psi(1)$ for $0 \leq c \leq 1$, and $\psi(1) \geq 1$ since $M \geq 2$. We show that $\psi(1) \leq C_{\mathcal{K}} T$. By (15) with $\sigma = e$ and $\theta = \theta(\alpha)$, $H_e(T\alpha) \leq e^{k-1} b(\theta') \prod_{i \geq 2} \theta_i e^{eT\langle \alpha, \theta \rangle}$. For the lower bound note that $H(T\alpha) e^{-T\langle \alpha, \theta \rangle} = T^{-(k-1)} e^{T I_F(\alpha)} f_T(\alpha)$. By Theorem 4.4 this is at least $\frac{1}{2} T^{-(k-1)/2} C_F(\alpha)$ for $T \geq T_1$, and for $1 \leq T \leq T_1$ it is continuous and positive on $[1, T_1] \times \mathcal{K}$. So in both cases $H(T\alpha) \geq c_{\mathcal{K}} T^{-(k-1)/2} e^{T\langle \alpha, \theta \rangle}$. Since $\theta(\alpha)$ and $b(\theta'(\alpha))$ are bounded on $\mathcal{K}$, this gives $\psi(1) \leq C_{\mathcal{K}} T$.

Now let $x = hq_{\max} \leq \frac{1}{2}$. Then $\mathbb{E}_w e^{xM} = e^{x + \psi(x)} \leq e^{x(1 + C_{\mathcal{K}} T)} = 1 + O(T^2/N)$, because $T^2/N \leq 1$. The Chernoff bound $\mathbb{P}_w(M - 1 \geq y) \leq e^{\psi(1) - y}$ gives

$$\mathbb{E}_w (M - 1)^2 = \int_0^\infty 2y \mathbb{P}_w(M - 1 \geq y) dy \leq \psi(1)^2 + \int_{\psi(1)}^\infty 2ye^{\psi(1) - y} dy = \psi(1)^2 + 2\psi(1) + 2,$$

so $\mathbb{E}_w M^2 = O(T^2)$. Finally $n_{\min} \geq \varepsilon_{\mathcal{K}} N$ with $\varepsilon_{\mathcal{K}} = \min_{\alpha \in \mathcal{K}} \min_i \alpha_i > 0$. So by (17), $\mathbb{E}_w \kappa_m$ lies between $e^{-q_{\max}^2 T^2 / N} (1 - \mathbb{E}_w M^2 / (2\varepsilon_{\mathcal{K}} N))$ and $\mathbb{E}_w e^{hq_{\max} M}$, and both are $1 + O(T^2/N)$, uniformly for $\alpha \in \mathcal{K}$ and $1 \leq T \leq N^{1/2}$. $\square$

A.6 Proof of Theorem 3.3(ii) for reducible faces

Irreducible admissible faces satisfy (H) by Lemma 2.2(b), and Theorem 4.6 covers them. So let F be admissible and reducible, with components $C_1, \dots, C_r$. Let a be the least positive rate q_{ij} , and take N_0 so large that $h \leq N^{-2/3}$ is as small as we need. Note that $T^2/N \leq 1$. As in the proof of Lemma 2.2, a path of positive probability with support F visits each component in one block of consecutive times, and it passes from one block to the next by one switch. We call the order of the blocks, the first state and these switches the pattern of the path. There are finitely many patterns.

Upper bound. Fix a pattern whose j -th block lies in the component D_j , starts at y_j and ends at x_j , and let $\text{supp } n = F$. Block j has length $N_j = \sum_{i \in D_j} n_i$. The paths with this pattern and composition n have total probability $\mu_{y_1} \prod_{j < r} hq_{x_j y_{j+1}}$ times the product over j of the weight of the block paths from y_j to x_j inside D_j with the right composition. This is at most the probability that $X_1, \dots, X_{N_j}$ have that composition for the chain started at y_j . For $C = D_j$ with $|C| \geq 2$, Lemma 3.2 with start y_j and $\theta = q - \lambda_C \mathbf{1}$ bounds it by $\kappa_1 h^{|C|-1} e^{-h\lambda_C N_j + \kappa_1 h^2 N_j}$, where κ_1 is the largest of these finitely many constants. For $C = \{i\}$ it is $(1 - hq_i)^{N_j-1} \leq e^{1/2 - h\lambda_C N_j}$. Since

$\sum_j (|D_j| - 1) + r - 1 = k_F - 1$, $\sum_j N_j = N$ and $\lambda_C \geq \lambda_F$ for every component, summing over the patterns gives $P_N(n) \leq \kappa_2 (T/N)^{k_F-1} e^{-T\lambda_F + \kappa_1}$.

Lower bound. By Lemma 2.2(a) we can number the components so that there are $y_1 \in C_1$ with $\mu_{y_1} > 0$ and $x_j \in C_j$, $y_{j+1} \in C_{j+1}$ with $q_{x_j y_{j+1}} > 0$ for $j < r$. Take any $x_r \in C_r$, choose j^* with $\lambda_{C_{j^*}} = \lambda_F$ and $|C_{j^*}| = k^*$, and put $m = \lceil N/T \rceil$, so that $1 \leq hm \leq 2$.

Claim A. There is $c_A > 0$ such that for every component C and all $y, x \in C$, the chain started at y stays in C for its first m visits, visits every state of C and has $X_m = x$ with probability at least c_A . Take a walk $y = v_0, \dots, v_J = x$ through every state of C with $q_{v_s v_{s+1}} > 0$ and $J \leq |C|^2$. The paths that make exactly these switches, at any J of the $m - 1$ transition times, have total probability at least $\binom{m-1}{J} (ha)^J (1 - hq_{\max})^m \geq (1 - Jh)^J a^J e^{-4q_{\max}/J!}$, since $1 \leq hm \leq 2$. Also $(1 - Jh)^J \geq 2^{-J}$ for small h .

Claim B. There is $c_B > 0$ such that for all $y, x \in C_{j^*}$ and $\ell \geq 2m$, the same event with C_{j^*} and ℓ visits has probability at least $c_B e^{-h\lambda_F \ell - h^2 \lambda_F^2 \ell}$. Let $w > 0$ satisfy $Q_{C_{j^*}} w = -\lambda_F w$ and let $B = I + hQ_{C_{j^*}} \geq 0$. Then $(B^s)_{zz'}$ is the probability of going from z to z' in s steps inside C_{j^*} , and $\sum_{z'} (B^s)_{zz'} \geq (B^s w)_z / \max w \geq (1 - h\lambda_F)^s \min w / \max w$. We use Claim A for the first m visits, from y back to y , then $s = \ell - 2m + 1$ steps inside C_{j^*} , then Claim A for the last m visits, ending at x . This gives at least $c_A^2 (\min w / \max w) (1 - h\lambda_F)^\ell$. Also $\lambda_F \leq q_{\max}$, because the Perron root of $Q_{C_{j^*}}$ is at least each diagonal entry. So $h\lambda_F \leq \frac{1}{2}$ and $\log(1 - h\lambda_F) \geq -h\lambda_F - h^2 \lambda_F^2$.

Give block $j \neq j^*$ the length m and block j^* the length $N - (r - 1)m$, which is at least $2m$ for $T \geq T_0 = r + 2$ and $N \geq N_0$. The paths that go from y_j to x_j in block j as in the claims and pass from block j to block $j + 1$ by the switch $x_j \rightarrow y_{j+1}$ all have support F . By the Markov property their total probability is at least $\mu_{y_1} (ha)^{r-1} c_A^{r-1} c_B e^{-T\lambda_F - q_{\max}^2}$. Their compositions are fixed by the restrictions to the blocks, so there are at most $(N + 1)^{k^*-1} \prod_{j \neq j^*} (m + 1)^{|C_j|-1}$ of them, and $m + 1 \leq 3N/T$. Since $\sum_{j \neq j^*} (|C_j| - 1) = k_F - k^* - (r - 1)$, dividing the probability by this number gives $M_F \geq c N^{-(k_F-1)} T^{k_F-k^*} e^{-T\lambda_F}$. $\square$

B A Laplace lemma with a parameter

This appendix proves the Laplace lemma that Appendix A uses for Theorem 4.4. On $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$ we use $\|\varphi\| = \max_i \text{dist}(\varphi_i, 2\pi\mathbb{Z})$. For $x \in \mathbb{R}^m$ we write $\|x\| = \max_i |x_i|$ and $|x|$ for the Euclidean norm. Repeated indices are summed from 1 to m .

Lemma B.1 (Laplace integrals with a parameter). *Let $\mathcal{K}$ be a compact metric space and $m \geq 1$. Let $\Psi, a : \mathcal{K} \times \mathbb{T}^m \rightarrow \mathbb{C}$ be such that the φ -derivatives of Ψ up to order 4 and of a up to order 2 exist and are jointly continuous on $\mathcal{K} \times \mathbb{T}^m$. Assume that for every $\alpha \in \mathcal{K}$*

- (i) $\Psi(\alpha, 0) \in \mathbb{R}$ and $\nabla_\varphi \Psi(\alpha, 0) = 0$,
- (ii) $\mathcal{J}(\alpha) = -\nabla_\varphi^2 \Psi(\alpha, 0)$ is real, symmetric and positive definite, and
- (iii) $\text{Re} \Psi(\alpha, \varphi) < \Psi(\alpha, 0)$ for every $\varphi \neq 0$ in $\mathbb{T}^m$.

Then, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$\int_{\mathbb{T}^m} e^{T\Psi(\alpha, \varphi)} a(\alpha, \varphi) d\varphi = e^{T\Psi(\alpha, 0)} \left(\frac{2\pi}{T} \right)^{m/2} \det \mathcal{J}(\alpha)^{-1/2} \left(a(\alpha, 0) + O\left(\frac{1}{T} \right) \right).$$

Proof. By compactness $B_\Psi = \max_{|\beta| \leq 4} \sup |\partial_\varphi^\beta \Psi|$ and $B_a = \max_{|\beta| \leq 2} \sup |\partial_\varphi^\beta a|$ are finite, the least eigenvalue of $\mathcal{J}(\alpha)$ has a positive minimum $h_{\min}$ over $\mathcal{K}$, and $\det \mathcal{J}(\alpha) \leq (mB_\Psi)^m$. Below, C

depends only on m , $h_{\min}$, B_Ψ , B_a and the numbers ε and η chosen below, and it may change from line to line. So every bound below is uniform on $\mathcal{K}$.

Localization. Choose $\varepsilon \in (0, 1)$ with $m^3 B_\Psi \varepsilon \leq \frac{3}{2} h_{\min}$. The set of (α, φ) with $\|\varphi\| \geq \varepsilon$ is compact, and by (iii) the continuous function $\operatorname{Re} \Psi(\alpha, \varphi) - \Psi(\alpha, 0)$ is negative on it. So it is at most $-\eta$ there for some $\eta > 0$, and the integral over this set has modulus at most $(2\pi)^m B_a e^{T(\Psi(\alpha, 0) - \eta)} \leq CT^{-1-m/2} e^{T\Psi(\alpha, 0)}$.

Scaling. Reduction modulo 2π identifies the cube $\|x\| \leq \varepsilon$ in $\mathbb{R}^m$ with the set $\|\varphi\| \leq \varepsilon$, and it preserves measure. Let $R(\alpha, \varphi) = \Psi(\alpha, \varphi) - \Psi(\alpha, 0) + \frac{1}{2} \varphi^\top \mathcal{J}(\alpha) \varphi$. By (i) and (ii), R and its derivatives of order 1 and 2 vanish at $\varphi = 0$. Taylor's theorem with integral remainder, which holds for complex values, gives on the cube $R = \frac{1}{6} \Psi_{ijk} \varphi^i \varphi^j \varphi^k + \omega_0$ with $|\omega_0| \leq C \|\varphi\|^4$, where Ψ_{ijk} are the third derivatives at $(\alpha, 0)$. It also gives $|R| \leq \frac{m^3}{6} B_\Psi \|\varphi\|^3 \leq \frac{1}{4} h_{\min} |\varphi|^2$, since $\sum_{|\beta|=3} 1/\beta! = m^3/6$ and $\|\varphi\|^3 \leq \varepsilon |\varphi|^2$. We substitute $\varphi = x/\sqrt{T}$ and put $w(x) = TR(\alpha, x/\sqrt{T})$. The integral over the cube is

$$T^{-m/2} e^{T\Psi(\alpha, 0)} \int_{\|x\| \leq \varepsilon \sqrt{T}} e^{-\frac{1}{2} x^\top \mathcal{J} x} e^{w(x)} a(\alpha, x/\sqrt{T}) dx,$$

and $|w(x)| \leq \frac{1}{4} h_{\min} |x|^2$.

Taylor expansion. Let $P(x) = \frac{1}{6} \Psi_{ijk} x^i x^j x^k$, $a_0 = a(\alpha, 0)$ and $a_i = \partial_{\varphi_i} a(\alpha, 0)$. Then $w = T^{-1/2} P + \omega_1$ with $|\omega_1| \leq CT^{-1} |x|^4$, and $|w| \leq CT^{-1/2} |x|^3$. Since $|e^w - 1 - w| \leq \frac{1}{2} |w|^2 e^{|w|}$, this gives $|e^w - 1 - T^{-1/2} P| \leq CT^{-1} (|x|^4 + |x|^6) e^{h_{\min} |x|^2/4}$. Taylor's theorem for a to orders 1 and 2 gives $|a(\alpha, x/\sqrt{T}) - a_0| \leq CT^{-1/2} |x|$ and $|a(\alpha, x/\sqrt{T}) - a_0 - T^{-1/2} a_i x^i| \leq CT^{-1} |x|^2$. Multiplying out,

$$e^w a(\alpha, x/\sqrt{T}) = a_0 + T^{-1/2} (a_0 P(x) + a_i x^i) + \omega(x), \quad |\omega(x)| \leq CT^{-1} (1 + |x|)^6 e^{h_{\min} |x|^2/4}.$$

Integration. Since $x^\top \mathcal{J} x \geq h_{\min} |x|^2$, the term ω contributes at most CT^{-1} to the integral, because $(1 + |x|)^6 e^{-h_{\min} |x|^2/4}$ is integrable. Outside the cube $|x| > \varepsilon \sqrt{T}$, so extending the integrals of the other 2 terms to $\mathbb{R}^m$ changes them by at most $C e^{-h_{\min} \varepsilon^2 T/4} \leq CT^{-1}$. On $\mathbb{R}^m$ we have $\int e^{-\frac{1}{2} x^\top \mathcal{J} x} dx = (2\pi)^{m/2} \det \mathcal{J}^{-1/2}$, and the odd polynomial $a_0 P(x) + a_i x^i$ integrates to 0 against this weight. This is why the error is $O(1/T)$ and not $O(T^{-1/2})$. So the integral over the cube is $e^{T\Psi(\alpha, 0)} T^{-m/2} [(2\pi)^{m/2} \det \mathcal{J}^{-1/2} a_0 + O(T^{-1})]$. Since $\det \mathcal{J} \leq (m B_\Psi)^m$, this error and the one from the localization are $O(1/T)$ relative to $e^{T\Psi(\alpha, 0)} (2\pi/T)^{m/2} \det \mathcal{J}^{-1/2}$. $\square$

C Proofs for the planar walk

We prove the bound on $o_N^{(4)}$ in Proposition 6.2 and Theorem 6.3. We identify $\mathbb{R}^2$ with $\mathbb{C}$, so $e_z = i^z$ and $Y_N = \sum_{t=0}^m \ell_t i^{Z_t}$ in the notation of Lemma 6.1. Let $\operatorname{bin}(m) = \binom{N-1}{m} (\beta h)^m p_0^{N-1-m}$ be the law of m , with mean $\bar{m} = (N-1)\beta h$. We write $\mathbb{P}_z$ and $\mathbb{E}_z$ for the skeleton walk of Lemma 6.1(c) started at z , $\mathbb{P}$ for its uniform start, and ξ for one of its increments. Put $\varkappa = s/\beta$, so $\varkappa < 1$ when $r > 0$. For a skeleton, let a, b, c, d be the numbers of runs going east, west, north and south. So $a + b + c + d = m + 1$, and the path uses all 4 directions exactly when $a, b, c, d \geq 1$. Throughout, $N = 2\nu$, and $[\cdot]$ is the indicator of the statement inside.

For $k \geq 1$ let $\chi_k(X) = \binom{X-1}{k-1} [X \geq 1]$, and let $\chi_0(X) = [X = 0]$. Then $\chi_k(X)$ counts the compositions of X into k positive parts.

Lemma C.1 (return counts). *Let Z be a skeleton with m switches and counts a, b, c, d . The number of compositions ℓ of N into $m + 1$ parts with $\sum_t \ell_t i^{Z_t} = 0$ is*

$$\Omega_N(a, b, c, d) = \sum_{X=0}^{\nu} \chi_a(X) \chi_b(X) \chi_c(\nu - X) \chi_d(\nu - X).$$

If $a, b, c, d \geq 1$, then $m \geq 3$ and $\Omega_N(a, b, c, d) \leq \frac{1}{2} \binom{N}{m-2}$.

Proof. The sum vanishes exactly when the runs going east and west have the same total length X , and those going north and south have total length $\nu - X$. The 4 compositions of these totals into runs are independent, which gives the formula. Now let $a, b, c, d \geq 1$. If $m = 3$, there is 1 run in each direction and $\Omega_N = \nu - 1 < \frac{1}{2} \binom{N}{1}$. Let $m \geq 4$. Pick t_z with $Z_{t_z} = z$ for each z , and fix the other $m - 3$ run lengths. If they add up to $N - j$, there are $\binom{N-j-1}{m-4}$ ways to choose them. The 4 free lengths add up to j , and $Y_N = 0$ fixes $\ell_{t_0} - \ell_{t_2}$ and $\ell_{t_1} - \ell_{t_3}$. So ℓ_{t_2} determines the other 3, and $\ell_{t_0} + \ell_{t_2}$ lies in $[2, j - 2]$ with a fixed parity. So there are at most $j/2$ solutions, and

$$\Omega_N(a, b, c, d) \leq \sum_{j=0}^{N-1} \frac{j}{2} \binom{N-1-j}{m-4} = \frac{1}{2} \binom{N}{m-2},$$

where the identity counts the subsets of $\{0, \dots, N - 1\}$ of size $m - 2$ by their second smallest element j . $\square$

By Lemma 6.1(b) and (d), $\varpi_N = \Omega_N(a, b, c, d) / \binom{N-1}{m}$ is the probability that $Y_N = 0$ given m and Z , so

$$o_N^{(4)} = \sum_{m \geq 3} \text{bin}(m) \mathbb{E}[\varpi_N \mathbf{1}\{a, b, c, d \geq 1\} \mid m]. \quad (18)$$

Lemma C.2 (skeleton returns). *Let $W_m = \sum_{t=0}^m i^{Z_t} = (a - b) + i(c - d)$. (a) $\mathbb{E}_z |W_m|^2 \leq m + 1$ for every start z . (b) If $s > 0$ and m is odd, then $\mathbb{P}(W_m = 0) \geq \kappa / (36(m + 1))$. (c) If $s = 0$ and $m \equiv 3 \pmod{4}$, then $\mathbb{P}(W_m = 0) \geq 1/(m + 1)$.*

Proof. (a) Since $\mathbb{E} i^\xi = \frac{\tau}{\beta} (i + i^{-1}) + \frac{s}{\beta} i^2 = -\kappa$, we get $\mathbb{E}_z |W_m|^2 = \sum_{t, t'} \mathbb{E}_z i^{Z_t - Z_{t'}} = m + 1 + 2 \sum_{k=1}^m (m + 1 - k) (-\kappa)^k$. The last sum alternates in sign, starts negative and has nonincreasing absolute terms, so it is at most 0.

(b) Write $m = 2m_1 + 1$ and split $W_m = W' + W''$, where W' sums over $t \leq m_1$ and W'' over $t > m_1$. The law of ξ is symmetric and the uniform law is stationary, so $(Z_{m_1}, \dots, Z_0)$ is again the walk from a uniform start, and $\mathbb{P}(W' = w, Z_{m_1} = z) = \frac{1}{4} \mathbb{P}_z(W_{m_1} = w)$. With probability κ , independently of $Z_0, \dots, Z_{m_1}$, the next switch is a reversal. Then given $Z_{m_1} = z$, the second half is the walk from $z + 2$, and $\mathbb{P}_{z+2}(W_{m_1} = -w) = \mathbb{P}_z(W_{m_1} = w)$. So $\mathbb{P}(W_m = 0) \geq \frac{\kappa}{4} \sum_z \sum_w \mathbb{P}_z(W_{m_1} = w)^2$. Let $R^2 = m + 1$. By (a) and Chebyshev's inequality, $\mathbb{P}_z(|W_{m_1}| \leq R) \geq \frac{1}{2}$. At most $9R^2$ Gaussian integers have modulus at most R , so by Cauchy-Schwarz the inner sum is at least $1/(36R^2)$. Summing over z gives (b).

(c) Now every switch is a turn, so $i^{Z_t} = i^{Z_{t-1}} \cdot i_{\epsilon_t}$ with independent fair signs ϵ_t . Hence $i^{Z_t} = i^{Z_0} i^t \sigma_t$, where $\sigma_0 = 1$ and $\sigma_t = \epsilon_1 \cdots \epsilon_t$ are independent fair signs for $t \geq 1$. Write $m = 4k - 1$. Then $i^{-Z_0} W_m$ has real part $\sum_{t \text{ even}} (-1)^{t/2} \sigma_t$, which is 1 plus a sum of $2k - 1$ fair signs, and an independent imaginary part that is a sum of $2k$ fair signs. So

$$\mathbb{P}(W_m = 0) = \binom{2k-1}{k} \binom{2k}{k} 2^{1-4k} = \prod_{j=1}^k \frac{(2j-1)^2}{(2j)^2} \geq \frac{1}{4} \prod_{j=2}^k \frac{2j-2}{2j} = \frac{1}{4k} = \frac{1}{m+1}.$$

$\square$

Lemma C.3 (balanced skeletons). *If $a = b \geq 1$, $c = d \geq 1$ and $m = 2(a + c) - 1 \leq N - 1$, then $\Omega_N(a, b, c, d) / \binom{N-1}{m} \geq \sqrt{m+1} / (2N^2)$.*

Proof. Put $k = a + c = (m + 1)/2$, so $2 \leq k \leq \nu$. Only $1 \leq X \leq \nu - 1$ contribute to Ω_N , so by Cauchy-Schwarz and Vandermonde's identity

$$\Omega_N = \sum_{X=1}^{\nu-1} \binom{X-1}{a-1}^2 \binom{\nu-X-1}{c-1}^2 \geq \frac{1}{\nu} \binom{\nu-1}{k-1}^2 = \frac{k}{\nu^2} \binom{\nu-1}{k-1} \binom{\nu}{k}.$$

Also $\binom{\nu-1}{k-1} \binom{\nu}{k} / \binom{2\nu-1}{2k-1} = \mathbb{P}(H = k - 1)$, where H is the number of red balls among $2k - 1$ balls drawn without replacement from $\nu - 1$ red and ν blue ones. If $k = \nu$, then $H = k - 1$. Otherwise the ratio $\mathbb{P}(H = j + 1) / \mathbb{P}(H = j) = (\nu - 1 - j)(2k - 1 - j) / ((j + 1)(\nu - 2k + 2 + j))$ decreases in j , is $\frac{\nu-k}{\nu-k+1} < 1$ at $j = k - 1$ and exceeds 1 at $j = k - 2$. So $k - 1$ is a mode of H . Since $\text{Var } H \leq (2k - 1)/4$, Chebyshev's inequality puts mass at least $\frac{3}{4}$ on the at most $3\sqrt{2k}$ integers within $\sqrt{2k - 1}$ of $\mathbb{E}H$, so $\mathbb{P}(H = k - 1) \geq 1/(4\sqrt{2k})$. As $\binom{N-1}{m} = \binom{2\nu-1}{2k-1}$, we get $\Omega_N / \binom{N-1}{m} \geq k/(4\nu^2\sqrt{2k}) = \sqrt{m+1}/(2N^2)$. $\square$

Proof of the bound on $o_N^{(4)}$ in Proposition 6.2. Note that $(\beta h)^m p_0^{N-1-m} = p_0^{N-1} w^m$. By (18) and Lemma C.1,

$$o_N^{(4)} \leq \frac{p_0^{N-1}}{2} \sum_{m=3}^{N-1} \binom{N}{m-2} w^m \leq \frac{p_0^{N-1} w^2}{2} ((1+w)^N - 1) = \frac{w^2}{2p_0} (1 - p_0^N),$$

since $1 + w = 1/p_0$. Dividing by c_N gives the bound on E_N . $\square$

Proof of Theorem 6.3. We first prove 3 bounds. Since $I_1 \leq I_0 \leq e^z$ for $z \geq 0$, we have $\psi(z) \leq 2ze^z$, and (8) gives $2S_N(u) \leq 8ue^{Nu}$. With Proposition 6.2 and $(1+w)^N \leq e^{Nw}$,

$$\frac{o_N}{c_N} \leq 8ue^{Nu} + 2w^2 e^{Nw}. \quad (19)$$

Since $o_N \geq o_N^{(0)}$, (8) with $z = Nu$ gives

$$\frac{o_N}{c_N} \geq 2S_N(u) \geq \frac{4}{N} \left(1 - \frac{z(z+2)}{2N}\right) \psi(z). \quad (20)$$

Third, we show that there are $c' > 0$ and T_0 , depending only on r and s , with

$$o_N^{(4)} \geq \frac{c'}{N^2 \sqrt{T}} \quad \text{whenever } T = Nh \geq T_0 \text{ and } \beta h \leq \frac{1}{2}. \quad (21)$$

Let $\mathcal{M}$ be the set of odd m if $s > 0$, and the set of $m \equiv 3 \pmod{4}$ if $s = 0$. Let $m \in \mathcal{M}$. The event $\{a = b \geq 1, c = d \geq 1\}$ is $\{W_m = 0\}$ minus $\{a = b = 0\}$ and $\{c = d = 0\}$. Each of these needs every switch to be a reversal, so it has probability at most $\frac{1}{2} \varkappa^m$, and it is empty if $s = 0$. So Lemma C.2 gives m_2 and $c_2 > 0$ with $\mathbb{P}(a = b \geq 1, c = d \geq 1 \mid m) \geq c_2/(m+1)$ for $m \in \mathcal{M}$ with $m \geq m_2$ (take $c_2 = \varkappa/72$ if $s > 0$ and $c_2 = 1$ if $s = 0$). On this event Lemma C.3 gives $\varpi_N \geq \sqrt{m+1}/(2N^2)$. Let $J = [\bar{m}/2, 2\bar{m}]$. By (18), if $\bar{m} \geq 2m_2$ then $o_N^{(4)} \geq \sum_{m \in \mathcal{M} \cap J} \text{bin}(m) c_2 / (2N^2 \sqrt{m+1}) \geq c_2 \mathbb{P}(m \in \mathcal{M} \cap J) / (2N^2 \sqrt{2\bar{m}+1})$. By the Chernoff

bound, $\mathbb{P}(m \notin J) \leq 2e^{-\bar{m}/8}$. Put $p = \beta h \leq \frac{1}{2}$. Then $\mathbb{P}(m \text{ odd}) = \frac{1}{2}(1 - (1 - 2p)^{N-1}) \geq \frac{1}{2}(1 - e^{-2\bar{m}})$. For $s = 0$ the roots of unity filter gives

$$\mathbb{P}(m \equiv 3 \pmod{4}) = \frac{1}{4} \sum_{k=0}^3 i^{-3k} (1 - p + p i^k)^{N-1} \geq \frac{1}{4} - \frac{3}{4} e^{-\bar{m}/2},$$

since $|1 - p + p i^{\pm 1}|^2 = 1 - 2p(1 - p) \leq e^{-p}$ and $|1 - 2p| \leq e^{-2p}$. So $\mathbb{P}(m \in \mathcal{M} \cap J) \geq \frac{1}{8}$ when $\bar{m}$ is large. As $\bar{m} \geq \beta T/2$, and $2\bar{m} + 1 \leq 3\beta T$ when $\beta T \geq 1$, this proves (21) with $c' = c_2/(16\sqrt{3}\beta)$.

(a) Let $h = \tau L/N$ and $T = \tau L$. Then $u, w = O(L/N)$, $Nu = s\tau L + O(L^2/N)$ and $Nw = \beta\tau L + O(L^2/N)$. So (19) gives $o_N/c_N = O(LN^{s\tau-1} + L^2N^{\beta\tau-2})$. If $\tau < \tau^*$, then $s\tau < 1$ and $\beta\tau < 2$, and this is at most $N^{-\gamma}$ for large N whenever $0 < \gamma < \min(1 - s\tau, 2 - \beta\tau)$. Now let $\tau > \tau^*$, so $s\tau > 1$ or $\beta\tau > 2$. If $s\tau > 1$, then $z = Nu \rightarrow \infty$ and $z^2/N \rightarrow 0$. As $I_0(z) + I_1(z) \sim 2e^z/\sqrt{2\pi z}$ [31, Eq. 10.40.1], we have $\psi(z) \sim \sqrt{2z/\pi} e^z$, and (20) gives $o_N/c_N \geq N^{s\tau-1}$ for large N . If $\beta\tau > 2$, then (21) and $c_N^{-1} = 4p_0^{-(N-1)} \geq 4e^{\beta T - \beta h}$ give $o_N/c_N \geq 4c'e^{-\beta h} N^{\beta\tau-2} (\tau L)^{-1/2}$, which is at least $N^{(\beta\tau-2)/2}$ for large N .

(b) Let $\tau < \tau^* < \tau'$. By (a), for large N the ratio o_N/c_N is below 1 at $h = \tau L/N$ and above 1 at $h = \tau' L/N < 1/\beta$. It increases in h by Proposition 6.2, so $\tau L < T_N^* < \tau' L$.

(c) Put $\tau_N = T_N^*/L$, so $\tau_N \rightarrow \tau^*$ by (b), and $w^* = \beta h_N^*/(1 - \beta h_N^*)$. As in (a), $u_N^*, w^* = O(L/N)$, $Nu_N^* = s\tau_N L + o(1)$ and $Nw^* = \beta\tau_N L + o(1)$. At the crossing Proposition 6.2 gives $2S_N(u_N^*) + E_N = 1$. If $s > 2r$, then $\tau^* = 1/s$ and $\beta\tau^* < 2$. So $E_N \leq 2w^{*2}e^{Nu^*} = N^{\beta/s-2+o(1)} \rightarrow 0$, and $2S_N(u_N^*) \rightarrow 1$. The 2 axes carry the same mass by symmetry. If $s < 2r$, then $\tau^* = 2/\beta$ and $s\tau^* < 1$. So $2S_N(u_N^*) \leq 8u_N^*e^{Nu_N^*} = N^{2s/\beta-1+o(1)}$, which is at most $N^{-\gamma}$ for large N whenever $0 < \gamma < 1 - 2s/\beta$. $\square$

D The large-concentration limit of the direct jump

We prove Proposition 7.4(b), with B_1, B_2, B_3 independent $\text{Beta}(a, a)$ as in part (a) and $a = \omega/3 \geq 1$. As D is a thin tube around the cyclic kernels, we expect $\mathbb{P}_\omega(D) \approx a^{3/2} \int t^4 e^{-cat^2} dt \approx 1/a$.

Put $\ell_j = \log(B_j/(1 - B_j))$, $g(y) = 2 \log \cosh(y/2)$ and $g_j = g(\ell_j)$. Then g is even and convex, $g \geq 0 = g(0) = g'(0)$, $B_j(1 - B_j) = \frac{1}{4}e^{-g_j}$, $\log(2B_j) = \frac{1}{2}(\ell_j - g_j)$ and $\log(2(1 - B_j)) = -\frac{1}{2}(\ell_j + g_j)$. As $dB_j/d\ell_j = B_j(1 - B_j)$, ℓ_j has density $e^{-ag(\ell_j)}/Z$ with $Z = 2\sqrt{\pi}\Gamma(a)/\Gamma(a + \frac{1}{2})$. So D is

$$M_1 \leq g_1 + g_2, \quad M_2 \leq g_1 + g_3, \quad M_3 \leq g_2 + g_3, \quad (22)$$

with $M_1 = \ell_1 + \ell_2$, $M_2 = \ell_3 - \ell_1$, $M_3 = -\ell_2 - \ell_3 = -M_1 - M_2$. With $\psi = g_1 + g_2 + g_3$,

$$\mathbb{P}_\omega(D) = Z^{-3} \int_D e^{-a\psi(\ell)} d\ell. \quad (23)$$

For real y ,

$$\frac{y^2}{4} - \frac{y^4}{96} \leq g(y) \leq \frac{y^2}{4}, \quad (g1)$$

$$g(y) \geq \frac{1}{10} \min(y^2, |y|). \quad (g2)$$

With $u = y/2$, the upper bound is $\cosh u \leq e^{u^2/2}$ (termwise), and the lower bound holds as $\chi = \log \cosh u - \frac{u^2}{2} + \frac{u^4}{12}$ has $\chi(0) = \chi'(0) = 0$ and $\chi'' = u^2 - \tanh^2 u \geq 0$. For (g2),

$g'' = \frac{1}{2}\text{sech}^2(y/2) > 0.2$ on $|y| \leq 2$, and for $|y| \geq 2$ convexity gives $g(y) \geq \frac{1}{2}|y|g(2) > 0.43|y|$. Let $v = (1, -1, 1)$ and $t = \langle \ell, v \rangle / 3$. The map $\ell \mapsto (t, M_1, M_2)$ has determinant 1 and inverse $\ell = tv + w$ with $w = \frac{1}{3}(M_1 - M_2, 2M_1 + M_2, M_1 + 2M_2) \perp v$. So $|\ell|^2 = 3t^2 + |w|^2$ with $|w|^2 = \frac{1}{3}(M_1^2 + M_2^2 + M_3^2)$, and with $|M|$, $|M|_\infty$ the 2-norm and max norm of (M_1, M_2) ,

$$\frac{1}{3}|M|^2 \leq |w|^2 \leq |M|^2 \leq 2|M|_\infty^2. \quad (24)$$

Proof of Proposition 7.4(b). Step 1: an a priori bound. On D , $M_1 \leq \psi$ and $M_1 = -M_2 - M_3 \geq -2\psi$, and the same for M_2 . So by (g1)

$$|M|_\infty \leq 2\psi(\ell) \leq \frac{1}{2}|\ell|^2 = \frac{1}{2}(3t^2 + |w|^2). \quad (25)$$

Step 2: localization. Put $\varepsilon = 1/12$ and let U be the set where $|t| \leq \varepsilon$ and $|M|_\infty \leq \varepsilon$. Off U , $|\ell| \geq \max(\sqrt{3}|t|, |M|/\sqrt{3}) \geq \varepsilon/\sqrt{3}$, and some $|\ell_j| \geq |\ell|/\sqrt{3}$. So (g2) gives $\psi \geq \gamma := 0.1 \min(|\ell|^2/3, |\ell|/\sqrt{3}) \geq \gamma_0 = \varepsilon^2/90$. Hence $e^{-a\psi} \leq e^{-(a-1)\gamma_0} e^{-\gamma}$ off U , and $\int_{D \setminus U} e^{-a\psi} = O(e^{-\gamma_0 a})$.

Step 3: the section of D at t . On $D \cap U$, (25) and (24) give $|M|_\infty \leq \frac{3}{2}t^2 + \varepsilon|M|_\infty$, so $|M|_\infty \leq 2t^2$. On $E = \{|t| \leq \varepsilon, |M|_\infty \leq 2t^2\}$ we have $|w|^2 \leq 8t^4$, so $\ell_j = \pm t + w_j$ gives $|\ell_j| \leq 2|t|$ and $|\ell_j^2 - t^2| \leq 7|t|^3$. Then (g1) gives $|g_j - \frac{1}{4}t^2| \leq 2|t|^3$, and with $|\ell|^2 = 3t^2 + |w|^2$ it bounds ψ . Hence

$$|g_i + g_j - \frac{1}{2}t^2| \leq 4|t|^3 \quad (i \neq j), \quad |\psi - \frac{3}{4}t^2| \leq 2t^4. \quad (26)$$

Let $\mathcal{T}(h) = \{M : M_1, M_2 \leq h, M_1 + M_2 \geq -h\}$, a triangle of area $\frac{9}{2}h^2$ in $\{|M|_\infty \leq 2h\}$. Let D_t be the section of $D \cap U$ at t and $h_\pm = \frac{1}{2}t^2(1 \pm 8|t|) \geq 0$. If $M \in D_t$, then $(t, M) \in E$, and (22) and (26) give $M_1, M_2, M_3 \leq h_+$. If $M \in \mathcal{T}(h_-)$, then $|M|_\infty \leq 2h_- \leq \min(t^2, \varepsilon)$, so $(t, M) \in E \cap U$, and (26) gives (22). So $\mathcal{T}(h_-) \subseteq D_t \subseteq \mathcal{T}(h_+)$, and

$$\frac{9}{8}t^4(1 - 8|t|)^2 \leq |D_t| \leq \frac{9}{8}t^4(1 + 8|t|)^2. \quad (27)$$

Step 4: integration. By Fubini (Jacobian 1), (26) and (27), $J = \int_{D \cap U} e^{-a\psi}$ lies between the integrals over $|t| \leq \varepsilon$ of $\frac{9}{8}t^4(1 \mp 8|t|)^2 e^{-a(\frac{3}{4}t^2 \pm 2t^4)}$. With $t = s/\sqrt{a}$ these are $a^{-5/2}$ times integrals whose integrands tend to $\frac{9}{8}s^4 e^{-3s^2/4}$ and are at most $\frac{9}{2}s^4 e^{-s^2/2}$, since $8|t| < 1$ and $2s^4/a \leq 2\varepsilon^2 s^2$. By dominated convergence and $\int s^4 e^{-3s^2/4} ds = \frac{3}{4}\sqrt{\pi}(\frac{4}{3})^{5/2}$, $J = a^{-5/2}(\sqrt{3\pi} + o(1))$, and the same holds over D . As Γ is log-convex, $\Gamma(a + \frac{1}{2})^2 \leq a\Gamma(a)^2$ and $\Gamma(a + 1)^2 \leq (a + \frac{1}{2})\Gamma(a + \frac{1}{2})^2$, so $Z = 2\sqrt{\pi/a}(1 + O(1/a))$. Then (23) gives $\mathbb{P}_\omega(D) = \frac{\sqrt{3}}{8\pi a}(1 + o(1)) = \frac{3\sqrt{3}}{8\pi\omega}(1 + o(1))$. $\square$

E Proof of the Fisher information theorem

We keep the notation of Section 8. Let $\sigma_i \in \{-1, 1\}$ be the direction of step i . Then $X = \sigma_1 + \dots + \sigma_N$, and J counts the $i < N$ with $\sigma_{i+1} \neq \sigma_i$. Put $Z = \mathbb{E}[J | K]$, so $e_N = \text{Var } Z / \text{Var } J$ by Proposition 8.1.

Lemma E.1 (exact moments). *For $N \geq 1$ and $0 < q < 1$,*

$$\begin{aligned} \mathbb{E}X^2 &= \frac{Np}{q} - \frac{\rho(1 - \rho^N)}{2q^2}, \\ \text{Cov}(J, X^2) &= -\frac{p}{q^2} [Nq(1 + \rho^N) - p(1 - \rho^N)], \\ \mathbb{E}X^4 &= \frac{3N^2p^2}{q^2} - \frac{Np(1 + 10\rho + \rho^2)}{2q^3} + \frac{\rho(1 + 2\rho)(2 + \rho)(1 - \rho^N)}{2q^4} - \frac{3Np\rho^{N+1}}{q^3}, \end{aligned}$$

and $\text{Var } X^2 = 2N^2p^2V/q^2$, with V as in Theorem 8.3.

Proof. Let $m_N(\phi, \chi) = \mathbb{E}e^{\phi X + \chi J}$. With $A_{ab} = \mathbb{P}(\sigma_2 = b \mid \sigma_1 = a)e^{\phi b + \chi \mathbf{1}[a \neq b]}$ and $v_a = \frac{1}{2}e^{\phi a}$ we have $m_N = v^\top A^{N-1} \mathbf{1}$. So $\sum_{N \geq 1} m_N z^N = U/D$ as formal power series, where $U(z) = zv^\top \text{adj}(I - zA) \mathbf{1}$ and $D(z) = \det(I - zA)$ have degree at most 2 in z . For a partial derivative ∂ of order m in (ϕ, χ) , induction and the quotient rule give $\partial(U/D) = U_\partial/D^{m+1}$ with $\deg U_\partial \leq 2m + 2$, as each derivative multiplies the numerator by D or a derivative of D . At $\phi = \chi = 0$ the matrix A has trace $1 + \rho$ and determinant ρ , so $D(z) = (1 - z)(1 - \rho z)$. Let $q \neq \frac{1}{2}$, so its roots 1 and $1/\rho$ are distinct. By partial fractions U_∂/D^{m+1} is a constant plus a combination of $(1 - z)^{-j}$ and $(1 - \rho z)^{-j}$ with $j \leq m + 1$, whose coefficients of z^N are $\binom{N+j-1}{j-1}$ and $\binom{N+j-1}{j-1}\rho^N$. So for $N \geq 1$ the sequence ∂m_N at 0 lies in $\mathcal{S}_m = \text{span}\{N^i, N^i \rho^N : i \leq m\}$. A sequence in $\mathcal{S}_m$ satisfies the recurrence with characteristic polynomial $(t - 1)^{m+1}(t - \rho)^{m+1}$, so it is 0 if it vanishes at $N = 1, \dots, 2m + 2$.

The moments $\mathbb{E}X^2$, $\mathbb{E}[JX^2]$ and $\mathbb{E}X^4$ are derivatives of m_N of order at most 4 at 0 , so they lie in $\mathcal{S}_4$. The claimed formulas for $\mathbb{E}X^2$, $\mathbb{E}X^4$ and $\mathbb{E}[JX^2] = \text{Cov}(J, X^2) + (N - 1)q \mathbb{E}X^2$ lie in $\mathcal{S}_2$. So it is enough to check them at $N = 1, \dots, 10$. The script `verify_sticky_fisher.py` computes the 3 moments exactly for $N \leq 12$ by a dynamic program with q as a symbol, and every difference is 0 . For fixed N both sides are rational in q with no pole at $\frac{1}{2}$, so the case $q = \frac{1}{2}$ follows. Finally, with R standing for ρ^N and ρR for ρ^{N+1} , the claim $\mathbb{E}X^4 - (\mathbb{E}X^2)^2 = 2N^2 p^2 V / q^2$ is an identity of rational functions of N , q and R , which the same script checks with `sympy`. $\square$

Proof of the closed form in Theorem 8.3. For $N \geq 2$, X^2 is N^2 on the constant paths and $(N - 2)^2$ on the paths that switch only at the last step, so $\text{Var } X^2 > 0$ and $V > 0$ by Lemma E.1, which also gives $\text{Cov}(J, X^2) = -\frac{Np}{q}[1 + \rho^N - p(1 - \rho^N)/T]$, so $\text{Cov}(J, X^2)^2 / \text{Var } X^2$ is the squared bracket over $2V$. We divide by $\text{Var } J = (N - 1)pq$. $\square$

Lemma E.2 (convexity). *Let f and g be convex and differentiable on $[t - \delta, t + \delta]$, with $0 \leq g - f \leq \eta$ and $g'' \leq \kappa$ there. Then $|f'(t) - g'(t)| \leq \eta/\delta + \kappa\delta/2$.*

Proof. By convexity and Taylor's theorem, $f'(t) \leq (f(t + \delta) - f(t))/\delta \leq (g(t + \delta) - g(t) + \eta)/\delta \leq g'(t) + \kappa\delta/2 + \eta/\delta$, and the lower bound is the same with $t - \delta$. $\square$

Lemma E.3 (a Bessel ratio). *For $t > 0$ let $R(t) = I_1(t)/I_0(t)$. Then $(\sqrt{1 + t^2} - 1)/t \leq R(t) < 1$. In particular $0 < 1 - R(t) \leq 1/t$.*

Proof. Since $I'_0 = I_1$ and $I'_1(t) = I_0(t) - I_1(t)/t$, we have $R' = 1 - R/t - R^2$, and $R(t) = t/2 + O(t^3)$ at 0 . At a first point t_0 with $R(t_0) = 1$ we would have $R'(t_0) \geq 0$, but the equation gives $-1/t_0$. So $R < 1$. Put $s = \sqrt{1 + t^2}$ and $\ell = (s - 1)/t$. Then $1 - \ell/t - \ell^2 = \ell/t = 1/(s + 1)$ and $\ell' = 1/(s(s + 1))$, so $\ell' \leq 1 - \ell/t - \ell^2$. For $d = R - \ell$ this gives $d' \geq -d/t - (R^2 - \ell^2) = -d(1/t + R + \ell)$. So Θd is nondecreasing, where $\Theta(t) = t \exp \int_0^t (R + \ell)$. As $\ell = t/2 + O(t^3)$, we have $d = O(t^3)$ and $\Theta d \rightarrow 0$ at 0 . Hence $d \geq 0$. Finally $1 - \ell = (t + 1 - s)/t \leq 1/t$. $\square$

Lemma E.4 (moments of the endpoint). (a) *For each $m \geq 1$ there is C_m with $\mathbb{E}W^{2m} \leq C_m T^{-m}$ for all $0 < q < 1$ and $T \geq 1$.*

(b) *If $0 < q \leq \frac{1}{2}$, then $\mathbb{E}W^4 \leq 3p^2/T^2 + 9/(2T^4)$.*

Proof. (a) Add a step σ_{N+1} to the walk. Since $\mathbb{E}[\sigma_{i+1} \mid \sigma_{\leq i}] = \rho\sigma_i$, the $\xi_{i+1} = \sigma_{i+1} - \rho\sigma_i$ are martingale differences. Summing $\sigma_{i+1} - \sigma_i = -2q\sigma_i + \xi_{i+1}$ over $i \leq N$ gives $X = (\sigma_1 - \sigma_{N+1} + \Xi)/(2q)$ with $\Xi = \sum_{i=2}^{N+1} \xi_i$. Also ξ_{i+1}^2 is $4q^2$ if step $i + 1$ repeats step i and $4p^2 \leq 4$ otherwise. So $\sum \xi_i^2 \leq 4q^2 N + 4J'$, where $J' \sim \text{Bin}(N, q)$ counts the switches among the $N + 1$ steps. Burkholder's inequality [28, Ch. 2] gives $\mathbb{E}\Xi^{2m} \leq c_m \mathbb{E}(4q^2 N + 4J')^m \leq c_m 8^m ((q^2 N)^m + \mathbb{E}J'^m)$, and $q^2 N \leq T$. The falling factorial moments of J' are at most T^j , and J'^m is a combination of the falling factorials

of order $j \leq m$ with nonnegative coefficients. So $\mathbb{E}J^m \leq b_m T^m$ for $T \geq 1$. As $|\sigma_1 - \sigma_{N+1}| \leq 2$, we get $\mathbb{E}X^{2m} \leq (2q)^{-2m} 2^{2m-1} (2^{2m} + \mathbb{E}\Xi^{2m}) \leq C_m q^{-2m} T^m$, and $\mathbb{E}W^{2m} = N^{-2m} \mathbb{E}X^{2m} \leq C_m T^{-m}$.

(b) We divide the formula for $\mathbb{E}X^4$ by N^4 . As $0 \leq \rho < 1$, the 2 terms of order T^{-3} are at most 0 and the one of order T^{-4} is at most $9/(2T^4)$. $\square$

Proof of the bound in Theorem 8.3. Let $\mathcal{P}$ be the orthogonal projection in L^2 onto the span of 1 and X^2 . Since X^2 is a function of K , $\text{Cov}(Z, X^2) = \text{Cov}(J, X^2)$. So $\text{Var } Z = \text{Cov}(J, X^2)^2 / \text{Var } X^2 + \mathbb{E}(Z - \mathcal{P}Z)^2$, and $e_N - R_N^2 = \mathbb{E}(Z - \mathcal{P}Z)^2 / \text{Var } J \geq 0$. Also $\mathbb{E}(Z - \mathcal{P}Z)^2 \leq \mathbb{E}(Z - G)^2$ for $G = y(1 - W^2/2) + \frac{1}{2}$. From now on $0 < q \leq \frac{1}{2}$ and $1 \leq T \leq N^{1/4}$, and C is an absolute constant that may change at each use. Then $p \geq \frac{1}{2}$, $T \leq y \leq 2T$ and $\text{Var } J = (N-1)pq \geq T/4$. It suffices to show $\mathbb{E}(Z - G)^2 \leq C(T^{-2} + T^4/N)$.

For $0 < k < N$ put $s_k = \sqrt{k(N-k)}$, $c_k = N/(2s_k)$ and $w_k = 2s_k u$. On $K = k$ we have $s_k^2 = N^2(1 - W^2)/4$, so $c_k = (1 - W^2)^{-1/2} \geq 1$, $w_k = y\sqrt{1 - W^2}$ and $c_k w_k = y$. Let $\mathcal{I}_k(z) = I_0(2z) + c_k I_1(2z)$ and $\tilde{S}_k(v) = 2v \mathcal{I}_k(s_k v)$, which approximates $S_{k, N-k}(v)$. By the uniform Bessel estimate for every bin in [2],

$$0 \leq 1 - \frac{S_{k, N-k}(v)}{\tilde{S}_k(v)} \leq \frac{Nv^2}{2} + v \quad \text{for all } v > 0. \quad (28)$$

Let $\Lambda_k(t) = \log S_{k, N-k}(e^t)$ and $\beta_k(t) = \log \tilde{S}_k(e^t)$. By Proposition 8.1, $Z = \Lambda'_K(\log u)$ on $0 < K < N$.

Step 1. Both Λ_k and β_k are convex, as logarithms of series in e^t with nonnegative coefficients. Let $|t - \log u| \leq 1$. Then $v = e^t \leq eu$ and $y \leq 2N^{1/4}$ give $Nv^2/2 + v \leq (e^2 y^2/2 + ey)/N \leq (2e^2 + 2e)/\sqrt{N} \leq \frac{1}{2}$ for $N \geq 1700$. As $-\log(1 - \varepsilon) \leq 2\varepsilon$ for $0 \leq \varepsilon \leq \frac{1}{2}$, (28) gives $0 \leq \beta_k - \Lambda_k \leq \eta$ there, with $\eta = (e^2 y^2 + 2ey)/N$. Next, $\beta_k(t) - t - \log 2$ is the logarithm of $\mathcal{I}_k(s_k e^t) = \sum_n a_n e^{nt}$ with $a_n \geq 0$. So β_k'' is the variance of n under the weights $a_n e^{nt}$, at most the mean of n^2 . As $(z \frac{d}{dz})^2$ maps $I_0(2z)$ to $4z^2 I_0(2z)$ and $I_1(2z)$ to $(4z^2 + 1)I_1(2z)$, this mean is at most $4z^2 + 1 \leq e^2 y^2 + 1 = \kappa$, since $z = s_k e^t \leq ey/2$. For $N \geq 4$, $\delta = \sqrt{2\eta/\kappa} \leq 1$ and Lemma E.2 gives

$$|\Lambda'_k(\log u) - \beta'_k(\log u)| \leq \sqrt{2\eta\kappa} \leq \frac{2e^2 y^2 + 2ey + 1}{\sqrt{2N}} \leq \frac{11(1 + y)^2}{\sqrt{N}}. \quad (29)$$

Step 2. Write $z = s_k e^t$, $w = 2z$ and $R = R(w)$. By the formulas for I'_0 and I'_1 ,

$$\beta'_k(t) = 1 + \frac{z \mathcal{I}'_k(z)}{\mathcal{I}_k(z)} = 1 + \frac{wR + c_k w - c_k R}{1 + c_k R} = w + \frac{1 + w(c_k - 1)(1 - R)}{1 + c_k R}.$$

At $t = \log u$ we have $w = w_k$, so $\beta'_k(\log u) = w_k + B_k$ with B_k the last fraction. By Lemma E.3, $0 < R < 1$ and $w(1 - R) \leq 1$. So the numerator of B_k lies in $[1, c_k]$ and is at most $1 + wc_k = 1 + y$. Hence $0 < B_k \leq 1 + y$. Subtracting $\frac{1}{2}$ from $1/(1 + c_k R) \leq B_k \leq c_k/(1 + c_k R)$ gives

$$-\frac{c_k - 1}{2} \leq B_k - \frac{1}{2} \leq \frac{(c_k - 1) + c_k(1 - R)}{2(1 + c_k R)} \leq \frac{c_k - 1}{2} + \frac{c_k^2}{2y},$$

by $c_k(1 - R) \leq c_k/w_k = c_k^2/y$. If $W^2 \leq \frac{1}{2}$ on $K = k$, then $c_k^2 \leq 2$ and $c_k - 1 \leq W^2$, since $(1 - v)^{-1/2} - 1$ is convex, vanishes at 0 and is below $\frac{1}{2}$ at $v = \frac{1}{2}$. So $|B_k - \frac{1}{2}| \leq W^2/2 + 1/y$ there.

Step 3. Let $\psi(a) = 1 - a^2/2 - \sqrt{1 - a^2}$ for $|a| \leq 1$. The series of $\sqrt{1 - v}$ is $1 - v/2 - \sum_{j \geq 2} b_j v^j$ with $b_j \geq 0$ and $\sum_j b_j = \frac{1}{2}$, so $0 \leq \psi(a) \leq a^4/2$. On $K = k$, $w_k = y(1 - W^2/2) - y\psi(W)$. So on

$0 < K < N$ we have $Z - G = (Z - \beta'_K(\log u)) + (B_K - \frac{1}{2}) - y\psi(W)$, and on the atoms $K \in \{0, N\}$ we have $Z = 0$, $W^2 = 1$ and $Z - G = -(y + 1)/2$.

Step 4. By Lemma E.4 and Markov's inequality, $\mathbb{E}W^4 \leq 8/T^2$, $\mathbb{E}W^8 \leq CT^{-4}$ and $\mathbb{P}(W^2 > \frac{1}{2}) \leq 16\mathbb{E}W^8 \leq CT^{-4}$. Also $\mathbb{P}(K \in \{0, N\}) = p^{N-1} \leq e^{-q(N-1)} \leq e^{1/2-T}$.

Step 5. By Step 3 and Minkowski's inequality, $\|Z - G\|_2$ is at most the sum of the L^2 norms of the 3 terms of Step 3 on $0 < K < N$ and of $\frac{y+1}{2}$ on the atoms. By (29) the first is at most $99T^2/\sqrt{N}$, as $(1+y)^2 \leq 9T^2$. For the second we use Step 2 where $W^2 \leq \frac{1}{2}$ and $|B_K - \frac{1}{2}| \leq 1+y$ elsewhere, so its square is at most $\mathbb{E}W^4/2 + 2/y^2 + C(1+y)^2T^{-4} \leq C/T^2$. The third is at most $\frac{y}{2}(\mathbb{E}W^8)^{1/2} \leq C/T$, and the last is $\frac{y+1}{2}\mathbb{P}(K \in \{0, N\})^{1/2} \leq CTe^{-T/2} \leq C/T$. So $\mathbb{E}(Z - G)^2 \leq C(T^4/N + T^{-2})$, and dividing by $\text{Var } J \geq T/4$ gives the bound. Only Step 1 used $N \geq 1700$, so $N_0 = 1700$ works. $\square$

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